

Pampanito FAQ

226 entries (001–226)

001: What was the purpose of the maneuvering room?

The maneuvering room was the control center for the submarine's propulsion. That's where they controlled the engine speed and the load on the engine. Speed changes and direction changes of the propellers took place here. The large levers switched the electricity from the submarine's generators to either charge the batteries or power the main propulsion motors. When submerged, the levers were changed to take power from the battery to operate the boat. The submarine's electric motors drove the twin propellers.

002: Were there crew bunks in the torpedo rooms?

Yes. Both the forward torpedo room and the after torpedo room had crew bunks located above the torpedoes. The torpedo rooms doubled as sleeping quarters for some crew members. In the forward torpedo room there are ten reload torpedoes in skids (trays), and the crew bunks are above them. In the after torpedo room, there are four reload torpedoes, and the crew bunks are also above the torpedoes. Sleeping in the torpedo room was an experience — as one crew member described it, his bedfellows were a torpedo on one side and a torpedo above him.

003: What kind of engines does the Pampanito have?

The Pampanito has four Fairbanks-Morse model 38D8-1/8 opposed-piston diesel engines. They are ten-cylinder engines with two crankshafts — one on top and one on the bottom — which is what makes them 'opposed piston.' Two engines are in the after engine room and two more are in the forward engine room. Together they produce about 5,400 horsepower. The diesel engines do not drive the propellers directly. Instead, they drive generators that produce electricity. That electricity either charges the batteries or powers the electric motors that turn the propellers. When submerged, the diesel engines cannot run because they need air; the submarine ran on battery power underwater.

004: Were submariners smarter than other sailors?

Submarine service attracted volunteers who were motivated, technically capable, and willing to serve in confined, high-risk conditions. Every submariner was a volunteer — no one was assigned to submarines against their will. Candidates had to pass a physical exam and a psychological screening to confirm they could handle the stress and close quarters of submarine life. Officers and enlisted men had to qualify in submarines, which meant learning every system on the boat well enough to operate it and save the ship in an emergency. That qualification process was demanding and not everyone made it through. Whether submariners were 'smarter' is debatable, but they were highly trained, self-selected volunteers who took great pride in their service.

005: How was food stored on board a submarine?

Food was stored throughout the boat to take advantage of every available space. The main dry goods storage room was directly below the galley deck — you accessed it through a trapdoor, and the shelves

were packed from top to bottom; a fully loaded boat filled that well all the way up. Canned food was also stowed outboard of the engines in the engine rooms, packed into the spaces around the machinery. Fresh food and frozen items were kept in a small refrigerator and freezer, also located beneath the crew's mess area under a separate trapdoor. At the start of a two-month patrol the submarine was loaded to capacity; canned goods were sometimes stacked on the deck plates and walkways until the crew ate through them and space opened up. Personal storage for sailors was extremely limited — not much room for two months at sea.

006: How were torpedo reloads handled at sea?

Each torpedo room carried spare torpedoes stored on steel skids — tracks running along the sides of the room. When a torpedo was fired and a tube needed to be reloaded, the crew first had to remove the bunks that folded down over the skids. Then three or four men would slide the torpedo along the skid track until it lined up with the open tube, and manually push and guide it in. It was heavy, awkward work in a very cramped space. The Pampanito carried a total of twenty-four torpedoes — six tubes and spare reloads in the forward torpedo room, and four tubes and spare reloads in the after torpedo room. Reloading at sea after a firing could take roughly fifteen to thirty minutes depending on sea conditions and crew experience.

007: How could a submarine run underwater if diesel engines needed air?

World War II submarines like the Pampanito were diesel-electric boats, not purely diesel-powered ones. The diesel engines could only run on the surface, where they had access to fresh air. On the surface, the diesels drove generators that produced electricity — and that electricity charged large storage batteries located below the crew berthing decks. When the submarine dived, the diesel engines shut down completely. Underwater, the boat ran entirely on battery power, using electric motors to turn the propellers. The batteries could propel the submarine silently for many hours, though speed and depth affected how long they lasted. When the batteries ran low, the submarine had to return to the surface or to periscope depth to run the diesels and recharge. During wartime patrols this made the boat vulnerable, so recharging was usually done at night. The maneuvering room controlled the switching between diesel power and battery power.

008: Why were WWII submarines usually on the surface at night?

Fleet submarines like the Pampanito ran on diesel engines while on the surface and electric motors while submerged. Because diesel engines need air to run, the boat had to surface regularly to recharge the batteries that powered the electric motors underwater. Nighttime was much safer for surfacing because aircraft couldn't spot the submarine as easily in the dark. On the surface at night, the crew could run the diesels to recharge the batteries, travel at higher speed (surface speed was about three times faster than submerged speed), ventilate the boat with fresh air, and receive radio messages. Staying submerged on battery power was necessary to avoid detection, but submarines had to surface at night to recharge and cover ground.

009: How did submarines navigate underwater without GPS?

WWII submarines navigated by a combination of dead reckoning, celestial observation, and fathometer (depth sounder) readings. Dead reckoning meant the navigator tracked the boat's known speed, heading, and time elapsed since the last confirmed position to continuously update an estimated position on the chart. Whenever the submarine was on the surface at night, the navigator could take star or celestial sights

with a sextant to get an accurate fix. Depth soundings from the fathometer could confirm position by matching the sea floor contour to navigation charts. When operating close to land, landmarks or lighthouses could also be used. Submarines accepted a degree of positional uncertainty underwater — accuracy improved each time they surfaced to take a proper celestial fix.

010: How did sonar work on a WWII submarine?

WWII submarines used passive sonar — hydrophones — to listen for the sounds of enemy ships. The sonar operator wore headphones and could hear screws turning, machinery noise, and the pinging of enemy active sonar. Passive sonar gave bearing (direction) but not range. Active sonar, called FM sonar or a fathometer, could send out a pulse and measure the echo return time to determine range, but using it revealed the submarine's position to the enemy, so submarines were very reluctant to ping actively while hiding. The sound heads (the hydrophone arrays) on fleet submarines like the Pampanito were retractable — they could be lowered below the hull keel for a clearer listening path, or raised when not in use. A skilled sonar operator could identify different types of ships, estimate their speed, and track multiple contacts.

011: Could the crew send radio messages to other submarines while submerged?

WWII submarines could not communicate by radio while submerged because radio waves do not penetrate seawater. The only form of underwater communication available was the underwater telephone, called Gertrude or the UQC, which transmitted voice through the water using sonar frequencies. However, Gertrude had very limited range — typically only a few miles — and using it risked revealing the submarine's position to the enemy. In practice, submarines operating together in wolfpacks communicated primarily by radio while on the surface at night. During submerged operations each boat was essentially on its own. Messages from fleet command were sent via low-frequency radio to submarines at periscope depth or near the surface, where the antenna could be raised just above the waterline.

012: How many ships did the Pampanito sink and what is she most famous for?

USS Pampanito (SS-383) completed six war patrols during World War II and is credited with sinking six Japanese ships and damaging four others. She is most famous for an extraordinary rescue mission in September 1944 during her third war patrol. Pampanito and other submarines in a wolfpack attacked a Japanese convoy in the South China Sea and sank several ships — not knowing that the enemy vessels were transporting British and Australian prisoners of war captured at the fall of Singapore. When the submarines returned to the area two days later, they found hundreds of survivors clinging to wreckage. Pampanito rescued 73 survivors and the entire group of submarines saved 159 Allied prisoners of war from the sea. It remains one of the most dramatic rescue operations in submarine history.

013: When was the Pampanito built and launched?

USS Pampanito (SS-383) was built at the Portsmouth Naval Shipyard in Kittery, Maine. Her keel was laid on March 15, 1943. She was launched on July 12, 1943 and commissioned into the United States Navy on November 6, 1943. After her commissioning she moved to the Pacific Fleet and began her first war patrol in mid-1944. After the war she was decommissioned and eventually became a maritime museum ship in San Francisco, where she is berthed at Fisherman's Wharf (Pier 45) as part of the San Francisco Maritime National Historical Park.

014: How did submarine families know if a sailor was still alive on patrol?

Families at home had very limited information about whether a submarine sailor was safe during a patrol. Submarines maintained radio silence for most of a patrol to avoid detection, so there were no regular communications home. Sailors could write letters, but they were held at the base until the submarine returned to port — the family couldn't receive mail until the boat came back. When a submarine arrived in port, sailors were sometimes allowed to send a brief cable home. If a submarine was overdue and presumed lost, the Navy would notify the families that the boat was missing. The phrase "on eternal patrol" was used to describe submarines and their crews that never came home. No news was generally considered good news — families knew that if the submarine were lost, the Navy would eventually notify them.

015: What is a depth charge and how did it work?

A depth charge is an anti-submarine weapon — essentially a canister filled with high explosive (TNT or Torpex) designed to sink to a set depth before detonating. Surface ships dropped them off the stern or launched them from projectors called K-guns, creating a pattern of explosions around a suspected submarine location. The fuse was set to detonate at a specific depth, calibrated before launching using a pressure-activated mechanism. The explosion didn't have to directly hit the submarine — the concussion and pressure wave from a nearby blast could rupture hull welds, crack pipes, shatter gauge glasses, and kill or injure crew members. A very close depth charge could crush the pressure hull entirely. The sound inside a submerged submarine under depth charge attack was described by survivors as deafening and terrifying — every loose object flew, lights shattered, and the whole hull shuddered with each explosion.

016: Were submarine crews paid more than regular Navy sailors?

Yes, submarine sailors received extra pay specifically because of the dangerous and demanding nature of submarine service. This was called submarine pay or hazardous duty pay. Officers received an extra 25% of their base pay, and enlisted men received an extra \$5 per month (increased to \$10 in later years of the war), which was meaningful money at the time. In addition, submariners received extra food allowances — because submarines were too small to carry a supply officer, the cooks could purchase higher-quality food directly, which is one reason submariners were famous for eating better than most sailors in the Navy. The combination of higher pay, better food, and a strong sense of camaraderie made submarine service attractive despite the obvious dangers.

017: How did the crew know where they were during a patrol?

WWII submarines navigated by a combination of dead reckoning, celestial observation, and fathometer (depth sounder) readings. Dead reckoning meant the navigator tracked the boat's known speed, heading, and time elapsed since the last confirmed position to continuously update an estimated position on the chart. Whenever the submarine was on the surface at night, the navigator could take star or celestial sights with a sextant to get an accurate fix. Depth soundings from the fathometer could confirm position by matching the sea floor contour to navigation charts. When operating close to land, landmarks or lighthouses could also be used. Submarines accepted a degree of positional uncertainty underwater — accuracy improved each time they surfaced to take a proper celestial fix.

018: How did the crew breathe fresh air inside a submerged submarine?

When a WWII submarine dived, it sealed itself off from the outside air and ran on a fixed supply. The crew breathed the same recirculated air throughout the boat. Carbon dioxide — exhaled by the men — was absorbed using soda lime canisters, a chemical process called CO₂ scrubbing. Oxygen was replenished from high-pressure oxygen flasks stored on board, slowly bled into the atmosphere as needed. Hydrogen gas, a dangerous byproduct of the charging batteries, was another hazard that required careful ventilation whenever possible.

The air grew progressively stale and thick on a long dive. In a serious depth-charge attack that forced the boat to stay down for many hours, CO₂ levels could rise to dangerously uncomfortable levels — heads throbbed, men moved slowly, speech slurred. The submarine's time underwater was thus limited by both battery charge and breathable air. As soon as conditions permitted, the boat surfaced and opened the main induction valve to flush the interior with fresh sea air.

019: How did a WWII torpedo work?

A WWII torpedo was a self-propelled underwater weapon about 21 inches in diameter and 20 feet long. The standard US weapon was the Mark 14 steam torpedo, driven by a turbine burning a mixture of compressed air, water (converted to steam), and alcohol fuel. It traveled at about 45 knots and had a range of around 4,500 yards on its high-speed setting.

A gyroscope inside kept it running straight on a preset heading at a preset depth, maintained by a hydrostatic valve that adjusted the rear fins. The warhead detonated on contact via a firing pin, or in theory via a magnetic influence exploder that was supposed to trigger beneath the hull of a ship.

Early in the war, US torpedoes were plagued by two serious defects: the magnetic exploder fired prematurely or not at all, and the contact exploder pins would shear on a direct hit, also failing to fire. These flaws went unacknowledged for over a year and cost countless opportunities. By 1943 the problems were finally corrected, and torpedo performance improved dramatically for the rest of the war.

020: What weapons did submarines carry besides torpedoes?

WWII US fleet submarines carried a deck gun as their main secondary weapon — typically a 4-inch or 5-inch gun mounted forward of the conning tower, plus 40mm and 20mm anti-aircraft guns on the bridge. The Pampanito carried a 4-inch .50-caliber deck gun.

The deck gun was used to sink small surface targets — sampans, fishing vessels, lifeboats transporting troops, and small coastal freighters — when using a torpedo would be wasteful. It was also used to finish off ships that had been torpedoed but refused to sink. Surface gun attacks required the submarine to remain exposed on the surface within range of return fire, so they were reserved for situations where the risk was acceptable.

Anti-aircraft guns provided defense against low-flying patrol aircraft, which were a constant threat to surfaced submarines, especially as the war progressed and Allied airpower increased in the Pacific. By late 1944 many submarine losses were to aircraft, making anti-aircraft capability critical.

021: How did the captain decide to fire torpedoes at a target ship?

A submarine attack began with detection — a sonar contact, a periscope sighting, or a radio tip about a convoy. The captain would track the target by taking repeated periscope observations, noting bearing,

estimated speed, and course. The Torpedo Data Computer (TDC), an electromechanical analog computer in the conning tower, continuously calculated the firing solution: the correct torpedo gyro angle to intercept the target.

The submarine maneuvered to close the range and achieve a favorable firing angle — ideally perpendicular to the target's course, known as a 90-degree track, giving the best chance of a hit. When the captain was satisfied with the solution, he ordered a spread of two to four torpedoes fired with a small angular spread timed to compensate for any course errors.

After firing, the submarine immediately went deep and turned away to evade the counterattack from destroyer escorts. The depth charges that followed could last for hours. A successful attack used four to six of the submarine's twenty-four torpedoes, so fire discipline was essential — every torpedo had to count.

022: Where is the Pampanito?

USS Pampanito (SS-383) is permanently moored at Pier 45, Fisherman's Wharf, San Francisco, California. She is operated as a museum ship by the Maritime Park Association and is part of the San Francisco Maritime National Historical Park.

Visitors can board and walk through the entire length of the submarine, from the after torpedo room to the forward torpedo room. A self-guided audio tour — the same one you may be listening to right now — is available and takes approximately 45 to 60 minutes. The tour features the voices of officers and crew who actually served on Pampanito and other US submarines during World War II.

Admission is charged; current hours and ticket prices are posted at the pier. The ship is ADA accessible to a limited extent on the exterior decks, but the interior requires climbing through narrow hatches and ladder access, as on the original vessel.

023: What class of submarine is the Pampanito and what does that mean?

USS Pampanito (SS-383) is a Balao-class fleet submarine, one of the most-produced submarine types of World War II. The Balao class was an evolution of the earlier Gato class, with the primary improvement being a thicker, higher-yield steel pressure hull that increased the safe test depth from 300 feet (Gato) to 400 feet. This extra margin was operationally significant — it gave crews more room to dive away from depth charges.

Balao-class submarines were approximately 311 feet long, displaced around 2,400 tons submerged, and carried a crew of 80 to 85 men. They were armed with ten torpedo tubes (six forward, four aft) and carried 24 torpedoes plus a deck gun and anti-aircraft guns. Over 120 Balao-class submarines were commissioned during the war, making them the backbone of the US Pacific submarine fleet.

024: What happened to the Pampanito after WWII ended?

After Japan's surrender in August 1945, USS Pampanito was decommissioned from active service in 1946. She spent years in the Pacific Reserve Fleet before being reassigned to the Naval Reserve as a training vessel, giving future submariners hands-on experience in a fleet boat.

In 1976 she was transferred to the city of San Francisco and the National Maritime Museum Association (later the Maritime Park Association), which undertook years of restoration work to preserve her in a

condition as close as possible to her wartime configuration. She was opened to the public as a museum ship and is today berthed permanently at Pier 45. In 1986 she was designated a National Historic Landmark.

Pampanito is one of only a handful of WWII fleet submarines still in existence and open for public tours, making her an irreplaceable piece of American naval history.

025: How deep could a WWII submarine dive?

The Balao-class submarines like the Pampanito had a designed test depth of 400 feet — the depth the Navy certified as safe for normal operations. The maximum "crush depth," where the hull would fail under sea pressure, was estimated at roughly 600 feet or more, but no one wanted to find out exactly where that line was.

In practice, submarines under depth-charge attack sometimes went well below their rated test depth in desperation, and some survived. The hull creaked and groaned under the added pressure, valves leaked, and lights flickered — but well-constructed boats sometimes made it back. Others did not. Gato-class submarines, the earlier generation, had a shallower test depth of 300 feet, making them more vulnerable.

The ability to dive quickly and deeply was one of the submarine's most important survival tools when attacked by surface ships.

026: How fast could submarines go?

On the surface, a fleet submarine like the Pampanito could make about 20 knots (roughly 23 mph) using her four diesel engines — fast enough to race ahead of most convoys and get into position for an attack. Transit speed on long patrols was typically slower, around 10 to 14 knots, to conserve fuel.

Submerged, the submarine ran on battery power and was considerably slower. Top speed underwater was about 8 to 9 knots, but that pace drained the batteries within an hour. A sustainable submerged patrol speed was 2 to 3 knots, which could stretch battery life to 48 hours or more before requiring surfacing to recharge.

This speed gap between surface and submerged operation was a defining tactical limitation of WWII diesel submarines. A submarine that found itself on the wrong side of a fast destroyer could not outrun it underwater — the only option was to dive deep and evade, or be very lucky.

027: What was the watch schedule like on a WWII submarine?

Life on a WWII submarine was organized around four-hour watches rotating around the clock: midnight to 4 AM, 4 to 8 AM, 8 to noon, noon to 4 PM, 4 to 8 PM, and 8 PM to midnight. The crew was divided into three watch sections, with about one-third of the men on duty at any time. The other two-thirds ate, slept, maintained equipment, or stood by.

Sleep came whenever it could — WWII submariners rarely slept a full eight-hour stretch. Bunks were shared in rotation, a practice called hot-racking: one man came off watch, and the bunk he took was still warm from the man who had just left for duty. There was no such thing as a personal bunk on a crowded fleet boat.

When the alarm sounded for general quarters — battle stations — every man in the crew went immediately to his designated post, regardless of what he had been doing. After a night attack and depth-charge

evasion, some men might go eighteen hours without sleep.

028: How long did it take to train a submarine crew?

Submarine training in WWII was compressed by wartime urgency. After completing 8 weeks of basic Navy training, enlisted volunteers attended the Navy Submarine School in New London, Connecticut — a course of roughly three months that covered diving, escape procedures, submarine systems, and watch-standing duties. Officers went through a separate three-month program tailored to their duties.

After reporting to an actual boat, a new crew member began the qualification process — earning the right to wear the submarine warfare insignia, the "dolphins." Qualification meant demonstrating that you understood and could operate every system on the submarine well enough to take over in an emergency. You had to be able to close any valve, operate any piece of machinery, and explain what every pipe and wire did. This typically took another four to six months aboard.

Submarine service was always voluntary. Any man who decided submarine duty was not for him was transferred ashore without prejudice — the Navy did not want reluctant men in an environment where everyone had to be completely committed.

029: How could submariners escape from a submarine sinking on the bottom?

Escape from a sunken submarine was deeply difficult and rarely successful. The main individual device was the Momsen lung, a small rubber rebreather carried by each sailor. To use it, crew members entered the escape trunk — a small compartment with a lower and upper hatch — flooded it to equalize pressure with the sea, then opened the upper hatch and ascended while exhaling continuously to prevent their lungs from over-expanding as water pressure decreased.

The method was practiced and did work at depths up to about 200 feet. Beyond that, the physiological risks of rapid ascent (air embolism, decompression sickness) became severe. Newer submarines also had built-in escape trunks large enough for one or two men at a time.

In practice, the vast majority of the 52 US submarines lost in WWII went down suddenly — explosions, depth charges, aerial bombs — giving the crew no time to use escape equipment. Of the more than 3,500 American submariners killed in the war, almost none escaped from boats sunk in combat. Escape training was critical, but survival was more often a matter of the submarine not getting hit in the first place.

030: How did submarine crews manage hygiene and bathing on a long patrol?

Fresh water on a WWII submarine was extremely scarce. Every drop was either carried in tanks at the start of patrol or produced by the sub's small evaporator, and the priority was always drinking water for the crew and cooling water for the engines. There was almost none left over for bathing.

Showering was a luxury most submariners went without for weeks at a time. When a wash was possible, it was a quick sponge bath or a brief rinse — not a real shower. Laundry was equally impossible; clothes were worn repeatedly and then stuffed away. The boat itself was permeated with the combined smells of diesel fuel, machine oil, cooking, and unwashed men in a confined space.

Veterans recall that they could tell when a submarine was returning to port not just by sight but by smell. After a 60-day war patrol, the men coming off the boat looked — and needed — a very long shower, a hot meal, and a few days' rest ashore before they looked anything like themselves again.

031: Were women allowed to serve on submarines in WWII?

No. During World War II, the US Navy did not allow women to serve on submarines or on any combat vessel. Women served in the WAVES (Women Accepted for Volunteer Emergency Service), working in crucial shore-based roles — as cryptographers, mechanics, yeomen, air traffic controllers, and in other positions that freed men for sea duty. Their contributions to the war effort were significant, but their assignments were on land or at naval stations, not aboard ships.

Practical realities reinforced the policy on submarines: a fleet boat on a 60-day patrol had no privacy whatsoever, minimal sanitation facilities, 80-plus men in a steel tube the size of an apartment building's hallway, and living conditions that were barely tolerable even for people who had volunteered for exactly that.

The US Navy finally began integrating women into the submarine service in 2010, starting with officers. Enlisted women followed in 2012. Today women serve throughout the submarine fleet, including on ballistic missile submarines.

032: Did submarines have a deck gun aboard?

WWII US fleet submarines carried a deck gun as their main secondary weapon — typically a 4-inch or 5-inch gun mounted forward of the conning tower, plus 40mm and 20mm anti-aircraft guns on the bridge. The Pampanito carried a 4-inch .50-caliber deck gun.

The deck gun was used to sink small surface targets — sampans, fishing vessels, lifeboats transporting troops, and small coastal freighters — when using a torpedo would be wasteful. It was also used to finish off ships that had been torpedoed but refused to sink. Surface gun attacks required the submarine to remain exposed on the surface within range of return fire, so they were reserved for situations where the risk was acceptable.

Anti-aircraft guns provided defense against low-flying patrol aircraft, which were a constant threat to surfaced submarines, especially as the war progressed and Allied airpower increased in the Pacific. By late 1944 many submarine losses were to aircraft, making anti-aircraft capability critical.

033: How did a submarine attack a ship?

A submarine attack began with detection — a sonar contact, a periscope sighting, or a radio tip about a convoy. The captain would track the target by taking repeated periscope observations, noting bearing, estimated speed, and course. The Torpedo Data Computer (TDC), an electromechanical analog computer in the conning tower, continuously calculated the firing solution: the correct torpedo gyro angle to intercept the target.

The submarine maneuvered to close the range and achieve a favorable firing angle — ideally perpendicular to the target's course, known as a 90-degree track, giving the best chance of a hit. When the captain was satisfied with the solution, he ordered a spread of two to four torpedoes fired with a small angular spread timed to compensate for any course errors.

After firing, the submarine immediately went deep and turned away to evade the counterattack from destroyer escorts. The depth charges that followed could last for hours. A successful attack used four to six of the submarine's twenty-four torpedoes, so fire discipline was essential — every torpedo had to count.

034: Can I visit the Pampanito?

USS Pampanito (SS-383) is permanently moored at Pier 45, Fisherman's Wharf, San Francisco, California. She is operated as a museum ship by the Maritime Park Association and is part of the San Francisco Maritime National Historical Park.

Visitors can board and walk through the entire length of the submarine, from the after torpedo room to the forward torpedo room. A self-guided audio tour — the same one you may be listening to right now — is available and takes approximately 45 to 60 minutes. The tour features the voices of officers and crew who actually served on Pampanito and other US submarines during World War II.

Admission is charged; current hours and ticket prices are posted at the pier. The ship is ADA accessible to a limited extent on the exterior decks, but the interior requires climbing through narrow hatches and ladder access, as on the original vessel.

035: How many people were in a WWII submarine crew?

A US fleet submarine like the Pampanito carried about 80 men — roughly 10 officers and 70 enlisted sailors. The Balao class, to which Pampanito belongs, had a maximum complement of 85, though wartime patrols often ran slightly under that.

In a vessel 311 feet long and less than 28 feet at its widest, those 80 men lived, slept, worked, ate, and stood watch in conditions of extraordinary crowding. There were never enough bunks for everyone at the same time, which is why the "hot bunk" system was used — a sailor coming off watch would climb into the bunk just vacated by someone going on watch, the mattress still warm.

Despite the cramped conditions, a submarine crew functioned as a highly trained team. Every man had to qualify in every major system aboard — not just his own job. In an emergency, you might need any sailor to operate any piece of equipment. The Navy's word for this was "qualified in submarines," and earning it was a point of pride.

036: What did the crew do for fun on a long patrol?

The crew's mess — the small dining area in the middle of the boat — doubled as the only real recreation space on a fleet submarine. When men weren't eating, sleeping, or standing watch, this is where they gathered.

Card games were the most popular pastime: poker, cribbage, acey-deucey (a Navy version of backgammon), and checkers. Many boats had a small record player or phonograph for music. Reading paperback books was common — crewmen shared novels and magazines until they fell apart. In the later years of the war, some submarines were equipped with a 16mm movie projector, and a rotation of films would be screened in the crew's mess.

Patrol days were long and mostly routine. Ninety percent of a patrol was spent transiting to and from the patrol area, or waiting submerged during daytime hours. Men on the day watch might have several hours of free time. The boredom was interrupted by drills, maintenance, and — when it came — the sudden

controlled chaos of a potential contact or attack. Veterans often recall that the monotony was harder to endure than the danger.

037: What was the toilet like on a submarine?

On a submarine, the toilet is called the “head” — standard Navy terminology. A fleet submarine had two heads: one for the officers and one for the crew. For 80 men, that was very limited.

Using the head on a submerged submarine was not simple. Because the boat was under pressure, the waste had to be blown outboard against sea pressure using compressed air — you couldn’t just flush. The procedure involved a specific sequence of valves and levers, and signs were posted in each head explaining the steps. Getting the sequence wrong could result in sewage being blown back into the boat, covering the unfortunate user. Most veteran submariners have a story about this happening to someone.

At diving depths, the procedure became more complicated because higher pressure was required. Some submarines had sanitary tanks that held waste until the boat surfaced, when it could be discharged normally. The combination of the small number of heads, the complicated procedure, and the need to queue during busy periods made the head situation one of the reliable inconveniences of submarine life.

038: How did a submarine surface from underwater?

To surface, a submarine “blew” its ballast tanks — forcing out the seawater that made the boat heavy enough to submerge in the first place. High-pressure air stored in large banks of cylinders on the submarine was routed into the main ballast tanks, pushing the water out through flood holes in the bottom. As the tanks filled with air, buoyancy increased and the boat rose toward the surface.

Once the submarine breached — broke the surface — the bridge watch would open the conning tower hatch as soon as it was safe, and the lookouts and officer of the deck would climb topside to begin watching for aircraft and other threats. Surfacing was a vulnerable moment because the boat wasn’t yet moving fast enough to dive quickly if a threat appeared.

The diesel engines would be started as soon as the induction mast was clear of the water, beginning the process of charging the batteries for the next submerged period. On the surface, the diesels ran constantly — the submarine used one or two engines to drive the boat and the others to charge batteries.

039: What was the procedure for an emergency dive?

An emergency dive — called a “crash dive” — was practiced constantly until the entire crew could execute it in their sleep. Typically triggered by a sighting of an enemy aircraft or surface ship, the goal was to get below the surface as fast as possible.

The diving alarm — two blasts of the klaxon, the distinctive “ah-OOH-ga” sound — was the signal. Every man on the bridge scrambled below in a fixed order: lookouts first, then the officer of the deck, then the captain last, pulling the hatch shut behind him. The main ballast tank vents were opened and water flooded in. Full negative tanks were flooded for additional weight. The diving planes were angled steeply. The diesel engines were secured and the electric motors engaged.

A well-drilled fleet submarine crew could go from surfaced to periscope depth — roughly 63 feet — in 30 to 60 seconds. Getting those extra few seconds could be the difference between escaping and being bombed. Submariners drilled on the crash dive over and over again until every motion was automatic.

040: What is a periscope and how was it used?

A periscope is an optical instrument — essentially a long tube with prisms and lenses — that allows a submarine to observe the surface while remaining underwater. On a fleet submarine, the periscope extended from the conning tower and could be raised up through the hull to just above the waterline, allowing the captain to see the horizon without exposing the boat.

Fleet submarines typically carried two periscopes: a thicker “search” periscope with higher magnification for long-range observation and low-light use, and a thinner “attack” periscope used during torpedo approaches. The attack scope presented a smaller surface signature — less wake, harder for a sharp-eyed lookout to spot.

The captain “shot” periscope bearings by raising the scope for only a few seconds at a time, taking a bearing and range estimate on the target, then immediately lowering it. Extended exposure risked detection. The data was fed into the Torpedo Data Computer in the conning tower, which calculated the torpedo firing solution. The rule was: keep the scope up as little as possible. A careless periscope look was an invitation to a depth charge attack.

041: What is a conning tower?

On a WWII fleet submarine, the conning tower was a small cylindrical compartment — barely large enough for a handful of men — located between the bridge above and the control room below. It was the nerve center of an attack: the periscopes, the Torpedo Data Computer (TDC), the radar display, and the main helm were all located here.

During a torpedo attack, the captain stood at the periscope in the conning tower while the TDC operator tracked the target and calculated the firing solution. The fire control party — captain, exec, TDC operator, and assistant — worked in very close quarters. It could get intense.

The word “conning tower” is also sometimes used loosely to refer to the entire raised structure above the main deck — the “sail” or “fairwater” — which contains the bridge at the top. Strictly speaking, the conning tower is the pressure hull compartment inside that structure, while the outer shell is called the fairwater. On the Pampanito, you can visit the actual conning tower compartment and see the periscopes and equipment in place.

042: How did they control the depth of a submarine?

A submerged submarine used two systems in combination to control its depth.

The first was buoyancy: by flooding or pumping out water from ballast and trim tanks, the submarine could be made slightly heavier or lighter than the water it displaced. When properly “trimmed,” the boat was in neutral buoyancy at depth, neither rising nor sinking on its own.

The second system was the diving planes — pairs of horizontal fins, one set near the bow and one near the stern. Like the elevators on an airplane, the planes changed the angle of the hull and drove the boat up or down when it was moving forward. The planesmen — usually two sailors sitting side by side in the control room — operated these controls constantly, making small adjustments to hold the ordered depth. Holding a steady depth in rough water or changing thermoclines required constant attention.

The two systems worked together. Ballast and trim tanks gave coarse buoyancy control. The planes gave fine dynamic depth control while the boat was underway.

043: What were the biggest dangers for submarine crews?

The submarine service was the most dangerous assignment in the US Navy during World War II. Roughly one in five US submariners who went on patrol did not return — a casualty rate exceeded by no other branch of the American armed forces.

The primary threat was enemy anti-submarine warfare: destroyer escorts equipped with sonar and depth charges. A depth charge had to explode within about 50 feet to be lethal, but a sustained pattern from multiple destroyers could keep a submarine deep for hours, exhausting batteries and crew alike, until a critical failure forced it up. Enemy aircraft — especially later in the war — caught submarines on the surface and bombed or strafed them.

Mines were another major threat, especially in shallow coastal waters. Other boats were lost to operational accidents: a circular run by a malfunctioning torpedo, a diving casualty in training, or a collision. And there was always the possibility of a patrol going simply “missing” with no known cause — about eight boats fall in that category. Whatever the cause, if you weren’t coming home, no one would know until you were long overdue.

044: What caused the most submarine losses in WWII?

Of the 52 US submarines lost in World War II, the single largest cause was enemy anti-submarine warfare — depth charges and bombs from Japanese surface ships and aircraft. Destroyers and destroyer escorts were the submarine’s natural enemy, armed with sonar to track submerged boats and depth charges to attack them.

Mines accounted for a smaller but significant number of losses, particularly in the early part of the war when charting of enemy minefields was incomplete. Some boats were lost entering or exiting enemy harbors and coastal waters.

A handful of submarines were lost to operational accidents — training accidents, diving casualties, or mechanical failures. A few were lost to friendly fire. And for about eight US submarines that disappeared during the war, the cause was never determined, and they remain “lost to causes unknown.” Their crews are carried on the Navy’s rolls as “on eternal patrol.”

Japanese anti-submarine tactics improved significantly as the war progressed. In 1942 and early 1943, very few US submarines were lost. By 1944 and 1945, as Japan’s ASW forces became better equipped and submarines operated increasingly in shallow, confined waters, the loss rate climbed.

045: Were American torpedo problems fixed during the war?

Yes, but it took nearly two years of war and enormous frustration to sort out. The standard US submarine torpedo in 1941 — the Mark 14 — had at least three serious defects, each of which took time to identify and correct.

The first problem was depth-keeping: the Mark 14 ran about 10 feet deeper than set, meaning torpedoes aimed at a ship’s waterline passed harmlessly under the keel. This was confirmed by testing in 1942 and corrected.

The second problem was the Mark 6 magnetic exploder — designed to detonate under a ship's hull using its magnetic field. It was far too sensitive and detonated prematurely or not at all. Submarines began deactivating it independently in 1943, and the Navy officially ordered it deactivated in October 1943.

The third problem was the contact exploder's firing pins: when a torpedo struck at a 90-degree angle (a direct hit), the impact was so forceful that the firing pin mechanism crushed instead of firing. This was identified and fixed in mid-1943 with stronger pins and modified angles.

By late 1943 and into 1944, all three fixes were in place. US torpedo reliability improved dramatically, and the submarine force's effectiveness surged accordingly.

046: Who was the captain of the Pampanito?

The Pampanito served under two commanding officers during her six war patrols in World War II.

Lieutenant Commander Charles B. Jackson Jr. commanded Pampanito on her first two patrols, beginning when the boat commissioned in November 1943. Under Jackson, Pampanito conducted patrols in the Western Pacific and established herself as an effective boat.

Lieutenant Commander Paul E. Summers took command before the third patrol and led Pampanito through patrols three, four, five, and six. Summers commanded the boat during its most famous action: the September 1944 patrol in which Pampanito, operating with a wolf pack, helped attack a Japanese convoy — and then discovered that the convoy was carrying British and Australian prisoners of war. Pampanito rescued 73 men from the water in one of the most dramatic humanitarian missions in submarine history. Summers was awarded the Navy Cross for this patrol.

Both commanders were decorated for their service. The Pampanito was awarded six battle stars for her WWII service.

047: What is the silent service?

"The Silent Service" is the traditional nickname for the United States submarine force. The name has two meanings, both rooted in how submarines operated in World War II.

The first meaning is tactical. Submarines depended on silence and stealth. They operated alone, deep, and invisible. Every sound the crew made could be heard by enemy sonar. When submerged and evading, crew members spoke in whispers, removed their shoes, and tiptoed through the boat. The machinery was quieted as much as possible. The submarine's survival depended on being quieter than the water around it.

The second meaning is institutional. Submarine operations were classified. Crews returning from patrol could not tell their families where they had been, what they had done, or what ships they had sunk. A submariner could not explain why he couldn't talk about his service; he could only say he couldn't. The men came home, said nothing, and went out again. For years after the war, much of the submarine record remained classified.

Together, those two meanings — tactical stealth and institutional secrecy — gave the submarine force its enduring name.

048: How old were the men who served on submarines?

Submarine crews during World War II were young by any measure. Enlisted men were typically 17 to 24 years old, with many in their late teens or early twenties. Officers were somewhat older — ensigns fresh from submarine school might be 22 or 23, while a commanding officer was usually in his late twenties or early thirties. By the standards of other military commands, submarine captains were very young men.

All submarine service was voluntary. The Navy screened heavily for physical fitness, mental aptitude, and temperament. Men had to pass a demanding physical examination — good eyesight, hearing, and psychological screening for tolerance of confined spaces. Then they had to complete submarine school and actually earn their qualifications at sea.

The youth of the crews is perhaps surprising given the demands placed on them. These were teenagers and young men in their early twenties operating some of the most complex fighting machinery of the era, in life-or-death situations, far from home on patrols lasting two months. Veterans who survived the war often speak of being astonished, in retrospect, at how young they were — and how little they worried about it at the time.

049: Were there African American sailors on WWII submarines?

During World War II, the US Navy was racially segregated. African American sailors were largely confined to the messman and steward branches — the cooks and personal servants for officers — across all seagoing branches, including submarines. Very few Black sailors served in technical or combat roles in the submarine force during the war.

The submarine service reflected the Navy's broader segregation policy, established by regulations and reinforced by long-standing tradition. Even those African American sailors who did serve aboard submarines in steward roles often faced discrimination in shore facilities and port towns.

President Truman's Executive Order 9981 in 1948 mandated integration of all armed forces, and the Navy moved toward full integration in the early 1950s. Today the submarine force is fully integrated throughout its ranks, and women have served on submarines since 2010.

050: How did submarines get fuel and supplies during a patrol?

US fleet submarines were largely self-sufficient for the duration of a war patrol — typically 45 to 75 days. They departed with full fuel tanks, a full torpedo load, and enough food to feed 80 men for the patrol. Replenishment at sea was possible but not routine.

The primary supply bases were Pearl Harbor, Hawaii, and — as the war advanced westward — Midway Atoll, Majuro, Guam, and other forward bases. Submarine tenders — large ships like USS Fulton or USS Proteus — were moored at these bases and served as floating machine shops, supply depots, and rest facilities for crews between patrols. When a submarine returned from patrol, the tender took care of everything: fuel, torpedoes, food, spare parts, and repair of any damage.

At-sea transfers from tenders or other submarines were technically possible and occasionally done for special missions but were logistically complex and risky. For a normal war patrol, the submarine carried everything it needed and relied on nothing until it returned to port.

051: What does it smell like inside a submarine?

Veterans who served on WWII submarines are remarkably consistent in their answer: diesel fuel and unwashed men, with various undertones depending on where in the boat you were.

The dominant smell was diesel. The engines were inside the boat, the fuel was in tanks throughout the hull, and every surface absorbed the smell of petroleum over time. Clothes, blankets, food — everything took on a diesel-and-oil character after a few days at sea. The engine rooms were the worst, but the smell permeated the entire boat.

Add to that: 80 men in a sealed metal tube for two months with almost no fresh water for bathing, no laundry, and cooking happening around the clock. When the submarine was charging batteries, hydrogen gas from the battery cells added a sharp chemical note. The galley contributed cooking smells. In the after battery room, where many of the crew bunked, a faint battery acid smell was always present.

Long-time veterans say they stopped noticing it after a day or two. The sailors on the dock welcoming a returning submarine noticed it very clearly from a distance.

052: How did submarine crews get mail from home?

While a submarine was at sea on patrol, there was no mail delivery. The boat was completely radio-silent except for receiving encoded operational messages, and those messages carried no personal correspondence. For 60 or 70 days at sea, a submariner had no news from home whatsoever.

Mail sent to submarines was held at the submarine base — usually addressed to the boat's name in care of Fleet Post Office. When the submarine returned from patrol and tied up to the submarine tender, weeks of accumulated letters and packages were waiting.

The volume of letters a man received on return from patrol was often very large, and it was common to spend the first day or two just reading. For crewmen with wives, children, or parents back home, those letters were a lifeline to the world they had left behind.

Families were told to write frequently because letters would arrive in batches — but they had no reliable way to know when their sailor was due back, or whether he was still alive. The uncertainty was one of the hardest parts for families.

053: What happened when a submarine returned to port?

Returning from a war patrol was one of the defining experiences of submarine service. After 60 or more days at sea, the boat tied up to its submarine tender, and the transition from patrol to port began immediately.

The first pleasures were simple and powerful: fresh milk, fresh fruit, fresh baked bread — food that had no place on a patrol, where everything was canned or frozen. A hot shower. A real bed that didn't move. A beer ration issued by the Navy. The submarine tender's crew took over the boat for repairs and provisioning, and the submarine's crew was free to rest.

A formal debriefing followed, where the officers reported the patrol's results and any intelligence observations. Then came a rest period — typically two weeks at a rest camp at a location like the Royal Hawaiian Hotel in Honolulu, or a camp on Guam or Midway. Swimming, food, drinks, and sleep.

After rest, the crew returned to the tender to supervise the reloading of torpedoes, provisioning, and repairs before the next patrol began. The cycle — patrol, rest, prepare, patrol again — continued until the war ended or the boat was lost.

054: What is a wolf pack?

A wolf pack is a coordinated group of submarines assigned to attack the same convoy or patrol area together. The tactic was pioneered by Germany's Admiral Karl Dönitz, who used large packs of U-boats in the Atlantic to overwhelm convoy defenses that could only track one submarine at a time.

The US Navy developed its own version, called "coordinated attack groups," typically two to four submarines working together. Unlike the German wolf packs, which were directed by a shore headquarters, American groups operated with more independence, coordinating by radio once contact was made. The idea was the same: while the escorts chased one submarine, the others could attack from different directions.

The Pampanito participated in a coordinated attack group on its famous third war patrol in September 1944, operating with USS Growler (SS-215) and USS Sealion II (SS-315). They attacked a Japanese convoy — and that attack led directly to the discovery of hundreds of British and Australian prisoners of war in the water, setting off the dramatic rescue for which the Pampanito is best known.

American wolf packs contributed significantly to the destruction of Japan's merchant fleet, cutting off the supply lines that kept Japan's island garrisons and war economy alive.

055: Did the Pampanito rescue people from the ocean?

Yes — it is one of the most dramatic rescue operations in submarine history. On its third war patrol in September 1944, the Pampanito was part of a wolf pack that attacked Japanese convoy HI-72 in the South China Sea. What the submarines did not know at the time was that two ships in the convoy — the Rakuyo Maru and Kachidoki Maru — were carrying over 2,200 British and Australian prisoners of war being transported to labor camps.

After the attack, Pampanito surfaced and began finding survivors in the water — gaunt, oil-covered men who turned out to be Allied POWs. Commander Paul Summers began pulling men aboard immediately. Over the course of about nine hours, Pampanito rescued 73 British and Australian survivors. USS Sealion II rescued an additional 54. Other submarines and surface ships reached the area and rescued more.

Of the 2,200 POWs who had been aboard those ships, approximately 159 were rescued. The rest perished — many drowned; others died in the days after rescue from the effects of years of brutal captivity. Pampanito carried her 73 survivors back to port, with the injured crowding every available space on the boat. The rescue is memorialized in the exhibits aboard the Pampanito today.

056: What happened to men who were killed on a submarine patrol?

If a sailor died while the submarine was at sea on patrol — from combat injury, accident, or illness — he was buried at sea. There was no option to preserve the body or return it to port: submarines had no morgue, and a 60-day patrol left no practical alternative. The captain would conduct a brief ceremony, and the man would be committed to the ocean.

For families at home, the process was agonizing in its uncertainty. The Navy did not broadcast results from individual submarines, and any message home would have compromised operational security. Families received only a telegram from the Navy when a man's status changed. If a submarine went missing and was overdue, the families of every man aboard received a "missing in action" notification — but no explanation and no certainty.

Of the 52 US submarines lost in World War II, most were lost with all hands, their crews still aboard somewhere in the Pacific. The submarine community calls them "on eternal patrol." In total, approximately 3,500 US submariners died in the war — the highest loss rate, per capita, of any branch of the American armed forces.

057: What happened to the Japanese submarine force in WWII?

Japan entered World War II with one of the largest submarine fleets in the world — over 60 ocean-going submarines, with more under construction. In some respects Japanese submarines were technically impressive, and Japanese torpedoes (the oxygen-powered Type 91 and Type 95) were among the best in the world.

But the Imperial Japanese Navy made a strategic choice that limited its submarines' impact: instead of using them primarily against Allied merchant shipping — the approach that strangled Japan's own supply lines — Japanese doctrine emphasized attacks on warships. They achieved notable successes (the heavy cruiser Indianapolis was sunk by submarine I-58 in July 1945, one of the worst naval disasters in US history), but Japanese submarines never seriously threatened Allied supply lines across the Pacific.

As the war progressed, Japanese submarines were increasingly diverted to supply missions — resupplying isolated island garrisons that could no longer be reached by surface ship. This consumed much of their offensive potential.

By the war's end, the vast majority of Japan's submarine fleet had been sunk by US destroyers and aircraft, aided by Allied signals intelligence that often revealed their positions.

058: How did US submarines compare to German U-boats?

American fleet submarines and German U-boats were the two most effective submarine forces of World War II, but they were built for very different purposes and operated in completely different theaters.

US fleet submarines like the Pampanito were much larger: about 310 feet long with a displacement of over 1,500 tons submerged, compared to roughly 220 feet and 870 tons for the standard German Type VIIC U-boat. The greater size gave US submarines much longer range (over 11,000 miles on the surface), better habitability, and air conditioning that made long Pacific patrols bearable in tropical heat.

German U-boats were purpose-built for the North Atlantic — shorter-range operations from European bases against dense convoy routes. They operated in large wolf packs of 10 to 20 submarines directed from shore headquarters, with tight coordination. Germany built nearly 1,000 U-boats during the war, compared to about 200 US fleet submarines.

Both forces were devastating in their own theaters. US submarines sank approximately 55% of all Japanese merchant tonnage, strangling Japan's ability to fuel and supply its forces. German U-boats sank over 3,000 Allied ships. Both paid heavily: Germany lost over 700 U-boats and roughly 30,000 men; the US lost 52 submarines and about 3,500 men.

059: How many men were on a submarine crew?

A US fleet submarine like the Pampanito carried about 80 men — roughly 10 officers and 70 enlisted sailors. The Balao class, to which Pampanito belongs, had a maximum complement of 85, though wartime patrols often ran slightly under that.

In a vessel 311 feet long and less than 28 feet at its widest, those 80 men lived, slept, worked, ate, and stood watch in conditions of extraordinary crowding. There were never enough bunks for everyone at the same time, which is why the “hot bunk” system was used — a sailor coming off watch would climb into the bunk just vacated by someone going on watch, the mattress still warm.

Despite the cramped conditions, a submarine crew functioned as a highly trained team. Every man had to qualify in every major system aboard — not just his own job. In an emergency, you might need any sailor to operate any piece of equipment. The Navy’s word for this was “qualified in submarines,” and earning it was a point of pride.

060: What is a crash dive?

A crash dive is an emergency dive — practiced constantly until the entire crew could execute it in their sleep. Typically triggered by a sighting of an enemy aircraft or surface ship, the goal was to get below the surface as fast as possible.

The diving alarm — two blasts of the klaxon, the distinctive “ah-OOH-ga” sound — was the signal. Every man on the bridge scrambled below in a fixed order: lookouts first, then the officer of the deck, then the captain last, pulling the hatch shut behind him. The main ballast tank vents were opened and water flooded in. Full negative tanks were flooded for additional weight. The diving planes were angled steeply. The diesel engines were secured and the electric motors engaged.

A well-drilled fleet submarine crew could go from surfaced to periscope depth — roughly 63 feet — in 30 to 60 seconds. Getting those extra few seconds could be the difference between escaping and being bombed. Submariners drilled on the crash dive over and over again until every motion was automatic.

061: How many American submarines were lost in WWII?

The United States lost 52 submarines during World War II — roughly 18% of all US submarines that went on war patrol. Each submarine lost took its entire crew of 60 to 80 men with it. In total, approximately 3,505 American submariners died in the war, representing the highest per-capita casualty rate of any branch of the US armed forces.

Of the 52 submarines lost, the majority were sunk by Japanese anti-submarine forces — destroyers, aircraft, and smaller patrol craft equipped with depth charges. A number were lost to mines, especially in the early war years when enemy minefields were poorly charted. For about eight submarines there is no confirmed cause: they simply went missing on patrol and were never heard from again.

Despite these losses, the US submarine force inflicted devastating damage on Japan: sinking approximately 55% of all Japanese merchant tonnage and one-third of the Japanese Navy — with fewer than 2% of Navy personnel. The 52 boats and their crews are commemorated with the phrase still used in the submarine community: they are “on eternal patrol.”

062: Did submarines ever rescue survivors from the water?

Yes, though it was rare and dangerous. A submarine on patrol was primarily a weapon of war, not a rescue vessel. Stopping on the surface to pull survivors from the water exposed the boat to air and surface attack. For most of the war, submarine captains could not risk their crew and boat on rescue operations, however grim that reality was.

The most famous exception in the Pacific was the rescue mission carried out by USS Pampanito (SS-383), USS Sealion II (SS-315), and several other submarines in September 1944. After attacking a Japanese convoy, the submarines surfaced and discovered hundreds of British and Australian prisoners of war in the water — men who had been aboard Japanese prison ships unknowingly caught in the attack. Pampanito rescued 73 survivors; Sealion II rescued 54. Other ships arriving later pulled more from the sea. Of the approximately 2,200 POWs who had been aboard the doomed ships, about 159 were rescued in total. The rest perished.

This remains one of the most dramatic rescue operations in the history of the submarine service, and it is central to the story the Pampanito tells today.

063: Did the Pampanito save the POWs?

On its third war patrol in September 1944, the Pampanito took part in one of the most remarkable rescue operations in submarine history. After attacking Japanese convoy HI-72 in the South China Sea, Pampanito and the other submarines in its patrol group surfaced to find hundreds of men in the water — emaciated, oil-covered survivors who turned out to be British and Australian prisoners of war.

The convoy had been unknowingly transporting over 2,200 Allied POWs aboard Japanese prison ships, and the submarines had attacked without knowing this. When Commander Paul Summers realized what was happening, Pampanito diverted all efforts to rescue. In about nine hours, the boat pulled 73 survivors aboard — men in desperate condition after years of brutal captivity and days in the open ocean. Every available space on the submarine was used to care for the injured.

USS Sealion II (SS-315) rescued an additional 54. Other ships reached the area and saved more. Of the 2,200 POWs, approximately 159 survived. Commander Summers received the Navy Cross for this patrol. Today the rescue is the centerpiece of the Pampanito's story as a museum ship.

064: How did you flush a toilet on a submarine?

Flushing a toilet — called the "head" on any Navy vessel — on a submerged submarine required a careful procedure. You could not simply flush as you would on land, because the submarine was under pressure from the surrounding water. Waste had to be blown outboard against sea pressure using compressed air.

The procedure involved a specific sequence of valves and levers: close the hull valve, open the blow valve, use compressed air to force the contents into the sea, then reverse the sequence carefully. Signs were posted in each head explaining the steps. Getting any part of the procedure wrong — opening valves in the wrong order or at the wrong time — could result in the contents being blown back into the boat and covering the user. Veteran submariners have a word for this mishap, and it is not polite.

At deeper depths the procedure was more complicated because greater air pressure was needed to overcome the sea pressure outside. Some submarines had sanitary retention tanks that held waste until the boat surfaced, when it could be discharged normally. Either way, the heads were few in number (one for

officers, one for crew, for 80 men), the procedure was fiddly, and long lines during busy periods were simply part of submarine life.

065: How did submarines run underwater without air for the engines?

When a submarine submerged, its diesel engines had to stop. Diesel engines need air to run — specifically, they need large quantities of fresh air to burn fuel. Underwater, there is no fresh air available, so the diesels were shut down as the boat went under.

Instead, the submarine ran on electric motors powered by large banks of lead-acid batteries, just like a very large flashlight. The batteries had been charged while the boat was on the surface, with the diesels driving generators rather than the propellers. A fully charged battery bank could power the submarine at slow speed for 24 to 36 hours, or at higher speed for a much shorter time.

When the batteries ran low, the submarine had to surface to run the diesels again, both to drive the boat and to recharge the batteries. This is why WWII submarines spent most of their time on the surface at night — they needed the surface time to recharge for the next day's submerged operations.

066: Why did submarines stay on the surface at night?

WWII submarines spent the majority of their time on the surface, surfacing at dusk and diving again at dawn. This was driven by two practical necessities: battery charging and speed.

Submarines ran on batteries while submerged, and those batteries had to be recharged by diesel generators on the surface. A full day submerged at patrol speed would drain the batteries significantly. Night was the safest time to operate on the surface — aircraft, the most dangerous threat to a surfaced submarine, could not spot you as easily. So submarines surfaced at night to charge their batteries and traveled as far as possible toward their patrol area or toward their next target.

Speed was the other reason. A fleet submarine on the surface could make 17 to 20 knots. Submerged, it could only manage 8 to 9 knots for brief periods, or 2 to 3 knots for extended running. Covering hundreds of miles to reach a patrol area was only practical on the surface. Submarines were really surface ships that could dive for evasion and attack, not true underwater vessels.

067: How many torpedoes did a submarine carry?

A Balao-class fleet submarine like the Pampanito carried 24 torpedoes in a full load. The forward torpedo room had six torpedo tubes and space for ten reload torpedoes — a total of sixteen ready forward. The after torpedo room had four tubes and four reloads — eight aft. Together that was exactly 24.

Once all 24 were fired or the patrol was over, the submarine returned to base to reload. Reloading torpedoes at sea from a tender was possible but rare — it was a slow, difficult operation in open water. Most patrols ended when the torpedo supply ran out.

068: What was the temperature like inside a submarine?

Heat was a constant problem, especially in the Pacific. The four diesel engines generated enormous heat, and the engine rooms could reach 120 degrees Fahrenheit or higher in tropical waters. Crewmen in the engine room wore as little as possible and dripped sweat constantly. The rest of the boat was hot too — there was no real air conditioning, just ventilation fans moving warm, humid air.

When submerged, the diesels stopped and the boat cooled slightly, but heat from battery charging, equipment, and 80 men's bodies still built up. By the end of a long dive, the air could be thick and uncomfortably warm. Cold-water patrols near Japan were a different story — crews sometimes bundled up. The tropics were the hardest.

069: How loud was it inside a submarine?

On the surface with diesels running, a fleet submarine was extremely loud. The four engines each produced 1,600 horsepower and roared constantly. The engine rooms were close to 100 decibels — a level that causes hearing damage over time. Many World War II submariners suffered significant hearing loss by the end of the war.

When submerged on batteries, the main motors hummed quietly, and the overall noise level dropped dramatically. Submerged running was much more bearable. But during a depth charge attack, the noise was terrifying — explosions rattling the hull, lights flickering, equipment failing. The crew learned to stay silent during those moments so sonar could hear enemy propellers and the captain could plan the next move.

070: How long was a typical war patrol?

A typical war patrol for a fleet submarine like the Pampanito lasted between 45 and 75 days. The duration was limited by fuel, food, and torpedoes. When the torpedoes were gone or food ran critically low, the submarine headed back to base. Mechanical breakdowns could also cut a patrol short.

The Pampanito completed six war patrols between June 1944 and July 1945, ranging from about 40 to 75 days each. Between patrols, the crew had a rest period — usually three to six weeks — while the boat was refueled, provisioned, repaired, and loaded with fresh torpedoes. At the peak of the Pacific war, when bases moved closer to Japan, patrols became more efficient and more dangerous.

071: What did the crew do when they spotted an enemy ship?

When a lookout or radar operator spotted an enemy vessel, the officer of the deck called "battle stations — torpedoes" over the intercom. Every man immediately went to his battle station. The captain went to the conning tower and took over at the periscope. The bridge watch dove below and the hatches were sealed.

The crew began a careful approach — tracking the target's speed, course, and distance. These measurements were fed into the torpedo data computer, which calculated the correct firing solution. The approach could take minutes or hours depending on daylight, sea conditions, and enemy escorts. When the solution was ready, the captain gave the order to fire, typically sending a spread of two to four torpedoes at the target.

072: Did submarines ever fight other submarines?

It happened, but it was rare. US submarines did sink several Japanese submarines during World War II, mostly using torpedoes when the enemy submarine was spotted on the surface, or by depth charges when detected submerged. However, submerged submarine-vs-submarine combat was extremely difficult — both boats were maneuvering blind with only sonar to detect each other.

The primary mission of US fleet submarines was attacking enemy surface ships — freighters, tankers, warships, and troop transports. Submarine-vs-submarine engagements were an occasional bonus, not the

main focus of patrols. The Japanese submarine force suffered heavily not from US submarines but from US destroyers and aircraft using depth charges and hedgehogs.

073: What was the largest submarine in WWII?

Japan built the largest submarines of World War II. The I-400 class — called "Sen Toku" submarines — were about 400 feet long and could carry three aircraft in a watertight hangar on deck. They were designed to launch planes to attack the Panama Canal or US cities but arrived too late to be used in those missions.

American fleet submarines like the Balao class, which includes the Pampanito, were about 312 feet long and 27 feet in diameter at the pressure hull — considered large, capable ocean-going warships. German Type VII U-boats, the most common submarine of the war, were much smaller at about 220 feet long. US fleet submarines were larger and had far greater range than their German counterparts.

074: Did submarines ever carry troops or special forces?

Yes. US submarines performed a wide variety of special missions beyond standard war patrols. Submarines landed scouts, agents, and Filipino guerrilla fighters behind Japanese lines — sneaking them ashore in rubber boats at night. They also evacuated personnel from occupied islands and supported resistance fighters throughout the Philippines by delivering weapons, medicine, and supplies.

The Pampanito was dedicated to attack patrols as a fleet submarine, but many submarines in the Pacific fleet spent time on these special missions. Some submarines were specially configured to carry swimmer-delivery vehicles or teams of commandos. The submarine's ability to approach enemy coastlines undetected made it a natural choice for covert operations.

075: What was a war patrol?

A war patrol was an officially designated combat mission assigned to a submarine by fleet command. The Navy formally recognized patrols as "war patrols" when the submarine entered a combat zone and operated against the enemy. Completing a successful war patrol earned the crew a combat patrol pin — a prestigious decoration.

Each patrol was assigned a geographic area, called a patrol zone or patrol area, within which the submarine was authorized to attack enemy shipping. Fleet command tried to route multiple submarines through areas where Japanese convoys were expected. The patrol ended when torpedoes were expended, supplies ran low, the submarine was damaged, or the time limit was reached. The Pampanito completed six war patrols, earning battle stars for each one.

076: How did the crew handle doing laundry?

Doing laundry on a submarine was nearly impossible. Fresh water was extremely precious — the boat's evaporators could produce only a limited amount, and it was reserved primarily for drinking and cooking. There was nothing to spare for washing clothes.

Most sailors wore the same clothes for days at a time. In the tropical Pacific, many men worked in shorts and little else to cope with the heat — which also reduced the laundry problem. Personal items might be rinsed in a bucket of water, but full laundry had to wait until the submarine returned to port. Once ashore,

the crew could use proper Navy laundry facilities. Submariners joked that the best part of coming back from patrol was a hot shower, a real meal, and clean clothes.

077: Did the Pampanito crew get shore leave?

Yes. Between each war patrol, the crew received rest and recuperation — called R&R. Submarine crews were considered elite and received some of the best leave facilities in the Navy. Pearl Harbor had rest camps with swimming pools, good food, and comfortable bunks. Australia — primarily Brisbane and Fremantle — was another popular rest destination for submarine crews early in the war.

A typical turnaround between patrols lasted three to six weeks. During that time, the crew got real food, clean beds, sports, movies, and time away from the boat. The Navy understood that the psychological and physical strain of submarine patrols was severe, and that well-rested crews would be more effective than exhausted ones. The short time ashore was eagerly anticipated after weeks at sea.

078: What ports did the Pampanito operate from?

The Pampanito operated primarily from Pearl Harbor, Hawaii and advanced bases in the Central Pacific. As the war progressed and American forces moved closer to Japan, the Navy established forward submarine bases at Midway and later at Guam and other recaptured islands in the Marianas. Operating from Guam cut roughly 3,000 miles off the trip to Japanese waters and allowed longer time on patrol.

All six of the Pampanito's war patrols were in the Pacific — mostly in the waters of the South China Sea, the Luzon Strait, and the East China Sea. Japanese waters were heavily patrolled by enemy aircraft and warships, making every approach dangerous. The Pampanito returned to the United States in the summer of 1945 for an overhaul when the war ended before she could go out again.

079: What was the Mark 14 torpedo?

The Mark 14 was the standard torpedo used by US fleet submarines during World War II. It was 21 inches in diameter and 21 feet long, with a 643-pound explosive warhead. Torpedoes were fired from 21-inch tubes and could be launched at ranges up to 4,500 yards at 31 knots, or at 46 knots for a shorter sprint of about 1,000 yards.

Unfortunately, the Mark 14 had serious problems early in the war. The depth-control mechanism made torpedoes run 10 feet deeper than set, causing them to pass harmlessly under targets. The exploder — both the magnetic and contact versions — often failed to detonate on impact. Submariners reported torpedo after torpedo hitting enemy ships and failing to explode. The Navy resisted fixing these problems for years, but by 1943 the depth problem and exploder failures were finally corrected. After that, the Mark 14 performed reliably and became a highly effective weapon.

080: What happened to the crew after the war?

When Japan surrendered in August 1945, the Navy began rapidly demobilizing. Most of the Pampanito's crew returned to civilian life. Many used the GI Bill to go to college, start businesses, or buy homes. They got married, started families, and worked to rebuild their lives after years of wartime service.

Some stayed in the Navy and had long careers, rising through the ranks. A few remained connected through the USS Pampanito Association, a veterans' group that kept former crewmembers in touch. Over the

decades, veterans returned to visit the boat after she became a museum ship at Fisherman's Wharf in San Francisco. They brought their children and grandchildren, shared their stories, and helped preserve the history of what they had done. The last surviving Pampanito veterans passed away in their 90s and 100s in the early decades of the twenty-first century.

081: Were Pampanito crew members awarded medals?

Yes. The Pampanito received the Navy Unit Commendation ribbon and earned battle stars for each of her six war patrols. Individual crew members earned a range of decorations including the Bronze Star, Silver Star, and Navy Cross for acts of valor in combat.

The most celebrated event was the rescue of Allied prisoners of war in September 1944. After sinking Japanese ships that turned out to be carrying British and Australian POWs, the Pampanito turned back to pull survivors from the water. The crew rescued 73 men — working in the dark, in enemy waters, under the constant threat of attack. The officers and men involved in the rescue were rightly honored. Their actions saved lives that would otherwise have been lost without any record of what had happened.

082: How much fuel did a submarine carry?

A Balao-class submarine like the Pampanito carried approximately 116,000 gallons of diesel fuel. This fuel was stored in tanks built into the outer hull, called fuel ballast tanks, which were flooded with seawater as fuel was consumed to maintain the submarine's balance and buoyancy.

With 116,000 gallons aboard, the Pampanito had a surface range of roughly 11,000 nautical miles at cruise speed — enough to travel from Hawaii almost to Japan and back. The Navy also positioned submarine tenders and forward bases closer to the combat zones to reduce the distance submarines had to travel, stretching that range further and allowing submarines to spend more time on patrol rather than in transit.

083: What happened if a submarine got stuck on the bottom?

Submarines could end up on the bottom either by accident — running aground in shallow water — or intentionally during a depth charge attack, when a captain might order the boat to the bottom to reduce noise and wait out the hunt.

If the submarine bottomed intentionally during a hunt, the crew would shut down all non-essential systems and wait silently, sometimes for hours, while the batteries slowly discharged and the air quality declined. If the submarine was damaged and trapped, the crew could use emergency escape equipment — breathing devices called Davis Submerged Escape Apparatus, or later the Momsen Lung — to swim to the surface. If the submarine broached in shallow water, compressed air could blow the ballast tanks and the engines could back the boat free. Grounding in deep water was nearly always fatal.

084: What was the diving alarm sound?

The diving alarm was a Klaxon horn that made a distinctive AAOOGA sound — two blasts to signal a normal dive or three blasts for a crash dive emergency. The sound instantly sent every man on the boat scrambling to dive stations.

Lookouts on the bridge dove down the conning tower hatch in seconds. The officer of the deck was the last man below and dogged — sealed — the hatch behind him. Below, men opened flood valves to fill the ballast

tanks with seawater, pulled in the radio antenna, shut vents, and angled the diving planes down. A well-trained crew could take a fleet submarine from cruising on the surface to fully submerged in 30 to 45 seconds. The AAOOGA sound of the Klaxon became one of the most iconic sounds of World War II and is still recognized worldwide as the sound of a diving submarine.

085: What was the battery room on a submarine?

Fleet submarines like the Pampanito had two battery rooms — called the forward battery and the after battery — located in the lower hull, directly beneath the berthing areas. Each battery room held 126 large lead-acid battery cells, for a total of 252 cells in the whole boat.

Each cell was enormous — about the size of a large trunk and weighing over 1,600 pounds. The full battery system weighed hundreds of tons and provided the electric power to run the main motors when the submarine was submerged. The batteries were recharged every time the submarine surfaced and ran the diesels. Battery capacity determined how long the submarine could stay submerged — typically 24 to 48 hours at slow speed before the batteries were exhausted and the boat had to surface to run the diesels and recharge.

086: How did the crew detect and track enemy ships?

WWII fleet submarines used several methods to detect and track enemy ships. When on the surface at night, the crew relied on radar — the SJ surface-search radar could detect ships at ranges of 15 miles or more. During daylight, lookout sailors with high-powered binoculars scanned the horizon from the bridge. Submerged, the men used passive sonar — hydrophones — to listen for the sounds of enemy screws, propellers, and machinery.

Once a target was located, the captain tracked it through the periscope, estimating the enemy ship's speed, course, and range. The crew fed these observations to the torpedo data computer (TDC), which calculated the firing solution needed to intercept the target. The approach was often slow and careful — closing the distance while staying undetected — before the captain gave the order to fire torpedoes.

087: How far could a submarine travel on one fuel load?

A Balao-class submarine like the Pampanito could travel far — roughly 11,000 nautical miles on one full load of fuel — enough to travel from Hawaii nearly to Japan and back. The submarine carried approximately 116,000 gallons of diesel fuel stored in tanks built into the outer hull. As fuel was consumed, seawater replaced it to maintain the submarine's balance.

At cruising speed — about 10 knots on the surface — this range enabled patrols lasting weeks in distant Pacific waters. The Navy also established forward bases at Pearl Harbor, Midway, and eventually Guam to reduce the distance submarines had to travel before reaching enemy waters, effectively extending the useful patrol time within the combat zone.

088: What kind of food did the crew eat on a submarine?

Submarine crews ate surprisingly well — better than almost anyone else in the Navy. Because the boats had no room to carry large quantities of food, they had to turn around after a limited number of days, and the Navy compensated with high-quality rations and a \$1.25-per-day subsistence bonus called "submarine pay."

At the start of a patrol, fresh food was plentiful: steaks, eggs, fresh vegetables, ice cream, and bread baked aboard. Sailors on submarines often ate steak for breakfast. As the patrol stretched toward six or eight weeks, fresh food ran out and the menu shifted to canned goods, powdered eggs, and preserved meats — still decent by wartime standards. The cook was one of the most important men on the boat; good food was crucial for morale. In the tight, stressful world of a submarine, a good meal was one of the few pleasures available.

089: What was a submarine tender?

A submarine tender was a large support ship designed to service and resupply a fleet of submarines. Tenders carried spare parts, torpedoes, food, fuel, repair equipment, and even recreational facilities. When submarines returned from patrol, they tied up alongside the tender rather than going all the way back to a shore base for routine maintenance.

The tender's crew included machinists, electricians, torpedo technicians, and medical personnel who could overhaul engines, repair hydraulics, reload torpedoes, and treat injured sailors. Living conditions on the tender were also much better than aboard a submarine — it was a welcome break. Key submarine tenders in the Pacific included USS Fulton, USS Holland, and USS Sperry. Tenders were vital to keeping submarines operational far from home ports, especially as bases moved forward toward Japan.

090: What does the SS in USS Pampanito SS-383 stand for?

The designation "SS" stands for "Submarine, Submersible" — the official US Navy hull classification for a conventional, diesel-electric attack submarine. The number 383 is Pampanito's hull number, assigned in the order submarines were authorized by Congress. USS Pampanito (SS-383) was the 383rd submarine authorized for construction, though it was not the 383rd built.

USS stands for "United States Ship." The "Pampanito" name comes from a type of Pacific Ocean fish — the pompano. Most American fleet submarines of World War II were named after fish. The SS designation distinguished conventional submarines from other underwater vessels like SSK (hunter-killer submarines) and SSBN (ballistic missile submarines) that appeared later in naval history.

091: How did the crew know survivors in the water were Allied prisoners?

When the Pampanito and other submarines in the wolf pack realized they had been shooting at Japanese ships carrying Allied prisoners of war, the situation was horrifying. The initial attack on September 12, 1944 sank the Rakuyo Maru and Kachidoki Maru — Japanese ships that, unknown to the Americans, were transporting hundreds of British and Australian POWs captured in Singapore.

The crew first noticed the survivors were white men — a shocking sight in Pacific waters. As they were pulled aboard, the POWs identified themselves in English. Many wore rags or nothing at all after days in the water under a tropical sun. They were desperately weak, covered in oil, and barely alive. The Pampanito rescued 73 men; together with the submarine USS Sealion, which also turned back, a total of 159 POWs were saved. The Rakuyo Maru alone had been carrying about 1,300 prisoners; most were lost.

092: Were there doctors on submarines?

No. Fleet submarines did not carry a medical doctor. Medical care was provided by a Pharmacist's Mate — an enlisted man trained as a Navy corpsman, equivalent to a medic or paramedic today. The Pharmacist's

Mate had a medical kit and some training, but was not a physician.

The Pharmacist's Mate was responsible for treating injuries, infections, dental emergencies, appendicitis, and any other medical crises that arose at sea. One famous case became part of submarine lore: in 1942, Pharmacist's Mate Wheeler "Wahoo" Lipes performed an emergency appendectomy on a sailor aboard USS Seadragon using the ship's wardroom table, improvised instruments, and ether as anesthesia. The patient survived. Such episodes highlighted both the limitations of submarine medical care and the resourcefulness of the men who served.

093: What was submarine school in Groton Connecticut?

The Naval Submarine School was — and still is — located in Groton, Connecticut, at the Naval Submarine Base New London, which despite its name is actually in Groton across the Thames River from New London. It has been the primary training center for US Navy submariners since 1916.

During World War II, candidates first had to complete basic Navy training, then volunteer and pass psychological and physical screening for submarine duty. Submarine school lasted about 12 weeks and covered diving procedures, torpedo operations, damage control, and every system on the boat. Graduates were assigned to submarines as qualified candidates — but the real training continued on the boat itself, where a sailor had to demonstrate competency in every compartment to earn his Dolphins pin. The submarine Dolphins — a qualification badge — could take six months to a year to earn after reporting to a boat.

094: What was the SJ radar?

The SJ radar was a surface-search radar introduced on US fleet submarines in 1942 and was one of the most important technological advantages American submarines had in the Pacific. The SJ stood for the Submarine-model J radar set. It transmitted microwave pulses from a rotating antenna on the periscope shears and detected the reflected echoes from ships or coastlines.

The SJ radar was revolutionary because it allowed submarines to find and track targets on the surface at night or in poor visibility — ranges up to 15 miles for large ships. Before radar, submarines had to rely entirely on lookouts, making night attacks much more dangerous and difficult. The radar allowed the Pampanito and other fleet submarines to stalk convoys in complete darkness, approach to firing range, attack, and escape before the enemy could locate them. An improved SD radar was used for air search — detecting aircraft — to give the submarine time to dive before a plane could reach attack position.

095: How did submarines help win the war?

US submarines played a decisive role in winning the war in the Pacific. Although submarines made up less than 2% of US Navy personnel, they sank approximately 55% of all Japanese shipping destroyed during the war — including 214 warships and over 1,100 merchant vessels.

Japan was an island nation almost entirely dependent on imports of oil, food, steel, and raw materials. As US submarines systematically destroyed Japanese merchant shipping, the Japanese war machine slowly starved. By 1945 Japan's tanker fleet was almost wiped out, fuel was critically short, pilots couldn't be trained because there was no aviation gasoline, ships couldn't move, and factories couldn't produce weapons. The submarine campaign against Japanese shipping was arguably the single most decisive factor

shortening the Pacific war. Some historians compare it to the German U-boat campaign against Britain — but unlike Germany, the United States succeeded.

096: How did submarines receive their patrol assignments?

Patrol assignments came from Commander Submarine Force Pacific Fleet — called ComSubPac — located at Pearl Harbor. Intelligence from codebreakers (who had broken Japanese naval codes), aerial reconnaissance, and reports from other submarines all fed into a detailed picture of Japanese convoy routes and schedules.

ComSubPac assigned each submarine a geographic patrol zone — a specific area of ocean where Japanese shipping was expected. The boat's captain received sealed orders that were opened only after leaving port. Radio communication during the patrol was one-way where possible — submarines received messages but rarely transmitted, to avoid giving away their position. If a submarine found a convoy or important target outside its assigned zone, it could pursue and attack at the captain's discretion. After the patrol, the captain filed a detailed patrol report, which was used to evaluate performance and plan future assignments.

097: Was the Pampanito ever damaged in combat?

The Pampanito experienced several close calls but was never seriously damaged. Like all submarines operating in Japanese waters, she was subjected to depth-charge attacks by enemy destroyers and patrol craft. Depth charges exploding nearby could crack gauges, pop light bulbs, rupture pipes, and rattle the crew badly, but no direct structural damage to the Pampanito was recorded in her patrol reports.

The Pampanito did experience mine threats during operations in waters that Japan had heavily mined, and encountered enemy aircraft that required emergency dives. The crew also had to navigate through areas with strong currents, shallow water, and enemy-swept channels. That the Pampanito completed all six war patrols and returned safely every time was a reflection of skilled seamanship, good luck, and the professionalism of her crew. 52 American submarines were lost during the war — making every safe return significant.

098: What was the Pampanito doing at the end of the war?

When Japan surrendered in August 1945, the Pampanito was stateside undergoing a refit and overhaul at Mare Island Naval Shipyard in Vallejo, California, preparing for what would have been her seventh war patrol. The news of Japan's surrender reached the crew while the boat was still in the shipyard — they would not need to go back to the Pacific.

After the surrender, many submarines were sent to Japan to participate in the formal occupation. The Pampanito was instead decommissioned in 1946 and placed in reserve at Mare Island. She sat in the reserve fleet for years while the Navy decided what to do with its large surplus of WWII-era submarines. Eventually she was transferred to the city of San Francisco, where she opened as a museum ship at Pier 45, Fisherman's Wharf in 1982. She is one of only four surviving Balao-class submarines open to the public.

099: How was the Pampanito decommissioned and preserved as a museum?

After decommissioning in 1946, the Pampanito sat in reserve at Mare Island for decades. In 1976 she was struck from the Naval Vessel Register, and in 1978 she was transferred to the National Maritime Museum Association (now the San Francisco Maritime National Historical Park) for preservation as a museum ship.

The Pampanito was towed to San Francisco and opened to the public on July 4, 1982, at Pier 45, Fisherman's Wharf. A nonprofit organization, the Maritime Park Association, manages the boat and funds her preservation. Restoring and maintaining a 75-year-old submarine is a constant challenge — the salt air, seawater, and age are always working against the volunteers and staff who keep her seaworthy. The Pampanito was designated a National Historic Landmark in 1986, recognizing her importance as a surviving example of WWII naval history.

100: What was the commissioning of a submarine?

Commissioning was the ceremony at which a Navy ship officially entered active service. The shipbuilder formally delivered the submarine to the Navy, a commanding officer took command, and the crew formally "manned the ship." The commissioning ceremony included the reading of orders, the playing of the national anthem, and often a speech by a senior officer.

From that moment, the ship became a US Navy vessel and the crew had official military status aboard her. The Pampanito was commissioned on November 6, 1943, at Portsmouth, New Hampshire, after being built at the Portsmouth Naval Shipyard. The commissioning was the end of the building process and the beginning of the boat's service life. Decommissioning — the opposite ceremony — takes the ship out of active service.

101: How long can a submarine stay submerged and what was the longest?

A WWII fleet submarine's submerged endurance was determined by battery capacity. Moving at 2 knots (barely creeping), the batteries on a Balao-class submarine could last roughly 48 hours. At higher speeds the batteries drained much faster — at maximum speed of about 8-9 knots submerged, batteries lasted only about an hour.

In practice, submarines rarely stayed down for more than 24 hours. The battery had to be recharged by surfacing and running the diesels, which required access to air. Additionally, CO₂ built up in the air after extended submersion, and oxygen was slowly depleted, making breathing increasingly difficult. Chemical canisters could absorb some CO₂ and oxygen could be bled in from tanks, but these were limited. In extreme cases during depth charge attacks, crews endured 30+ hours submerged before surfacing was safe. The record during the war was reportedly over 40 hours submerged in an emergency situation.

102: Were there discipline problems on submarines?

Serious discipline problems were rare on submarines, partly because the selection process screened out men who showed behavioral issues, and partly because the close quarters meant that troublemakers made life miserable for everyone. There were no mutinies recorded on US WWII submarines.

Minor discipline issues — arguments, petty theft, insubordination — certainly happened. A captain on a submarine had nearly absolute authority at sea and could confine a man to the boat or restrict privileges. However, the culture on submarines was also more informal than on surface ships. Officers and men ate together, worked in close proximity, and depended on each other for survival. This mutual dependence usually created strong unit cohesion rather than resentment. When serious problems did arise, they were typically handled administratively when the boat returned to port.

103: What was the Pacific Fleet submarine strategy in WWII?

The core US submarine strategy in the Pacific was unrestricted submarine warfare — attacking all Japanese shipping without warning. This was in sharp contrast to prewar international law (which required warning before sinking merchant ships), but the Navy Department authorized unrestricted submarine warfare on December 8, 1941, the day after Pearl Harbor.

The strategic goal was to cut off Japan from the raw materials its economy required: oil from the Dutch East Indies, rubber from Malaya, steel from China, rice from Southeast Asia. Without these imports, Japan's military machine would grind to a halt. Submarine forces were organized into wolf packs that could coordinate attacks on convoys, and patrol zones were assigned based on Ultra intelligence — decoded Japanese naval communications — that told commanders exactly where convoys would be. The strategy worked: by 1944, Japanese oil imports were less than a third of pre-war levels, crippling their war effort.

104: How was the Pampanito resupplied at sea?

During a war patrol, the Pampanito was on her own — there was no mid-patrol resupply. Everything the crew needed for 45–75 days had to be loaded before departure: food, torpedoes, fuel, spare parts, and medical supplies. The crew loaded as much as physically possible into every available space — under bunks, in passageways, in torpedo rooms, and in every storage locker.

Occasionally a submarine could transfer torpedoes from a submarine tender if one was operating nearby — but this was rare and difficult in open water. The primary means of between-patrol resupply was the return to a submarine base or tender, where the boat was completely restocked over two to three days. The Pampanito operated primarily from Pearl Harbor and later from Guam, where forward-positioned tenders provided fuel, food, torpedoes, and repair services between each of her six war patrols.

105: What was the hull number of the Pampanito?

The Pampanito's hull number is SS-383. "SS" is the Navy hull classification for a conventional (diesel-electric) submarine, and 383 was the sequential number assigned when the boat was authorized by Congress. The full official designation is USS Pampanito (SS-383).

The name "Pampanito" comes from a Pacific Ocean fish, the pompano. American fleet submarines of World War II were traditionally named after fish. Her keel was laid on March 15, 1943, she was launched on July 12, 1943, and commissioned into US Navy service on November 6, 1943, all at the Portsmouth Naval Shipyard in Kittery, Maine.

106: Were there doctors or medical care on submarines?

Fleet submarines did not carry a doctor. The medical officer was a Pharmacist's Mate — an enlisted Navy corpsman (medic). On the Pampanito, the Pharmacist's Mate had a small medical kit and first-aid training, but was not a physician.

For minor injuries, infections, and illnesses this was workable. For serious emergencies, the limitations could be life-threatening. The boat had no operating room, no X-ray equipment, and no means to get a sick man to a hospital in less than days or weeks. The Navy recognized this and gave Pharmacist's Mates extra training in emergency procedures. The most famous example of submarine medical improvisation was in 1942, when a Navy corpsman performed an emergency appendectomy on a sailor aboard USS Seadragon using a mess table, improvised instruments, and ether as anesthesia — and the patient survived.

107: How did submarines handle medical emergencies at sea?

Medical care at sea on a submarine was limited but resourceful. Submarines carried no doctor — medical care was provided by a Pharmacist's Mate, a trained Navy corpsman. The medical supplies included medications, bandages, instruments for minor surgery, and dental tools.

For serious emergencies — appendicitis, serious injuries, infections — the crew had to improvise or cut the patrol short and race for port or a rendezvous with a submarine tender. The most famous example was a 1942 appendectomy performed aboard USS Seadragon by Pharmacist's Mate Wheeler Lipes using the wardroom table and improvised instruments. In extreme cases, sick or badly injured men were transferred to other ships at sea using a rubber boat — a dangerous operation in both enemy waters and rough seas. The limitations of submarine medicine were accepted as one of the hazards of the service.

108: Did the Pampanito ever hit a mine?

The Pampanito never hit a mine. However, mines were a constant threat. Japan heavily mined the waters around the home islands and key shipping lanes, and US submarines had to navigate these minefield zones carefully. Passing through a minefield — surfaced or submerged — required careful navigation, slow speeds, and a great deal of luck.

Several American submarines were lost to mines during World War II. The Pampanito was aware of and careful about mine threats during her six war patrols, particularly during operations in the South China Sea and around Japanese home waters. The fact that she survived all six patrols without being lost to a mine, depth charge, or aircraft was a testament to her crew's skill and caution.

109: When was the Pampanito decommissioned?

The Pampanito was decommissioned on December 15, 1945, at Mare Island Naval Shipyard in Vallejo, California, shortly after Japan's surrender ended World War II. She was placed in the reserve fleet — "mothballed" — where she sat for decades while the Navy maintained her in case of future need.

She was struck from the Naval Vessel Register in 1976, ending her formal Navy status. In 1978 she was transferred to the National Maritime Museum Association in San Francisco. After restoration work, she opened to the public on July 4, 1982, at Pier 45, Fisherman's Wharf, where she remains today as a museum ship and National Historic Landmark.

110: How did radar work on a WWII submarine?

WWII submarines used two types of radar: the SJ surface-search radar and the SD air-search radar. Radar works by transmitting pulses of radio energy and detecting the echoes that bounce back from ships, land, or aircraft. The time delay between the transmitted pulse and the returning echo tells the operator how far away the target is; the direction of the antenna gives the bearing.

The SJ surface-search radar operated in the microwave frequency range (around 3,000 MHz) and could detect large ships at ranges up to 15 miles. The rotating antenna was mounted on the periscope shears — the raised structure above the conning tower. The SD air-search radar detected aircraft at ranges of about 10 miles, giving the submarine time to dive before a plane could reach attack range. By 1943 nearly all US fleet submarines were equipped with both types of radar, giving them a major tactical advantage in night and poor-visibility attacks.

111: Who donated the Pampanito to the museum?

The Pampanito was not donated by a private individual. She was transferred by the US Navy to the National Maritime Museum Association (now part of the San Francisco Maritime National Historical Park) in 1978, after she was struck from the Naval Vessel Register in 1976.

The National Maritime Museum Association, along with the Golden Gate National Recreation Area and the National Park Service, worked together to preserve and open the Pampanito to the public. Today the Maritime Park Association, a nonprofit organization, manages the Pampanito and funds its ongoing restoration and operation. Admission fees, donations from visitors, and grants help maintain the submarine for future generations. The Pampanito was designated a National Historic Landmark in 1986.

112: What was the longest any submarine stayed submerged?

A WWII fleet submarine's submerged endurance was determined by battery capacity. Moving at 2 knots (barely creeping), the batteries on a Balao-class submarine could last roughly 48 hours. At higher speeds the batteries drained much faster — at maximum speed of about 8-9 knots submerged, batteries lasted only about an hour.

In practice, submarines rarely stayed down for more than 24 hours. The battery had to be recharged by surfacing and running the diesels, which required access to air. Additionally, CO₂ built up in the air after extended submersion, and oxygen was slowly depleted, making breathing increasingly difficult. Chemical canisters could absorb some CO₂ and oxygen could be bled in from tanks, but these were limited. In extreme cases during depth charge attacks, crews endured 30+ hours submerged before surfacing was safe. The record during the war was reportedly over 40 hours submerged in an emergency situation.

113: What was the lifeguard mission in WWII?

The "lifeguard" mission was a special assignment where US submarines were stationed near Japanese-held islands or along the flight paths of bombing raids to rescue downed Allied airmen. Beginning seriously in 1944, as American air operations over Japan intensified, submarines were assigned to wait on the surface or nearby while B-29s and carrier aircraft flew their missions overhead.

When an aircrew ditched or bailed out, they radioed their position and a waiting submarine moved to retrieve them. Submarines rescued hundreds of airmen this way. Famous lifeguard rescues include USS Tang (SS-306), which rescued 22 aviators in one day, and USS Finback (SS-230), which famously rescued a young Navy pilot named George H.W. Bush in September 1944 after his aircraft was shot down over Chichi Jima.

The Pampanito was not specifically assigned a lifeguard role — she was primarily a combatant submarine on attack patrols. But the lifeguard mission shows the range of tasks US submarines performed beyond simply sinking ships.

114: Did submarines lay mines in WWII?

Yes. US submarines played an important role in minelaying operations during World War II. Submarines could secretly plant mines in Japanese harbors, shallow coastal waters, and shipping lanes that surface ships could never safely reach. The mines were carried in the torpedo tubes and fired (or dropped) like torpedoes, sinking to the bottom or floating at set depths.

Several Gato and Balao-class fleet submarines were modified for minelaying missions, carrying a combination of mines and torpedoes. Notable minelaying operations included laying mines in the approaches to Japanese home ports, which sank or damaged ships that would have been impossible to reach by direct attack. Submarine-laid mines were credited with sinking dozens of Japanese ships.

The USS Pampanito (SS-383) was not specifically used for minelaying — she was deployed as a front-line attack submarine and completed six war patrols focused on torpedo attacks against merchant and naval targets. Minelaying was a specialized assignment given to specific submarines on specific patrols.

115: How did a depth charge attack feel inside a submarine?

A depth charge attack was one of the most frightening experiences a submarine crew could endure. When a depth charge exploded nearby, the shock wave struck the hull like a massive hammer blow — loud crashing, jolting the entire boat violently, knock men off their feet and sending loose gear flying. Light bulbs shattered, cork insulation showered down from the overhead, gauges cracked, and pipes could spring leaks.

The sounds were terrifying: a sharp CLICK as the charge detonated, followed instantly by an enormous BANG, then a deep rumbling. When charges were very close, the lights went out momentarily. Men who had been through attacks described the experience as sitting inside a steel drum being struck by a sledgehammer — over and over, sometimes for hours. The psychological toll was immense: not knowing how close the next one would be, whether the hull would hold, or whether the next one would be the last.

Submariners developed intense silent discipline during attacks — no unnecessary movement, no talking, only essential whispers. Every man knew that the slightest noise could travel through the water and help the enemy re-acquire the submarine. After surviving an attack, crews reported the silence afterwards as almost dreamlike.

116: What was hot bunking on a submarine?

Hot bunking (also called hot racking) was the practice of two or more sailors sharing a single bunk on alternating watch schedules. Because a WWII fleet submarine had more crew members than available bunks — especially for enlisted men — bunks had to be shared. While one watch section was on duty, another was sleeping in their bunks, and when duty rotated, the off-going watch took over the "warm" (hot) bunks just vacated.

On Balao-class submarines like the Pampanito, the torpedo rooms served double duty: the long flat skids that held reload torpedoes also served as bunks for the torpedo gang. When the spare torpedoes were fired or used, proper bunk frames could be installed. Some enlisted men had no assigned bunk at all during the first part of a patrol and slept wherever space was available.

Hot bunking was accepted as normal submarine life. Sailors didn't think much of it — they were so tired after watch that sleeping in a warm bunk didn't bother them. The practice required that men keep their personal gear in small individual lockers, bringing only what was needed for immediate use.

117: How did the deck gun work on a submarine?

The deck gun on USS Pampanito was a 4-inch/50-caliber naval gun mounted on the forward deck, with a smaller 40mm anti-aircraft gun added later in the war. To use it, the submarine first surfaced. The gun crew

— typically six to eight men — rushed up from below through the hatches, took their stations, and began firing within seconds. Speed was critical because a surfaced submarine was vulnerable.

Shells were stored in a below-deck ready locker near the gun. A trained crew could fire ten or more rounds per minute. The gun was used primarily against small merchant vessels, patrol boats, and larger targets when torpedoes were unavailable or would be wasted. Gun attacks were riskier than torpedo attacks because the submarine was exposed on the surface within visual range of the target.

As air patrols increased later in the war, deck gun attacks became more dangerous — an aircraft could appear before the crew could dive. By 1944-45, submarines relied more heavily on torpedoes and used the deck gun less. The gun required regular maintenance because of constant exposure to saltwater spray, which caused rapid corrosion.

118: How did the Pampanito get its name?

USS Pampanito (SS-383) was named after the pampanito, a species of fish found in Pacific ocean waters — a small, silvery member of the pompano family. The United States Navy had a long tradition of naming submarines after fish and sea creatures, a practice that began with the earliest American submarines and was formalized for the fleet submarine program of World War II.

When a new submarine was ordered from the shipyard, the Bureau of Ships assigned a name from an approved list of fish names. The names were typically chosen to be distinctive (easy to distinguish by radio), not too similar to other ships already in service, and reasonably short. The assignment of "Pampanito" to SS-383 was essentially administrative — the name was available and appropriate. There was no particular connection between the pampanito fish and the submarine's mission or crew.

This naming tradition continues today — US Navy submarines are still named after underwater creatures, geographic features, or American cities and states, depending on the submarine's class.

119: How did submarine crews handle a fire onboard?

Fire was one of the most feared emergencies on a submarine. In a sealed steel hull with limited oxygen and no easy escape route, a fire could quickly become catastrophic. Submarine crews trained extensively for fire emergencies, drilling until the response was automatic.

The primary fire-fighting tools were CO2 extinguishers and foam applicators positioned throughout the boat. When a fire broke out, crew members donned Emergency Breathing Devices (EAB) — rubber masks that provided clean air from a central manifold system — so they could work in smoke without being incapacitated. Watertight doors between compartments were shut to prevent smoke and fire from spreading. The affected compartment was isolated, and the fire was attacked from inside while fresh air supply was cut.

Battery fires were particularly dangerous because the lead-acid batteries could produce hydrogen gas, which is explosive. Electrical fires in the engine room or battery compartment required immediate isolation of circuits. If a fire broke out while the submarine was on the surface, the fastest solution was sometimes to dive — seawater flooding over the hull could extinguish topside fires, and submerging reduced airflow to the fire. After any fire, the crew had to assess damage before the boat could safely continue its mission.

120: Did American WWII submarines have a snorkel?

No. American fleet submarines in World War II did not use snorkels. The snorkel was a Dutch invention that the Germans adopted for their U-boats starting in 1944. It was a retractable mast that allowed a submarine to run its diesel engines while remaining just below the surface, sucking in air and exhausting fumes through the mast rather than having to fully surface.

American submarines worked differently — they surfaced at night to run their diesel engines, recharge batteries, and ventilate the boat. This required coming fully to the surface, making them detectable by radar and lookouts. The strategy relied on operating in areas where enemy surface patrols were manageable, and on careful watch-keeping to spot threats before they became dangerous.

The US Navy did adopt snorkels after World War II once the technology's value became clear. The post-war GUPPY (Greater Underwater Propulsion Power) conversion program modified many fleet submarines, including adding snorkel capability. The USS Pampanito, however, was preserved without these modifications, so she appears today essentially as she was during her active wartime service in 1944-1945.

121: What was the role of submarines in the Battle of Midway?

At the Battle of Midway in June 1942, US submarines played a supporting but largely unsuccessful role. Twelve submarines were deployed in a defensive screen around Midway Atoll and along anticipated Japanese approach routes. The plan was for the submarines to intercept Japanese ships, but the fast-moving carrier air battle unfolded differently than expected.

USS Nautilus (SS-168) fired torpedoes at the already-damaged Japanese carrier Kaga, but the torpedoes failed to explode — a common problem with defective Mark 14 torpedoes early in the war. USS Grouper (SS-214) fired at the battleship Mikuma but also failed to score hits. Despite positioning some 12 submarines across the expected approach routes, the Japanese fleet's track and the rapid air battle left the submarines unable to effectively engage.

The submarines at Midway were largely bypassed by the engagement but did force Japanese ships to conduct evasive maneuvers that consumed time and fuel. After the battle, the decisive result came from the dive bombers that sank four Japanese carriers — not the submarines. However, the lessons of coordination between submarines and surface/air forces were absorbed and improved upon throughout the rest of the war.

122: What was radio direction finding and how did it threaten submarines?

Radio direction finding (RDF), also known as HF/DF or "Huff-Duff" (High Frequency Direction Finding), was technology that could locate a radio transmitter by triangulating the direction of its signal from multiple receiver stations. If a submarine broadcast a radio message, shore stations or ships with RDF equipment could detect the signal, measure its bearing from two or more positions, and plot the intersection to find the submarine's location — sometimes within miles or even less.

This technology posed a serious threat to submarines because Japanese shore stations and some warships were equipped with RDF. A submarine that transmitted a long message could be located quickly enough for aircraft or destroyers to be dispatched to attack it. American submariners learned to keep radio transmissions brief — compressed coded messages sent in seconds rather than minutes — to minimize the risk.

Ironically, the United States also used RDF against Japan with great success. The code-breaking operation at Station HYPO in Pearl Harbor combined decrypted Japanese signals (Ultra intelligence) with RDF bearings to track the movements of the Japanese fleet. This intelligence was critical to the American victory at Midway and many other operations throughout the Pacific War.

123: How did the crew deal with claustrophobia on a submarine?

The submarine service screened carefully for men who could handle confined spaces. During the selection process, volunteers underwent psychological evaluation and observed responses to simulate enclosed environments. Men who showed signs of claustrophobia or extreme anxiety were disqualified. Even so, some sailors didn't fully discover their own reaction to confinement until they were underway.

On patrol, the best cure for claustrophobia was keeping busy. The watch schedule — four hours on, eight hours off — kept men occupied with their duties for much of the day. Off-watch time was filled with maintenance, training, card games, reading, movies (when a projector was available), and sleep. Veterans noted that the interior of a fleet submarine was larger than popular imagination suggests: men could stand upright in most compartments, and the forward and after torpedo rooms had relatively open spaces.

Experienced submariners generally became habituated to the environment after the first patrol. The sense of purpose — being part of an elite force doing important work — helped counteract psychological stress. Men who genuinely couldn't adapt were transferred off submarines without stigma; the Navy preferred motivated and capable volunteers to unwilling men in potentially dangerous situations.

124: What was the galley like on a submarine?

The galley on USS Pampanito was a compact but well-equipped kitchen located just off the crew's mess. Despite the limited space, it was stocked to feed 70-80 men three hot meals a day, plus "mid-rats" (midnight rations) for the mid-watch section coming off duty at midnight.

Fleet submarines were famous throughout the Navy for having the best food. This was partly a deliberate morale policy: the Navy believed that good food made up for the hardships of submarine duty. Fresh meat, vegetables, and fruit were loaded at the start of each patrol and consumed first, before the boat shifted to canned and dehydrated food. The cook made ice cream when milk was available, baked fresh bread and pies, and generally went to great lengths to vary the menu.

The ship's cook was the most important morale figure after the captain and executive officer. A good cook was valued above almost anyone else on the crew. Cooking in a small galley while the boat pitched and rolled on the surface — and without being able to open a window — required skill, creativity, and physical toughness. The galley was also the social heart of the boat, where off-watch crew gathered, got coffee, and talked.

125: How did submarine crews get fresh water?

Fresh water on fleet submarines like Pampanito came primarily from a distillation plant (evaporator) that used engine heat to boil seawater and condense the steam into drinkable fresh water. The evaporator ran continuously when the engines were operating, producing several hundred gallons per day, but this had to serve all the boat's needs — drinking, cooking, and minimal washing.

Water was strictly rationed. Drinking and cooking water was always available, but bathing was another matter. Fresh water showers were rare luxuries — most crew members bathed infrequently, using small amounts of fresh water or bucket baths with salt water and special salt-water soap that would lather without fresh water. Laundry was done in salt water. By the end of a long patrol, the entire crew smelled strongly of sweat, diesel fuel, and machinery oil — this was accepted as normal.

When submarines visited a tender or returned to port, the first priority for many men was a long hot fresh water shower. Fresh water tanks could also be topped off from a submarine tender at sea, though this required rendezvousing with a surface ship, which was unusual on a standard war patrol.

126: Was the Pampanito ever close to being sunk?

Yes, on multiple occasions. During Pampanito's third war patrol in September 1944, she participated in a wolf pack attack on a Japanese convoy in the South China Sea. After the attack, Japanese escorts pursued the attacking submarines with depth charges. Pampanito went deep and ran silent for hours while depth charges exploded around her. Some charges were close enough to damage cork insulation and rattle the crew severely.

Even more dangerous was the surface time immediately following the attack, when Pampanito and USS Sealion (SS-315) stopped to rescue British and Australian POWs from the water. For nearly two hours, Pampanito sat fully surfaced in daylight in enemy-controlled waters while survivors — many barely alive — were pulled aboard by hand. A submarine stopped on the surface was completely vulnerable to aircraft or ship attack. The crew worked frantically knowing they were exposed. Fortunately, no Japanese forces appeared during the rescue.

Throughout her six war patrols, Pampanito survived multiple depth charge attacks without sustaining serious damage to her pressure hull, a testament to solid construction, the skill of her crews, and a measure of luck that every submarine depended on.

127: How did submarines attack a convoy?

Attacking a Japanese convoy required patience, tracking skill, and precise timing. The submarine first detected the convoy — typically by radar at night or by periscope during the day — and began tracking its course and speed. The captain's goal was to get ahead of the convoy and position the submarine for the best firing angle, ideally from the side (a beam shot) or slightly ahead.

While tracking, the crew set battle stations. In the conning tower, the approach officer and tracking party fed bearings, ranges, and target speeds into the Torpedo Data Computer (TDC), which continuously calculated a firing solution. When the geometry was right, the captain ordered a spread of torpedoes — typically three to six — fired in sequence against individual ships in the convoy.

After firing, the submarine immediately dove deep or changed course to evade. The convoy's escort ships (destroyers and frigates) would charge toward the torpedo wakes and begin dropping depth charges while the attacked ships scattered. The submarine dodged, went silent, and waited — sometimes for hours — before cautiously returning to periscope depth to assess damage. A successful convoy attack might sink one to three ships; attacking a well-escorted convoy was one of the most dangerous operations a submarine undertook.

128: How did the crew handle seasickness on a submarine?

Yes, seasickness was a real problem on fleet submarines — particularly early in a patrol and for men new to submarine duty. On the surface in rough Pacific seas, a submarine rolls and pitches quite noticeably. The hull is round-bottomed with relatively limited width, making it susceptible to rolling motion. Some men were quite uncomfortable in rough weather.

The best cure for surface seasickness was simply to dive: once submerged, submarines are isolated from wave action and become extremely stable. Even in a severe storm, the crew submerged below the wave layer experienced none of the motion. Veterans said that many sailors who became seasick on surface ships found submarine duty easier because so much time was spent submerged.

Most sailors got their "sea legs" quickly — the regular motion of operating on the surface became normal within a few days, and the body adapted. Ginger tablets and other remedies were available in the pharmacist's locker. Men who were chronically prone to severe seasickness were generally not suited for submarine duty and were screened out during training or after their first patrol, as a seasick crew member at battle stations was a danger to the boat.

129: Did the Pampanito sink Japanese warships?

Pampanito's confirmed sinkings were primarily Japanese transport and merchant vessels — not warships. Her most famous victims were the transport RAKUYO MARU and the transport KACHIDOKI MARU, both sunk during her third war patrol in September 1944. These ships were tragically carrying British and Australian prisoners of war, which led to Pampanito's famous rescue mission. She also sank several freighters and tankers in other patrols.

This pattern was typical for US submarines. The primary strategic objective of the Pacific submarine campaign was to strangle Japan's economy and war effort by sinking merchant ships, tankers, and supply vessels — cutting the flow of oil, food, and raw materials to the Japanese home islands. Attacking warships was a secondary priority and was usually much more dangerous, as warships were faster, better armed, and accompanied by anti-submarine escorts.

Pampanito did participate in attacks on Japanese naval vessels. During one patrol, she fired at what was believed to be a light cruiser, without confirmed results. Warship sinkings by submarines did occur throughout the war — USS Darter (SS-227) and USS Dace (SS-247) sank Japanese cruisers at the Battle of Leyte Gulf — but the real measure of submarine success was tonnage of merchant shipping sunk.

130: How did submarines handle the threat of aircraft?

Aircraft were among the most dangerous threats to surfaced submarines. A patrol plane or carrier aircraft could spot a submarine from altitude and attack before the crew had time to react. The primary defense was simple: stay submerged during daylight in enemy-controlled waters whenever possible, and surface only at night. The general rule was: if in doubt, dive.

When a submarine was caught on the surface by aircraft, the crash dive was the only reliable defense. The bridge watch immediately sounded the diving alarm (AOO-GAH), and every man on the bridge dropped through the hatch as fast as possible — sometimes literally falling down the ladder — while the officer of the deck pulled the hatch shut behind him. From alarm to periscope depth could be accomplished in 30-40 seconds on a well-trained crew.

For aircraft that attacked before a dive was complete, fleet submarines carried anti-aircraft guns — a 40mm Bofors and 20mm cannon guns — mounted on the bridge fairwater. These were used to fight off low-flying aircraft during the few seconds of exposure. Late in the war, as American and Allied air superiority over the Pacific grew, the threat to US submarines from aircraft diminished. Japanese submarines, by contrast, faced constant attack from US aircraft and lost many boats to air attack.

131: What were wolfpack tactics in WWII submarines?

Wolfpack tactics were coordinated attacks by multiple submarines against a single convoy or task group. The concept was pioneered by Germany's U-boat service under Admiral Dönitz, where groups of 10-20 U-boats would converge on an Atlantic convoy and attack simultaneously from multiple angles, overwhelming the escorts.

American submarines used a more limited version of wolf packs — typically groups of two to four submarines operating in the same patrol area and coordinating attacks via radio. These US wolf packs were called "coordinated attack groups." One famous example is the wolf pack that included USS Pampanito, USS Sealion, and USS Barb during the September 1944 patrol in the South China Sea, which sank the Japanese convoy carrying POWs.

American wolfpack tactics differed from German practice in several ways: US packs were smaller, had better radar and communications equipment, operated in waters where the Japanese anti-submarine effort was less intense than the Allied escort forces in the Atlantic, and benefited from Ultra intelligence about convoy routes. The coordinated approach allowed submarines to attack from several angles at once, forcing escorts to split their response and making it harder to prosecute all the submarines simultaneously.

132: Were submarines used to land spies or special forces?

Yes. US submarines frequently carried out special missions to land agents, OSS personnel (Office of Strategic Services, the WWII predecessor to the CIA), Filipino guerrillas, and other intelligence operatives on enemy-held shores throughout the Pacific. Navy submarine special missions also included resupplying guerrilla forces, evacuating important personnel, and delivering critical supplies to isolated outposts.

Submarines were the ideal vehicle for these "cloak and dagger" operations: they could approach enemy coastlines silently at night, surface briefly to offload or recover personnel via rubber boats, and disappear before daylight. Several submarines became specialists in these missions. USS Narwhal (SS-167) and USS Nautilus (SS-168), converted from minelayers, were the most extensively used special mission submarines, making dozens of trips to the Philippines.

The USS Pampanito served as a standard attack submarine and did not participate in documented special forces missions — her six war patrols were focused on attacking Japanese shipping. However, several fleet submarines were temporarily diverted from combat patrols to support special operations when required, reflecting the versatility that made submarines uniquely valuable throughout the Pacific campaign.

133: What did the captain do during a depth charge attack?

During a depth charge attack, the captain's primary job was to keep the submarine alive while maintaining crew discipline and calm. He would typically be in or near the conning tower, receiving reports from sonar on the position of the attacking ships and directing the helmsman and diving officer on course, speed, and depth changes to evade the next pattern of charges.

The captain had to read the enemy's attack pattern — decide when to change depth, when to slow down to reduce noise, and when to sprint away at high speed to open the distance. Some captains ordered the boat deep, below the set depth of the charges. Others made radical course changes to get out from under the attacking escort. The captain had to make rapid decisions with incomplete information, often in total darkness with men bracing against the hull for the next explosion.

Veteran submarine captains like Ben Oakley, who commanded Pampanito's third war patrol, were known for their steady, controlled demeanor during attacks. The crew took its cue from the captain — if he appeared calm and decisive, the men maintained their discipline. Panic or poor judgment in those moments could cost everyone's life.

134: How did the crew communicate with each other inside the submarine?

Fleet submarines had a shipboard intercom system called the "JV" or "JC" system — a network of sound-powered telephones that ran between all compartments and key stations. These phones required no electrical power source: the voice vibrated a diaphragm that generated a small current, which drove the earpiece in the listener's phone. Sound-powered phones worked even if the main electrical systems were damaged or shut down.

Specific phone circuits connected key stations: the bridge to the conning tower, the engine rooms to the maneuvering room, the torpedo rooms to the conn, and the control room to all compartments. When battle stations were called, each station manned its phones and reported readiness up the chain to the officer of the deck.

For voice communication between adjacent compartments, crew members simply passed through the watertight doorways or shouted through them when open. A loudspeaker 1MC public address system allowed general announcements throughout the boat. During quiet running under depth charge attack, however, all voice communication was kept to whispers, and finger taps on pipes or hand signals replaced verbal orders to minimize any sound that might travel through the water to the enemy's sonar.

135: What did crew members wear on a submarine?

Clothing aboard WWII submarines was notably informal compared to the rest of the Navy. Because submarines operated in tropical Pacific waters, the interior was warm, and formal uniform regulations were relaxed. Crew members typically wore dungarees (denim work trousers and shirts) while on duty inside the boat. Many sailors wore t-shirts or went shirtless in the hottest spaces like the engine rooms.

There was little room to carry or stow large amounts of clothing. Each man had a small personal locker for his gear. Dress uniforms were brought on patrol to wear in port, but on patrol, utility clothing was the norm. Shoes were sometimes forgone inside the submarine — many men wore sandals or went barefoot, especially in sleeping areas.

In cold weather or on the bridge at night, heavy foul-weather gear and watch coats were worn, as bridge watches in the open air were cold and wet with spray. Officers wore the same informal clothing as enlisted men inside the boat — one of the leveling aspects of submarine culture was that rank formalities were reduced in tight quarters. When submarines returned to port, dress uniforms were donned for official ceremonies or liberty ashore.

136: What did it mean to be qualified in submarines?

"Qualified in submarines" was the Navy's formal certification that a sailor had demonstrated the knowledge and ability to operate every critical system on the boat — not just his own specialty, but every other man's job as well. Qualification was not automatic: arriving on a submarine did not make you "qualified." It was earned.

A new crew member received a qualification card that listed every system, piece of equipment, and emergency procedure on the submarine. He was required to trace every pipe, cable, valve, and piece of machinery; demonstrate to senior petty officers that he understood what each system did and how to operate it; and pass an oral board (examination) conducted by officers.

The process typically took six months to a year. Until qualified, a sailor was informally known as a "non-qual" (or sometimes a "nub") — not yet a full-fledged submariner. Once finished, he was awarded the coveted Submarine Warfare Insignia — the gold (officers) or silver (enlisted) dolphins pin. Earning your dolphins was one of the proudest moments in a submariner's career. The tradition continues today: every sailor, regardless of rate or rank, must qualify on submarines before being considered a full member of the crew.

137: What was the submarine combat patrol insignia?

The Submarine Combat Patrol Insignia (also called the "patrol pin" or "combat pin") was a special award given to submariners who completed at least one officially designated "successful" combat war patrol. A successful patrol was one where the submarine either sank or damaged enemy vessels, or completed a special mission determined to be of equivalent importance.

The pin was a small gold (for officers) or silver (for enlisted men) badge bearing a submarine design with a star and dolphins (or a similar combat symbol, depending on the year issued). Each additional successful patrol earned an additional small gold star to be affixed to the ribbon or medal.

This was separate from the Submarine Warfare Insignia (the dolphins pin earned upon becoming qualified). The combat patrol pin was specifically for wartime action. It was a mark of distinction — a submariner who wore many stars on his combat pin had been through a great deal of danger. The Pampanito completed six war patrols, all of them officially designated as successful, so her crew members were eligible for up to six stars on their combat patrol pins.

138: How did the Pampanito crew rescue the POWs from the water?

On the night of September 15, 1944, USS Pampanito received a radio message from USS Sealion reporting survivors in the water at the site where the two submarines had torpedoed a Japanese convoy two days earlier. Pampanito reversed course and raced to the scene. What the crew found was horrifying: hundreds of emaciated British and Australian prisoners of war were floating in the South China Sea, clinging to wreckage after two days in shark-infested water.

The rescue was physically demanding and emotionally overwhelming. Crew members hung over the side, reaching into the water to grab men too weak to climb up themselves. Lines were thrown and men were hauled aboard. Many survivors were so weakened by years of captivity, starvation, tropical disease, and days in the ocean that they could barely hold on. Some died even as they were being pulled from the water.

Pampanito rescued 73 men. The submarine was so crowded that the crew gave up their bunks and personal space, feeding and caring for the survivors using the last of the fresh food supplies. The medical

implications were severe — many survivors had serious wounds, infections, and malnutrition. USS Sealion rescued an additional 54. Together, 127 Allied POWs were brought to safety, in what became the largest rescue of Allied prisoners by a US submarine in the entire Pacific War.

139: What was the role of the executive officer on a submarine?

The executive officer (XO), also called the "exec," was the second-in-command of the submarine and the captain's right hand. While the captain set strategic direction and made key tactical decisions, the XO managed the day-to-day operation of the boat and functioned as the ship's administrator.

The XO was responsible for: maintaining the watch bill (the schedule of all watches and duty assignments), coordinating training and qualification of new crew members, enforcing discipline, overseeing the department heads (engineering, navigation, weapons, supply), and managing the boat's administrative paperwork including the patrol report. When general quarters (battle stations) was called, the XO typically took charge of the approach party and fire control in the conning tower while the captain conned the boat.

The XO was also the enforcer of procedure — when the captain was resting, the XO ran the boat. A good exec who could anticipate the captain's intentions and keep the crew sharp and disciplined was essential to a successful submarine. Many famous WWII submarine commanders served as exec before getting their own command. The relationship between captain and exec was crucial: a submarine with a good command team, working in sync, performed far better than one with friction between them.

140: What were battle stations on a submarine?

"Battle stations" (or "general quarters") was the highest state of combat readiness on a submarine. When the captain ordered battle stations, a klaxon alarm sounded and every man on the crew immediately reported to his designated battle station — a specific position on the boat assigned to him for conducting combat operations.

Key battle stations included: the captain at the periscope in the conning tower, the approach officer and fire controlman at the Torpedo Data Computer (TDC), the helmsman and diving officer in the control room, sonar operators at their equipment, a battle stations watch in the torpedo rooms ready to reload tubes, maneuvering room operators at the engine and motor controls, and lookouts and officers on the bridge (if on the surface). All watertight doors were checked closed. Reports came in by sound-powered telephone: "Forward room manned and ready," "After room manned and ready," and so forth, until the officer of the deck acknowledged all stations ready.

The difference between a successful attack and a missed opportunity — or between escaping a depth charge attack and being sunk — often came down to how fast and how smoothly the crew transitioned to battle stations. Submarines drilled this evolution constantly, and a well-trained crew could go from off-watch sleep to fully manned battle stations in under two minutes.

141: What was the role of the pharmacists mate on a submarine?

Fleet submarines in World War II did not carry a ship's doctor. Instead, they relied on a hospital corpsman (pharmacist's mate, or PhM) who was responsible for all medical care aboard the boat. This individual had received intensive medical training — far beyond the basics — specifically to handle the wide range of emergencies that could occur on a 70-day patrol far from any medical facility.

The pharmacist's mate performed routine sick call (treating injuries, infections, dental emergencies, and illness), managed the medical supplies and the pharmacist's locker, and in serious emergencies performed procedures including emergency appendectomies, wound treatment, and amputations when necessary. The most famous example was Pharmacist's Mate 1st Class Wheeler Lipes, who performed an emergency appendectomy on USS Seadragon (SS-194) in 1942 using improvised instruments and spinal anesthesia — with the patient surviving.

When Pampanito rescued the British and Australian POWs in September 1944, the pharmacist's mate was overwhelmed with casualties. The survivors presented with malnutrition, infected wounds, tropical diseases, and shock. The medical challenge of caring for 73 severely ill men in the limited space of a submarine was enormous. Pharmacist's mates on submarines routinely demonstrated a level of medical skill and resourcefulness that far exceeded their official training.

142: Did any American submarines get captured by the enemy?

No. Not a single US submarine was captured by Japanese forces during World War II. Every submarine that was lost in the Pacific was destroyed — sunk by depth charges, mines, aerial bombs, or other causes — with no known case of a surviving submarine being boarded and taken as a prize by the enemy.

This was partly by design. Standing orders for submarine commanders were that the boat must be scuttled (intentionally sunk by the crew) before allowing it to fall into enemy hands. The cryptographic equipment on board — code machines, code books, and intelligence materials — was to be destroyed or jettisoned at sea before any surrender. In practice, the situations that sent submarines to the bottom were almost always violent and rapid, leaving no opportunity for the Japanese to capture a boat intact.

One incident came close: USS Tullibee (SS-284) was accidentally sunk by her own torpedo in 1944, leaving one survivor who was captured by the Japanese. But the submarine itself was gone. In the Atlantic, the Germans actually captured a British submarine (HMS Seal) intact, and the Allies famously captured the German U-505 by boarding in 1944 — but nothing comparable happened to US submarines in the Pacific.

143: What was a ballast tank on a submarine?

A ballast tank was a large tank built into the outer hull of the submarine, used to make the boat dive beneath the surface or surface again. In simple terms: fill the ballast tanks with seawater and the submarine sinks; blow the seawater out with compressed air and the submarine floats back up.

Fleet submarines like Pampanito had main ballast tanks located along the sides of the hull, between the outer pressure hull and the streamlined outer casing. These tanks had flood valves on the bottom and vent valves on the top. To dive, the vent valves were opened, letting the tanks flood with seawater while air escaped from the top. The submarine's overall density increased until it was heavier than water, and the boat descended. To surface, high-pressure air was blown into the tanks, forcing the water back out through the bottom flood valves. The submarine became lighter than water and rose.

There were also trim tanks and variable ballast tanks used for fine-tuning the submarine's fore-and-aft balance (trim) while submerged. Managing trim was a constant task for the diving officer — fuel was consumed, water was used, and torpedoes were fired, all changing the boat's balance. The diving officer and chief of the watch had to constantly adjust water between tanks to keep the submarine level and at the correct depth.

144: What was a war patrol report?

A war patrol report (officially the "Patrol Report" or "War Patrol Report") was a detailed official document prepared by the commanding officer after each war patrol, describing everything that happened during the mission. These reports became critical intelligence documents and historical records.

A typical patrol report included: the date and location of the patrol area, all contacts with enemy vessels (including date, time, position, course, speed, and identification), all torpedo attacks (number of torpedoes fired, estimated hits, observed results, and damage assessment), any surface engagements using the deck gun, enemy anti-submarine attacks received (depth charges, aircraft, escort attacks), observations of enemy shipping patterns and routes, weather conditions, and administrative matters such as crew performance and mechanical issues. The captain also included his own tactical assessments and recommendations for future patrols.

These reports were closely analyzed by submarine command at Pearl Harbor and the Naval War College. Commanders who consistently reported aggressive tactics and accurate results were promoted; those with poor performance — too cautious, too many misses — were relieved of command. The patrol reports were also cross-referenced with Japanese records after the war as part of the Joint Army-Navy Assessment Committee (JANAC) review that officially credited ship sinkings.

145: How did submarines avoid enemy minefields?

Avoiding minefields was one of the most nerve-wracking navigation challenges for WWII submarine commanders. Mines could not be detected by sonar — they appeared as passive, silent threats. The primary defense was intelligence: US submarines received detailed mine charts from submarine command at Pearl Harbor showing known or suspected enemy minefields, and commanders were expected to navigate around them.

Navigating through mine-infested waters required precise position-keeping. Submarines transited at very slow speed, often at periscope depth or fully surfaced at night, with the captain personally supervising the approach. In narrow straits like the Lombok Strait, Luzon Strait, or the approaches to Japanese ports, submarines carefully followed swept channels or chose deep-water routes where mines were less likely to be anchored successfully.

Despite precautions, several US submarines were lost to mines during the war, including USS Grunion (SS-216), which may have been sunk by her own circular-running torpedo or a mine. Submarines often had no warning: if a submarine hit a mine, it was almost always fatal with complete loss of crew. The uncertainty of not knowing whether a charted route was still clear was a constant background anxiety during any patrol in restricted, mine-threatened waters.

146: What was a fleet boat?

"Fleet boat" was the informal nickname for the Gato, Balao, and Tench class fleet submarines that formed the backbone of US submarine operations in World War II — the same general type as USS Pampanito. The term distinguished these large, ocean-going submarines from smaller coastal or minelaying submarines and from experimental or specialized types.

Fleet boats were designed to operate with the fleet — meaning they had the speed (nominally 20+ knots on the surface), range (11,000+ nautical miles), and endurance (75 days) to operate across the entire Pacific

Ocean alongside surface task forces. In practice, they were used independently on long-range war patrols, but the "fleet" designation reflects the original requirement that they be able to keep up with fast-moving surface forces.

They were the largest and most capable conventional submarines the United States had ever built, and they proved devastatingly effective against Japanese shipping. Fleet boats like Pampanito were approximately 311-312 feet long, displaced about 1,500 tons surfaced, carried 24 torpedoes, and had a crew of around 70-80 men. By contrast, earlier submarines like the S-class boats were much smaller, slower, and shorter-ranged — not suited for the vast distances of Pacific warfare.

147: How did submarines recharge their batteries?

Fleet submarines like Pampanito used a diesel-electric propulsion system. When on the surface, the diesel engines drove generators that both propelled the submarine and charged the large banks of storage batteries located below the crew quarters. When submerged, the diesels couldn't run — they require air — so the submarine ran on battery power alone.

Recharging the batteries required coming to the surface and running the diesel engines. A full charge took several hours. Submarines typically surfaced after dark to recharge, since daylight on the surface in enemy waters was dangerous. The crew followed a routine: diesel engines started, generators clutched in, battery charging began, while simultaneously the boat was ventilated with fresh air and any routine maintenance was done topside. A watch was kept on the bridge with radar scanning for surface or air threats.

The batteries had a limited capacity: at slow submerged speed (2-3 knots), they lasted roughly 24-48 hours. At higher submerged speed, they drained in hours. The balance between conserving battery power while submerged and finding safe opportunities to surface and recharge was one of the fundamental strategic constraints of submarine warfare in the diesel-electric era. This is why the snorkel (absent on US WWII submarines) was such an important development — it allowed recharging while remaining nearly submerged.

148: What was the difference between a Gato and a Balao class submarine?

The Gato and Balao class submarines were nearly identical externally and very similar in most respects — the Balao was essentially an improved version of the Gato, incorporating lessons learned early in the war. The most important difference was hull strength and diving depth.

The Gato class (built 1941-43) had a test depth of 300 feet and could go somewhat deeper in emergencies. The Balao class (built 1942-45) used stronger, higher-yield steel in the pressure hull, giving it a test depth of 400 feet. This extra 100 feet of depth was significant — it allowed Balao-class submarines to dive below the set depth of many Japanese depth charges, which were often set to explode at shallower depths. USS Pampanito is a Balao-class submarine (look for the SS-383 hull number range: Gatos were roughly SS-212 to SS-284, Balao class SS-285 to SS-416).

The Tench class, which followed, was a further improvement with more internal space and better equipment. All three classes were visually similar to the untrained eye, and the crews operated them in essentially the same way. The deeper diving capability of the Balao gave her crews slightly better odds of surviving depth charge attacks in the later years of the Pacific war when Japanese anti-submarine techniques were improving.

149: What were the daily duties of the captain of a submarine?

On a WWII fleet submarine, the captain's daily routine revolved around keeping the boat combat-ready, maintaining the crew's morale and discipline, and pursuing the mission. A typical patrol day included reviewing the night's radio traffic from submarine command, studying charts and intelligence on enemy shipping in the patrol area, conducting or attending training drills, reviewing the engineering plant status with the chief engineer, and conducting periscope watches in the patrol area.

The captain did not stand a regular watch — he was always "on call" and expected to be at the conn whenever a contact was made, whenever conditions were dangerous (restricted waters, poor visibility, aggressive enemy activity), or when any decision of consequence had to be made. In practice, experienced captains slept in brief intervals near their battle station, ready to be awakened instantly.

The captain also managed the human side of the boat. He knew every man by name and maintained personal contact with the crew, which was easier on a submarine than on any larger warship. He dealt with crew concerns, discipline issues, and morale — the latter being crucial on a 70-day patrol. The evening meal was sometimes a briefing time when the captain told the crew where they were and what was happening. A commanding officer who kept his crew informed and treated them with respect earned extraordinary loyalty from men willing to follow him into mortal danger.

150: How did submarines handle the threat of enemy destroyers?

Destroyers were the primary anti-submarine threat to US submarines in the Pacific. They were fast, maneuverable, and equipped with sonar and depth charge throwers. When a submarine was detected, a destroyer could run down on it at 30+ knots and deliver a pattern of depth charges with deadly precision.

The submarine's primary defense was to dive deep, change course frequently, and reduce speed to minimize noise. By going deep (the Balao class could reach 400 feet or beyond in emergencies), submarines could get below the depth setting of many Japanese depth charges, which were often preset before the attack and couldn't be changed mid-drop. Radical course changes made the submarine's position harder to predict. Slowing to 1-2 knots minimized propeller noise, making it harder for enemy sonar to track the boat.

Experienced commanders also used the ocean's thermal layers — changes in water temperature at different depths that bend or scatter sonar waves — to hide from active sonar pings. Going deep and creeping under a thermocline layer could make a submarine effectively invisible to sonar above the layer. The cat-and-mouse battle between a submarine and a destroyer hunting it was a battle of nerve, technique, and endurance. Some attacks lasted many hours; a few went on for more than a day before the destroyer withdrew or ran out of depth charges.

151: What was the sound of a torpedo being fired?

Firing a torpedo from a submarine was a distinctive sensory experience. The crew in the torpedo room heard and felt a sharp, compressed-air WHUMP — a quick shudder as high-pressure air (or water-ram, depending on the firing system) ejected the torpedo from the tube. Immediately after, the outer door of the tube thudded shut. In most cases, the submarine barely moved from the force, but in shallow water or when multiple torpedoes were fired in quick succession, a slight shudder ran through the hull.

The conning tower crew listened for the "fish running hot, straight, and normal" confirmation from the torpedo room: the torpedo's own turbine engine starting up and running correctly. If the torpedo ran erratically or circled back (occasionally happened with defective torpedoes), the submarine had to take emergency evasive action.

After the torpedo left the tube and crossed the distance to the target — typically 1 to 3 minutes at 46 knots — the crew listened for the explosion. A hit produced a deep, rolling BOOM heard clearly through the hull, sometimes followed by secondary explosions as the target's boilers or ammunition detonated. A miss produced no sound. The tension of those silent minutes between firing and impact (or non-impact) was one of the defining emotional experiences of submarine warfare.

152: What was commissioning day like for a new submarine?

What was the commissioning of a submarine?

Commissioning was the ceremony that officially brought a new warship into the United States Navy. For a submarine, commissioning day was a milestone in the boat's history: the first time the crew formally took possession of the vessel, raised the national ensign, and placed the ship "in commission" as an active unit of the US Navy.

The ceremony typically took place at the shipyard where the submarine was built. The crew was assembled on deck or on the pier in dress uniform. A Navy official read the commissioning orders, the commanding officer officially accepted the ship, and the first watch was set — symbolically beginning the submarine's naval service. The Commissioning Pennant was broken at the masthead and the ensign raised for the first time. Family members and shipyard workers were often present.

After commissioning, the submarine underwent sea trials and shakedown exercises to correct any deficiencies found during testing, before proceeding to a fleet command for combat assignment. USS Pampanito was commissioned on November 6, 1943, at the Portsmouth Naval Shipyard in Kittery, Maine, and she completed her shakedown before heading to the Pacific to begin combat operations.

153: What happened to Japanese submarines at the end of WWII?

What happened to the Japanese submarine force in WWII?

Japan entered the war with roughly 60 front-line submarines and built many more during the conflict, eventually deploying over 170 submarines. By the end of the war in August 1945, Japan's submarine force had been devastated. The Japanese lost approximately 130 submarines during the war — more than 75% of their wartime submarine strength — to US destroyers, destroyer escorts, aircraft, mines, and accidents.

Japan's submarine force had been misused strategically: instead of deploying them primarily against Allied merchant shipping (as the US used its submarines against Japan), the Imperial Japanese Navy assigned submarines to resupply isolated island garrisons, carry supplies to besiege outposts, and conduct fleet support roles rather than commerce raiding. This reflected a fundamental difference in doctrine. The few Japanese submarines that did conduct attacks on merchant shipping had some successes, but the potential for a Pacific "unrestricted submarine warfare" campaign against Allied supply lines was largely squandered.

At the war's end, surviving Japanese submarines were surrendered to American and Allied forces. Most were eventually scuttled at sea by the US Navy in 1946 as part of "Operation Road's End," since no Allied

nation wanted to keep them and the cost of returning them to Japan was not justified.

154: What happened if a crew member died while on patrol?

What happened to men who were killed on a submarine patrol?

If a sailor died on a WWII submarine patrol — from illness, accident, or combat — the body could not be preserved or brought back to port. After any appropriate ceremony the captain deemed possible under the circumstances, burial at sea was the only option. The body was committed to the ocean. A full burial at sea with all honors was conducted if conditions permitted; in the middle of a combat patrol in enemy waters, a simpler and quicker ceremony was necessary.

The captain was required to document the death fully in the ship's log, recording the name, rate, circumstances of death, date, location, and any witnesses. Official notification was eventually sent to the Navy Department, which notified the sailor's family. Because a submarine's mission and location were classified, families often received only a minimal official notification that their loved one had died at sea.

The submarine's crew dealt with loss in the same stoic way they handled all hardships. On a small boat where every man knew every other, the death of a shipmate was deeply personal. Morale had to be maintained, and the mission continued. Veterans of WWII submarine service rarely spoke in detail about losing shipmates — it was one of the experiences that remained too difficult to discuss long after the war ended.

155: How did submarines get their orders for each war patrol?

Patrol assignments came from Submarine Command Pacific (ComSubPac), headquartered at Pearl Harbor under Admirals Withers, Lockwood, and their staff. When a submarine returned from a patrol and the crew had rested, the captain and executive officer met with the submarine command staff. They reviewed the previous patrol, received intelligence briefs on Japanese shipping traffic and known escort strengths in various areas, and then received their next patrol order.

The patrol order was a written document specifying the submarine's patrol area (designated by a grid zone in the Pacific), any special missions or restrictions, radio frequencies and schedules, liaison with other submarines or wolf packs in the area, and the expected duration of the patrol. The specific targets — what to attack — were left entirely to the captain's judgment once in the assigned area. Commanders had broad discretion on how aggressively to pursue contacts, when to attack, and when to evade.

Ultra intelligence (decoded Japanese radio transmissions from the code-breaking program at Pearl Harbor) was sometimes incorporated directly into patrol orders: a submarine might be directed to a specific location and time where a Japanese convoy was expected based on decoded traffic. This intelligence was among the most closely guarded secrets of the Pacific War.

156: What was the control room used for on a submarine?

The control room was the nerve center of the submarine — the space from which the boat was dived, surfaced, and steered. It was located amidships, directly below the conning tower. While the conning tower handled periscope observations and fire control during attacks, the control room handled the mechanical business of controlling the submarine's depth, trim, and position.

Key equipment in the control room included: the helm (steering wheel) for controlling the submarine's course, the bow plane and stern plane controls for adjusting depth, the ballast tank controls (flood and blow valves), the diving officer's station, and the Christmas Tree — a panel of red and green indicator lights showing whether every hull opening (hatches, torpedo tube outer doors, ventilation valves) was open (red) or shut (green). Before diving, the diving officer needed all green lights — "green board" meant safe to submerge.

The diving officer supervised the control room team, constantly adjusting trim by transferring water between tanks to keep the submarine level at the ordered depth. It was demanding, precise work: a poorly trimmed submarine in a combat situation could cost seconds that meant the difference between escaping and being sunk. The executive officer (XO) often stationed himself in the control room during battle stations while the captain used the periscope in the conning tower above.

157: What did the forward torpedo room look like?

The forward torpedo room was in the bow of the submarine, and it was both a weapons magazine and a crew berthing space. Six torpedo tubes were built into the forward bulkhead, each 21 inches in diameter and about 25 feet long. The tubes pointed forward and could be aimed only by steering the entire submarine toward the target.

Spare torpedoes — typically six to eight reloads — rested on heavy steel skids on the deck and in overhead racks. Each torpedo weighed over 3,000 pounds and was nearly 21 feet long. Handling these weapons in the confined space required a specialized team, and torpedo reloads at sea were slow, dangerous work done by men using chain hoists, rollers, and muscle power.

A small number of the crew (typically 8-10 men) bunked in the forward torpedo room. Their racks were mounted between and above the torpedo skids. This meant enlisted torpedomen slept inches from live warheads. When torpedoes were loaded or fired, the men working in the room had to navigate around each other in a space with almost no free floor area. Despite the cramped conditions, the torpedo room crew took great pride in their weapons and their ability to load and fire rapidly under pressure.

158: How did the crew dispose of garbage on a submarine?

Garbage disposal on a WWII submarine was a serious operational concern — not just a matter of cleanliness, but of tactical security. Any garbage or oil floating on the surface could reveal a submarine's position to enemy aircraft or ships.

Fleet submarines used a "garbage disposal unit" (GDU), also called the "Hanson" launcher — a dedicated tube that allowed garbage sealed inside weighted metal cans to be ejected from the submarine while submerged or on the surface. The can sank, taking the garbage to the bottom. This was the standard procedure while in enemy waters: all garbage was carefully collected, packed into the cans with sufficient weight to sink, and ejected through the GDU.

The crew also had strict policies against throwing any loose material over the side in enemy waters — no coffee grounds, food scraps, or waste paper. Even small amounts of floating debris could become a trail leading enemy forces to the boat or revealing that a submarine had recently been in the area. Oil leaks were another concern; a sheen of diesel fuel on the surface was a telltale sign that a submarine had been forced deep, and crews worked hard to fix any external leaks quickly.

159: What happened to enemy survivors after a submarine attack?

The policy for US submarines regarding enemy survivors was difficult and evolved during the war. The official position was generally that submarines should not stop to rescue enemy sailors from sinking ships — the danger of remaining on the surface near a burning or sinking vessel, with the risk of anti-submarine aircraft or escorts arriving, was too great. Submarines were the most irreplaceable and strategically valuable assets in the Pacific Fleet; losing one to rescue enemy survivors was considered unacceptable.

In practice, some submarine commanders did have their boats approach survivors briefly to ask about the nature of their cargo, the names of their vessels, or to take a prisoner for intelligence purposes. Others fired warning shots to alert survivors to lifeboats. But picking up large numbers of enemy survivors was not generally done.

This was in sharp contrast to Pampanito's famous rescue of Allied POWs in September 1944. When the Pampanito and Sealion sank the Japanese convoy carrying British and Australian prisoners, US submarine command authorized the submarines to break from mission and rescue all survivors they could find — precisely because these were Allied servicemen, not enemies. The rescue of 127 POWs was an extraordinary act in a war when combat necessity normally prevented any rescue operations.

160: How did submarines handle fires or flooding emergencies?

Fire and flooding were among the most feared emergencies on a submarine. In a sealed pressure hull with no escape routes while submerged, either could be fatal within minutes. Submarine crews drilled emergency procedures constantly.

For flooding, the first response was to isolate the affected compartment by closing the watertight doors — heavy steel hatches that could be dogged shut to contain flooding to one section of the boat. Pumps were activated to dewater the compartment if possible. If flooding was severe, the captain had to decide whether to surface immediately (dangerous in enemy waters) or attempt to control the flooding while submerged. Emergency surfacing was always possible by blowing all ballast tanks, but this exposed the submarine on the surface.

Fire on a submerged submarine was equally dangerous because burning consumes oxygen and generates toxic gases in an enclosed space. CO2 fire extinguishers were the primary tool, backed by Aquafoam for electrical fires. Crew members wore EABs (Emergency Air Breathers) — small chemical oxygen sets that provided a few minutes of clean air. The electrical and fuel systems presented the greatest fire risks. Several US submarines were lost or nearly lost to battery explosions and fires caused by seawater flooding into battery compartments and generating hydrogen gas or chlorine gas — one of the most dangerous scenarios the crew could face.

161: What was a periscope feather?

A periscope feather was the small white wake — a V-shaped bow wave — created on the surface of the ocean when a submarine's periscope was raised while the boat was moving at periscope depth. The periscope head, typically only 1.5 to 2 inches in diameter, would push against the water as the submarine moved forward, creating a small spray or ripple on the surface.

The feather was a serious tactical concern. A trained lookout on an enemy ship or an aircraft crew scanning the ocean could spot a periscope feather at considerable distance, especially in calm seas. Once spotted,

the feather immediately revealed the submarine's position, course, and approximate speed.

Experienced submarine captains minimized feather exposure by: limiting periscope observation to brief "looks" — raising the scope for only a few seconds at a time; slowing the submarine to the minimum steerage speed (typically 3-4 knots) before raising the scope, which reduced the feather size; avoiding periscope observations in calm, flat seas when the feather was most visible; and keeping the periscope as low as possible, with only the very tip above the surface. The standard operating technique was called "a quick sweep" — up for a few seconds, a rapid 360-degree scan, then back down. A captain who kept his scope up too long in enemy waters was making himself easy to find.

162: How did the crew deal with boredom on a long patrol?

A 60-75 day war patrol involved long stretches of routine — transiting to the patrol area, hours of waiting for contacts, and repeated watch cycles. Boredom and monotony were real challenges for crew morale, and submarine commanders took them seriously.

The crew's mess was the social center of the submarine. Card games (cribbage, pinochle, acey-deucey) ran almost continuously. Many boats had a small library of paperback books that circulated through the crew. Some submarines carried board games, and a few men played chess. Movies were shown on some boats using a small 16mm projector — a reel might be watched dozens of times over a patrol.

Food was a major morale booster. Submarine crews ate better than any other branch of the Navy: fresh food was loaded for the first two weeks, supplemented by an extraordinary range of canned goods, frozen meats, and freshly-baked bread. The cook was one of the most important men aboard for morale purposes.

Practical jokes were common — submarine culture was intensely informal compared to surface ships, and hazing/humor kept men loose. Some men spent free hours writing letters (to be mailed when they returned to port), reading, sleeping, or maintaining their equipment beyond what was required. The constant underlying tension of operating in enemy waters, paradoxically, made the quiet moments more valuable rather than less.

163: What was the role of the TDC operator on a submarine?

The TDC (Torpedo Data Computer) operator, typically a senior torpedoman or fire control officer, was one of the most critical positions during a torpedo attack. The TDC was an electromechanical analog computer that continuously calculated the firing solution — the gyro angle setting to be transmitted to each torpedo so it would arrive at the target's predicted position.

The TDC operator maintained the attack solution by constantly entering updated data from the captain's periscope observations: target bearing, estimated target angle on the bow (the angle between the target's course and the line of sight to the sub), estimated range, and estimated target speed. The TDC automatically solved the geometry of the intercept problem and set the torpedo gyro angles.

The operator had to track multiple inputs simultaneously: updating bearing from the captain's calls ("mark" each time the periscope was on target), reading the target's course from the bearing rate, and confirming that the firing solution made sense. If the TDC's solution diverged from what the captain was seeing, the operator had to quickly diagnose the discrepancy and correct it — often while the attack was already underway. A bad TDC solution meant torpedoes that missed. The Torpedo Data Computer and its skilled

operator were what separated a successful attack from a waste of expensive weapons and a potentially fatal exposure of the submarine's position.

164: What happened if a submarine ran aground?

Running aground was a nightmare scenario for any submarine commander — it immobilized the boat, made it vulnerable to enemy attack, and was often the precursor to the loss of the vessel. Groundings were most likely in restricted, poorly-charted waters such as harbor approaches, shallow coastal areas, or island passages where navigation was difficult and bottom depth could change rapidly.

When a submarine ran aground, the first priority was to free it. The captain would order emergency ballast blowing, all engines to full power in reverse, and crew members shifting aft to change the trim. If the boat had gone hard aground, pumping out all ballast and variable tanks to reduce weight was tried. In some cases, a submarine could back itself off a grounding with engine power alone; in others, it remained fast.

The situation became desperate quickly. A stationary submarine visible from the surface or detectable as an anomaly on sonar was highly vulnerable to air and surface attack. If enemy forces arrived before the sub could free itself, the captain faced the impossible choice of surrendering the boat (against all orders and tradition) or scuttling. Several submarines were lost in circumstances involving groundings. USS Grunion (SS-216) disappeared in 1942 in shallow Alaskan waters under circumstances that may have involved a grounding or a circular-running torpedo. Navigation in restricted waters was taken extremely seriously, and captains were generally cautious about transiting poorly-charted areas.

165: What did a Japanese convoy look like?

A Japanese convoy typically consisted of merchant ships — freighters, tankers, and troop transports — traveling in formation, escorted by naval vessels tasked with anti-submarine defense. The size varied enormously: small coastal convoys might have 3-5 ships with 1-2 escorts, while major strategic convoys could have 10-20 or more merchant vessels with multiple destroyer escorts, submarine chasers, and occasionally air cover.

The convoy formation was usually several columns of merchant ships, with escort vessels positioned ahead, on the flanks, and astern. The escorts would zigzag and conduct active sonar searches while the merchants maintained a fairly steady course. Japan's convoy system was poorly organized early in the war — ships often sailed independently, without escort, making them easy prey. As the war progressed and losses mounted, Japan improved its escort doctrine and convoy organization, which made US submarine attacks more dangerous and complicated.

From a submarine commander's perspective, attacking a convoy required threading between or around the escorts to get a shooting position on the high-value targets (large tankers and troop transports were the priority). After firing torpedoes, the submarine had to evade the inevitable counter-attack by the surviving escorts, who would immediately attack the submarine's last known position with depth charges. The Pampanito attacked convoys on her third, fourth, and fifth war patrols, contributing to the destruction that steadily throttled Japan's ability to supply its far-flung empire.

166: How did the Pampanito earn her battle stars?

USS Pampanito (SS-383) earned six battle stars during World War II — one for each of her six official war patrols, all six of which were designated as "successful" combat patrols.

Her most significant achievements came on the third and fifth patrols. On the third patrol (August–September 1944), Pampanito was part of a coordinated attack group with USS Sealion and USS Growler that attacked a major Japanese convoy carrying Allied POWs. The group sank multiple ships including the passenger-cargo vessel Rakuyo Maru and the tanker Zuiho Maru. After the attack, Pampanito returned to rescue 73 British and Australian prisoners of war — the largest number of POWs rescued by a US submarine in the Pacific War.

Over her six patrols, Pampanito was officially credited with sinking five Japanese ships and damaging others, contributing to the systematic destruction of Japanese merchant shipping that strangled Japan's ability to supply its empire. She operated in some of the most dangerous waters in the Pacific, including the South China Sea and the waters around Japan itself. The six battle stars reflect not just confirmed sinkings, but the full scope of her combat operations — torpedo attacks, convoy interdiction, and the extraordinary mercy mission that followed her most dangerous patrol.

167: What was the flying bridge on a submarine?

The "flying bridge" on a fleet submarine referred to the open platform at the very top of the conning tower fairwater (the streamlined structure built around the periscope shears), above the main bridge level. In some usages, "bridge" referred to the entire open top of the conning tower fairwater; the "flying bridge" was the highest platform, giving the officers and lookouts the maximum height above the water for better visibility.

On the main bridge, the officer of the deck (OOD) conn'd the submarine on the surface — giving orders for course, speed, and diving. Lookouts were stationed in the periscope shears or on the bridge wings, scanning the horizon with binoculars in all directions. On Pampanito and similar Balao-class submarines, the bridge was equipped with a compass repeater, engine order telegraphs, and communication to the conning tower below via voice tube or sound-powered telephone.

Bridge watches on the surface were wet, cold, and uncomfortable in rough weather — the bridge was open to the sea, and spray regularly soaked the watch standers. Rain, wind, and darkness made four-hour watches exhausting. When a dive was ordered, the bridge watch was the last to leave — the OOD was required to visually confirm all lookouts were below before dropping down the hatch himself, pulling it shut behind him as the boat went under.

168: How deep could the Pampanito dive?

USS Pampanito is a Balao-class submarine. The Balao class had an official test depth of 400 feet — the depth to which the submarine was certified safe by the Navy. In practice, Balao-class submarines could (and sometimes did) go significantly deeper than 400 feet in emergencies, though the risk of hull failure increased with every foot below test depth.

The improvement over the earlier Gato class (test depth 300 feet) was achieved by using higher-yield steel (HY-42 steel) in the pressure hull construction. This extra hundred feet of diving depth was operationally important: it allowed Balao-class submarines to dive below the set depth of many Japanese depth charges, which were often preset by the attacking destroyer crew before the run and couldn't be changed mid-attack.

The "crush depth" — the depth at which the hull would actually implode — was estimated at roughly twice the test depth, meaning around 800 feet, though this was never tested deliberately. In reality, any

submarine diving well past its test depth was gambling with the lives of its crew. The depth gauge, the groaning of the hull under compression, and the sound of leaking fittings were constant reminders during deep dives of exactly how thin the steel walls were between the crew and instant death at the bottom of the Pacific.

169: What was the magnetic exploder problem with US torpedoes?

The magnetic exploder (called the Mark 6 exploder) was one of the worst weapons failures in US naval history, and it cost the lives of many American submariners who attacked ships and got no results in the first two years of the war.

The concept was clever: instead of requiring a direct contact hit, the torpedo's magnetic influence exploder would detect the magnetic signature of an enemy ship's hull and detonate beneath it, breaking the keel with a massive water hammer effect — far more destructive than a simple hull puncture. The weapon was so secret it was not adequately tested before the war.

In actual combat, the Mark 6 failed in multiple ways. The exploder was set too sensitively and detonated prematurely (before reaching the target), or failed to detonate at all due to manufacturing tolerances, magnetic anomalies in tropical waters, or simple design flaws. Submarine commanders reported torpedo after torpedo running hot, straight and normal — striking an enemy ship and failing to explode. These were called "duds." The Bureau of Ordnance initially blamed the submarine commanders for poor targeting, an infuriating accusation that delayed fixing the problem. Admiral Lockwood, Commander Submarines Pacific, eventually ran direct tests in Hawaii and confirmed the defects. The magnetic exploder was officially deactivated in 1943, and the simpler contact exploder replaced it — but not before the confidence, morale, and tactical results of the entire submarine force had suffered for nearly two years.

170: Who was the most successful US submarine commander of WWII?

By official postwar damage assessments (JANAC — Joint Army-Navy Assessment Committee), Commander Dudley W. "Mush" Morton in USS Wahoo (SS-238) is often cited as one of the most aggressive and legendary submarine commanders of the war, though the records for confirmed tonnage sunk are held by others.

By total tonnage officially credited, Commander Richard H. O'Kane in USS Tang (SS-306) is generally considered the top US submarine ace of WWII, credited with sinking 24 ships totaling approximately 93,824 tons. O'Kane had previously served as Morton's executive officer aboard Wahoo, and brought that aggressive style to his own command. He was captured after USS Tang was sunk by one of her own circular-running torpedoes in 1944, survived a Japanese prison camp, and received the Medal of Honor.

Other top-credit commanders included Eugene Fluckey (USS Barb), Samuel Dealey (USS Harder — known as "The Destroyer Killer"), and Slade Cutter (USS Seahorse). These men shared a common characteristic: extreme aggressiveness in pressing attacks at close range, tenacity in following convoys despite counterattacks, and a willingness to accept risk that less decorated commanders avoided. The submarine force as a whole, however — not individual commanders — is credited with sinking over 55% of all Japanese shipping lost during the war.

171: How did the crew celebrate after a successful attack?

After a successful torpedo attack and the subsequent evasion of the inevitable depth charge response, submarine crews celebrated in ways appropriate to their confined circumstances. The first celebration was often simply survival — enduring a depth charge attack and emerging undamaged was itself worthy of relief bordering on elation.

When the danger had passed and the boat was clear, the cook would be called upon to prepare a special meal — something more elaborate than the routine fare. Ships that had sunk multiple vessels in a single patrol sometimes had a modest "victory dinner." The captain might pass the word through the boat with a brief announcement of the results, and the crew's mess would buzz with men comparing notes on what they'd heard — the explosion sounds, the secondary detonations, the sounds of a ship breaking up on the sonar.

Officers sometimes kept a running tally of ships sunk — some boats displayed pennants or flags on their periscope shears representing confirmed kills when they entered port. In port, the real celebration began: liberty in Pearl Harbor (or later Guam, Midway, or other advance bases), food, alcohol, and rest. Submarine sailors in port were known for their enthusiasm — they had money accumulated over a patrol (submarine duty pay was a premium), they were alive when many of their comrades had not come back, and they had roughly two weeks before the next patrol began.

172: What did submariners do when they returned to port after a patrol?

Returning to port after a 60–75 day war patrol was one of the most anticipated moments in a submariner's life. The first order of business after tying up at the submarine base was a medical inspection and rest. The crew was typically sent to a rest facility — in Pearl Harbor, this was the Royal Hawaiian Hotel, which the Navy had leased for submarine rest and recreation. The "Pink Palace" on Waikiki Beach became legendary among submariners: two weeks of the finest available food, beachside rest, swimming, and whatever recreation the men chose.

During the rest period, the submarine itself was turned over entirely to the relief crew — a permanent yard crew that handled repairs, maintenance, torpedo reload, provisioning, and any upgrades ordered by submarine command. The regular crew did essentially nothing related to their submarine during this period, by design.

The captain and XO were busy during the rest period: writing and submitting the patrol report, meeting with submarine command staff for debrief, receiving intelligence for the next patrol, and processing promotions, decorations, and disciplinary matters from the patrol. When the rest period ended, returning crew members were expected to be ready — physically recovered, mentally reset, and prepared for another two months in enemy waters. The knowledge that each patrol might be the last provided an intensity to the liberty period that peacetime shore leave never matched.

173: Why were submarines called the Silent Service?

"The Silent Service" was the popular nickname for the US submarine force in World War II, and it reflected two distinct characteristics of submarine operations.

The first was tactical: submarines operated by stealth and silence. They hunted while running as quietly as possible, avoided detection by minimizing machinery noise, and executed their attacks without warning. The silence of a submarine's approach — a vessel that could kill from an invisible position — was one of its defining tactical qualities.

The second meaning was institutional: during the war, the submarine service deliberately suppressed information about its operations. Unlike the surface Navy or the Army Air Forces, which publicized their battles and celebrated their victories with press coverage and public fanfare, submarine operations were kept secret. The locations of patrols, the names of ships sunk, even the names of submarine commanders were kept out of the newspapers. The reasons were operational security — telling the enemy which shipping routes were being interdicted would allow them to divert convoys — and also to protect code-breaking intelligence. If Japan knew that certain convoys were being targeted based on decoded radio traffic, it would suspect its codes were broken.

As a result, submariners returned from patrols that had devastated Japanese shipping and received almost no public recognition. The men who served in the Silent Service did so knowing that their extraordinary courage and sacrifice would go largely unacknowledged until long after the war ended.

174: How did submarine construction change during the war?

The US submarine construction program underwent a massive acceleration during World War II. Before the war, submarines were built in small numbers at a measured pace — a single fleet submarine took 18-24 months or more from keel-laying to commissioning. After Pearl Harbor, the Navy and shipyards restructured everything to build faster.

Construction time dropped dramatically: by mid-war, some fleet submarines were being commissioned in as little as 12-15 months. Shipyards like Portsmouth Naval Shipyard, Electric Boat in Groton, Manitowoc Shipbuilding in Wisconsin (which built submarines inland and floated them down the Mississippi River), and Mare Island Naval Shipyard all expanded capacity and adopted modern production techniques including prefabricated hull sections.

The designs themselves evolved. The Gato class (1941-43) gave way to the improved Balao class (1942-45) with deeper diving capability, and then the Tench class (1944-46) with more internal volume and improved equipment. Each class incorporated lessons from combat experience — crew habitability improvements, better radar, improved fire control, reduced construction time. By 1945, the US had placed over 200 fleet submarines in service, transforming the small pre-war force into the strategic weapon that sank over 5 million tons of Japanese shipping and contributed decisively to Japan's defeat.

175: What was a snorkel on a submarine?

A snorkel (or snort) was a retractable breathing tube that allowed a diesel-electric submarine to run its diesel engines while submerged just below the surface. It consisted of an air intake mast and an exhaust mast that extended above the waterline just enough to draw in air and expel exhaust gases, while the main hull of the submarine remained underwater.

The snorkel was a German invention, developed by the Dutch Navy in the 1930s and adopted and perfected by the German U-boat arm in 1943-44 as Allied anti-submarine warfare became increasingly effective. By snorkeling, a submarine could recharge its batteries and run on diesel power without having to surface entirely — massively reducing its radar and visual signature. A snorkeling submarine could be detected only by radar (the small mast was difficult to pick up), by its exhaust signature, or visually, making it far harder to find than a fully surfaced boat.

American WWII fleet submarines including Pampanito were not equipped with snorkels during the war — the US Navy chose not to adopt the technology during the Pacific conflict, in part because US submarines still

found adequate opportunities to surface safely at night for battery charging. The snorkel became standard on post-war submarines, and its adoption (along with faster battery technology) was a key step in the evolution toward the nuclear submarines that made the entire surface/submerge cycle unnecessary.

176: What was periscope depth on a submarine?

Periscope depth was the shallow depth at which a submarine could raise its periscope above the surface of the water to observe the surrounding area without fully surfacing. For fleet submarines like Pampanito, periscope depth was typically around 55-62 feet — deep enough that the submarine was completely below the surface (safe from ramming and surface waves), but shallow enough that the periscope head could be extended upward through the water.

Operating at periscope depth was a delicate balancing act for the diving officer. At that depth, ocean swells affected the submarine noticeably, and maintaining a steady depth required constant adjustment of the planes and ballast. If the boat was too shallow, the periscope might be swamped by waves or the periscope shears might briefly break the surface, creating a visible wake. If too deep, the periscope would be underwater and blind.

The periscope head itself was typically about 1.5-2 inches in diameter — just a thin tube above the surface — making it difficult to spot visually at distance, especially in choppy seas. The submarine's attack periscope was thin and low; the larger search periscope was used for longer-range observation. A submarine at periscope depth was in a position of controlled risk: able to see and potentially attack, but also potentially visible to enemy lookouts or radar, and much less protected than at deep running depth.

177: What was the officer of the deck responsible for on a submarine?

The officer of the deck (OOD) was the captain's direct representative and the individual responsible for the safe navigation and operation of the submarine at any given moment. When the captain was off watch, the OOD had command authority over all routine decisions — course, speed, dive or surface timing in safe waters, and watch standing. He was the one person who could order the ship to take action without waiting for the captain, within standing orders.

On the surface, the OOD was posted on the bridge, scanning the horizon with binoculars, managing the watch, and making sure the boat was proceeding safely. When submerged in the patrol area, the OOD held the conn in the conning tower, monitoring the periscope watch cycle, tracking sonar contacts, and deciding when to report developments to the captain.

The OOD was always a qualified officer (ensign through lieutenant commander). Junior officers served as OODs first under supervision, then independently once they had demonstrated competence. The position was a training ground for future commanding officers — the skills of ship-handling, decision-making under pressure, and crew management that made a good OOD were exactly the skills that made a good submarine captain. Every submarine CO started as a junior OOD, learning the craft watch by watch.

178: What was a conning tower on a submarine?

The conning tower was a small, heavily reinforced compartment built into the top of the pressure hull, accessed by a vertical ladder from the control room below. On fleet submarines like Pampanito, it was roughly 8-9 feet in diameter, shaped like a cylinder, and designed to be the command center for torpedo attacks.

Key equipment in the conning tower included: the attack periscope (thin, low-signature, used for final attack observations), the search periscope (larger, used for long-range horizon scanning), the Torpedo Data Computer (TDC), the helm (wheel for steering), radar repeaters, and sound-powered phone connections to every compartment. During an attack, the captain would be at the periscope, the XO or approach officer at the TDC, a helmsman at the wheel, and additional personnel tracking the fire control solution.

The conning tower was structurally separate from the main hull — it had its own flood and vent valves, and in theory could be sealed independently. Above the conning tower was the bridge access trunk, a watertight hatch leading up through the fairwater (the streamlined "sail" structure) to the open bridge on top. The conning tower was accessible only through the control room below and the bridge above, making it a well-protected, central command space shielded from the worst of any depth charge attack.

179: How were new submarine crew members trained?

New submarine sailors went through a structured pipeline before joining a fleet boat. First, enlisted men attended basic submarine school at New London, Connecticut (the "Submarine Capital" of the US Navy). The course covered basic submarine systems, emergency procedures, physics of pressure and buoyancy, torpedo handling, and the theory of submarine operations. Officers attended a separate officer basic course at the same facility.

After formal school, new crew members joined their assigned submarine as "non-quals" — men not yet certified. The qualification process aboard the boat was the real education: each man had to trace and demonstrate knowledge of every system, from the diesel engines to the trim pumps to the weapons systems. Senior hands guided the process, checking off each system after the trainee could describe its function and demonstrate its operation. The oral board — a formal examination by officers — was the final gate before awarding the submarine warfare insignia (dolphins).

Practical experience came fast in wartime. A new crew member who joined during refit between patrols might be on a combat patrol within a month of arriving on the boat. The compressed timeline of wartime training meant that some men qualified remarkably quickly, while others took longer. Throughout, the culture of the submarine service emphasized self-sufficiency: every man was expected to be able to fight the boat in any compartment if the regular operator was incapacitated.

180: What was a sound-powered telephone on a submarine?

A sound-powered telephone (SPT) was a communication device that required no electrical power supply to operate. The user's voice vibrated a diaphragm in the handset, which generated a small electrical current through electromagnetic induction. That current traveled through wires to the receiving handset, where it was converted back to sound — no battery, no amplifier, no external power source needed.

Sound-powered phones were essential on submarines because they functioned even when the main electrical systems were shut down, damaged by depth charges, or deliberately powered off for silent running. The system was reliable, simple, and nearly impossible to jam or intercept.

The submarine's sound-powered telephone network connected every key station: bridge to conning tower to control room to engine rooms to torpedo rooms to maneuvering room. Specific circuits (labeled by letter and color) served different functions — the JV circuit for the primary command circuit, JA for battle stations, 21MC for captain's orders, and so forth. During battle stations, every station manned its phones and reported readiness sequentially ("Forward torpedo room, manned and ready..." up the chain to the

officer of the deck. This roll call confirmed the entire crew was at their posts before any attack began. The sound-powered phone system was one of the most critical pieces of non-combat equipment on the boat.

181: How did submarines identify enemy ships at night?

Identifying enemy ships at night — distinguishing the type, size, and value of a target — was one of the more challenging skills in submarine warfare, and it was critical because the target identification determined whether to attack and how many torpedoes to fire.

The primary tool was the SJ radar, which could detect surface targets at ranges of 8-10 miles and greater. The radar return gave the target's bearing, range, and a rough size indication — a large return meant a big ship, a smaller return meant a smaller vessel or an escort. By tracking the radar return over several minutes, the crew could determine the target's course and speed.

When close enough, the periscope was used even at night. The Fleet-era attack periscope incorporated a "night glass" capability — a wider-aperture lens that gathered more light, making it possible to distinguish ship silhouettes in starlight or moonlight. Experienced captains could identify ship types from their silhouettes. The ONI (Office of Naval Intelligence) published silhouette recognition books that submarine crews studied before patrol — being able to instantly recognize a Maru-class freighter from a destroyer escort from a tanker by profile was a trained skill.

Acoustic information from the sonar operator also helped — propeller count and rhythm gave a rough sense of the number and type of ships in a convoy before visual or radar contact was made.

182: What was the after torpedo room used for on a submarine?

The after torpedo room was at the stern of the submarine and mirrored the forward torpedo room in function — it was both a weapons space and a crew berthing compartment. Unlike the forward room's six tubes pointing forward, the after torpedo room had four torpedo tubes pointing aft, allowing the submarine to fire at targets while running away from them.

The four after tubes were a tactically important capability. If a target appeared astern of the submarine after a forward attack, or if the submarine needed to attack an escort that was pursuing it from behind, the after tubes provided an option. Some attack solutions were worked specifically for an "up the kilt" shot — firing aft torpedoes directly at a pursuing destroyer.

Like the forward room, spare torpedoes rested on skids in the after torpedo room, held in place by cradles and securing chains. A small crew of torpedomen bunked in the space on racks mounted above and between the weapons. The aftermost compartment of the submarine, it was also the farthest from the conn — communication was by sound-powered telephone, and torpedo room personnel had to be sharp at interpreting both routine commands and emergency instructions rapidly. The after torpedo room chief was responsible for tube readiness, torpedo maintenance, and the safety of the crew members who lived and worked in that compartment.

183: How did the Navy select men for submarine duty?

Submarine duty was voluntary in the US Navy — no one was assigned to submarines against their will. Men had to volunteer, and the Navy then screened volunteers through a selection process designed to identify those physically and psychologically suited for the unique demands of submarine service.

Physical requirements included: normal or corrected-to-normal vision (severe visual impairment disqualified), no chronic ear problems (pressure changes during diving were a concern), no significant heart or lung conditions, and an ability to pass a pressure test in the escape training tank at New London — where recruits proved they could equalize pressure in their ears and handle the physical sensations of increasing water pressure.

The psychological screening was less formalized but no less real. Instructors and evaluators watched for signs of claustrophobia, poor stress response, or extreme anxiety. Men who showed these traits during training exercised in the mock-up tanks or drills were quietly directed elsewhere.

Beyond formal screening, submarines sought intelligent, adaptable sailors who could learn complex systems rapidly. The qualification requirement — that every man learn every other man's job — meant that below-average learners struggled to earn their dolphins and sometimes transferred off the boat. The combination of volunteer selection, physical screening, and the rigorous qualification process produced submarine crews that were, on average, among the most capable and self-sufficient men in the Navy.

184: What happened to the crew of a submarine that was sunk?

Of the 52 US submarines lost in World War II, virtually all were lost with all hands or with only rare survivors. The survival rate from sunken submarines was extremely low — the combination of depth, violent sinking, enemy action, and the Pacific Ocean made escape nearly impossible in most cases.

When a submarine was sunk in shallow water by depth charges, there was occasionally a slim possibility of using the escape apparatus — a rubber hooded suit (the Momsen lung or escape set) that allowed one or two men to breathe from a small air supply while ascending from a flooding compartment. The forward and after escape trunks were small flooding chambers that could cycle one man at a time to the surface. However, these devices worked only at relatively shallow depths (under about 100 feet) and required the submarine to be on the bottom intact with compartments still holding air.

In practice, most wartime submarine sinkings occurred under conditions where escape was impossible: at depth under sustained depth charge attack, through catastrophic flooding or explosion, or at depths far beyond where any escape device could function. Of the approximately 3,506 US submariners killed in WWII, the vast majority went down with their boats. The overall casualty rate for US submarine service was among the highest of any branch — roughly 22% of those who deployed on combat patrols were killed, the highest loss rate in the US Navy.

185: How did submarines attack targets at night on the surface?

Night surface attacks became one of the most successful and distinctive tactical developments of US submarine warfare in the Pacific. Instead of submerging to use the periscope, a submarine would approach a target convoy on the surface at night, using the low silhouette of the hull to remain nearly invisible against the dark water while taking advantage of its higher surface speed.

Surface attacks used the SJ radar as the primary targeting tool. The radar could track targets precisely enough to develop a fire control solution — bearing, range, course, and speed — without any visual contact. The Torpedo Data Computer received this radar data to calculate gyro angles, and torpedoes were fired using a radar-derived solution, sometimes without the captain ever seeing the target visually.

The advantages were significant: the submarine was faster on the surface than submerged, allowing longer pursuit of a convoy; radar provided accurate ranges that periscope rangefinding could not match at night; and the submarine avoided the battery drain of sustained submerged approach. The risk was reciprocal — the submarine was on the surface, visible to radar-equipped escorts and potentially to visual lookouts. Night surface attacks required nerve, fast decision-making, and a captain willing to close to relatively short range before firing. Many of the most devastating US submarine attacks on Japanese convoys in 1944-45 were executed precisely this way.

186: What did the lookouts on a submarine watch for?

Lookouts on the bridge of a surfaced submarine were among the most critical crew members at any given moment — their eyes were the first line of defense against the threats that could send the boat to the bottom. Typically three or four lookouts were posted on the bridge during surface running, each assigned a sector of the horizon to scan continuously. They were equipped with high-powered 7x50 binoculars, chosen for their light-gathering ability in darkness.

The primary threat lookouts watched for was aircraft — a single Japanese or Allied patrol plane could get within gun range before radar detected it, especially if the aircraft was flying low or the radar was temporarily off (to avoid detection). Lookouts shouted "Aircraft! [bearing]!" immediately upon sighting, giving the OOD the few seconds needed to initiate a crash dive. "Clear the bridge!" sent every man below in a controlled rush.

Lookouts also watched for ships — enemy surface vessels, other submarines (friendly and hostile), and surface debris. In enemy-controlled shipping lanes, a smudge on the horizon at dusk could be a smoke stack leading to a convoy. Lookouts were trained to distinguish silhouettes, estimate distances, and report bearing and elevation immediately without delay. An alert lookout who spotted a distant destroyer before it closed within gun range might save the boat; a distracted one who missed an approaching aircraft might doom it. Bridge watch duty was exhausting, cold, and wet — but profoundly consequential.

187: How were submarines refueled in the Pacific?

Fleet submarines in the Pacific were refueled primarily at fixed bases rather than at sea. The main base was Pearl Harbor, Hawaii, which served as the central hub for submarine operations throughout the war. Submarines returned to Pearl after each patrol to refuel, rearm, provision, and undergo maintenance and repair.

As the war progressed into 1944-45, the US established advance submarine bases closer to the patrol areas. Midway Atoll became an important forward base, as did Guam and other captured islands that were developed into submarine support facilities. Operating from Guam reduced the transit time to and from Japanese waters significantly, allowing longer time on patrol and shorter logistics tails.

Submarine tenders — large support ships like USS Proteus and USS Fulton — also provided fuel, torpedoes, provisions, and repair services. A tender anchored in a harbor or lagoon could support dozens of submarines simultaneously, giving each boat the equivalent of a small shipyard's worth of support. Tenders were particularly important at advance bases that hadn't yet developed full shore-based facilities.

At-sea replenishment of submarines from surface vessels (as was done for surface warships) was not a standard practice in WWII, primarily because the fuel transfer required both vessels to be relatively

stationary and the process was difficult and dangerous. The standard model was: patrol for 60-75 days, return to base, refuel and refit, depart for the next patrol.

188: What was a target's angle on the bow?

Angle on the bow (AOB) was one of the most critical measurements in setting up a torpedo attack. It was the angle between the target ship's heading (where the ship was pointed) and the direct line of sight from the submarine to the target — essentially, "how much of the target ship's course is angled toward you or away from you?"

An angle on the bow of zero degrees meant the target was headed directly toward the submarine (bow-on). An AOB of 90 degrees (starboard or port) meant the target was crossing at a perfect right angle. An AOB of 180 degrees meant the target was running directly away. The angle was measured from the target's bow — "port 45" meaning the target was heading 45 degrees to the left of the line of sight from the submarine.

AOB was estimated visually through the periscope by the captain, who compared the silhouette of the target ship to a standard recognition table — the more foreshortened the target appeared, the more bow-on or stern-on it was; the more the full profile was visible, the closer to 90 degrees crossing. Accurately estimating AOB was a trained skill that separated experienced submarine captains from novices. An error in AOB of just 10-15 degrees could mean the torpedo fired with a slightly wrong lead angle, causing a miss. The Torpedo Data Computer continuously used the AOB estimate as one of its key inputs to generate the final firing solution.

189: How did submarines communicate with Pearl Harbor during a patrol?

Radio communication between submarines on patrol and submarine headquarters at Pearl Harbor followed a carefully managed protocol designed to balance the need for information against the risk of radio direction finding (RDF) — the Japanese ability to detect and locate a transmitting submarine.

Submarines received traffic continuously. Pearl Harbor broadcast on a low-frequency schedule that submarines could receive while submerged or surfaced, using a trailing wire antenna. A submarine could stay current with headquarters orders, intelligence updates, and convoy position reports by simply copy the broadcast without ever transmitting — remaining completely silent and undetectable.

Transmitting from a submarine was avoided as much as possible. When a captain needed to send a contact report (reporting a sighting of enemy vessels), a sinking, or an emergency message, he would compose the shortest possible encoded transmission, surface briefly if necessary for better antenna height, transmit rapidly, and dive immediately. The message was encrypted using the Navy's communications codes, so the content was unreadable by the Japanese — but the transmission itself, even encoded, could be located by RDF at a range of hundreds of miles if Japanese shore stations triangulated simultaneously.

Ultra intelligence — decoded Japanese traffic — was sometimes used to vector submarines to specific convoy routes. These orders came via the broadcast, keeping the submarine entirely passive. The combination of one-way broadcast reception and rare, brief transmissions kept submarines as electronically invisible as possible.

190: What was the forward engine room on a submarine?

Fleet submarines like Pampanito had two engine rooms — the forward engine room and the after engine room — each containing two large diesel engines for a total of four main engines. The forward engine room was located amidships, just aft of the crew's forward living spaces.

Each engine room contained two Fairbanks-Morse Model 38D-8½ opposed-piston diesels (on Pampanito's class). These engines drove electrical generators: on the surface, the generators powered the propulsion motors and charged the batteries simultaneously. The forward and after engine rooms were essentially identical in layout and function, giving the submarine redundancy — if any one engine or even a full engine room was disabled, the other could continue to provide power.

The engine rooms were among the noisiest and hottest spaces on the boat. Working in them while running at full speed was a physically demanding environment — the engines were enormous, the heat from four large diesel engines operating at high load was intense, and the noise level made normal communication impossible without shouted words directly into the ear. Engine room personnel wore ear protection.

The engines required constant maintenance: oil changes, fuel injection system service, cooling water monitoring, and mechanical adjustments. The machinist's mates and motor machinist's mates assigned to the engine rooms were among the most technically skilled enlisted men aboard, responsible for keeping the prime movers of the submarine running reliably on patrols that could last 60-75 days from any repair facility.

191: How did submarines navigate to their patrol areas from Pearl Harbor?

The transit from Pearl Harbor to a patrol area in the central or western Pacific was itself a significant navigation challenge — distances measured in thousands of miles, with no landmarks and no GPS. Submarines relied on a combination of celestial navigation and dead reckoning.

On the surface at night, the navigator used a sextant to shoot star sights (measuring the altitude of known stars above the horizon) and calculated the submarine's precise position from those astronomical observations using spherical trigonometry and pre-computed tables. A skilled navigator could fix the submarine's position to within a mile or two using star sights taken at morning and evening twilight. Sun lines were taken at noon for a noon latitude fix.

Submerged during the day (when in potentially hostile waters), the boat navigated by dead reckoning: tracking course, speed, and time from the last known position to estimate current position. The submarine's log (a device measuring speed through the water) and the gyrocompass provided the inputs. Dead reckoning accumulated error over time — currents, compass drift, and small speed errors all added up — which is why surface position fixes by celestial navigation were essential every night.

Approaching the patrol area, the captain used more frequent periscope observations to cross-check the navigator's estimate against recognizable landmasses, landmarks, or coastal soundings. The accuracy required to navigate safely through narrow straits or mine-suspected channels demanded meticulous navigation work throughout the entire transit.

192: How did submarine captains decide when to surface?

The decision to surface was one of the most consequential a submarine captain made regularly — it was the moment of greatest vulnerability and also of operational necessity. Batteries needed recharging, fresh air was needed to ventilate the boat, and the higher surface speed allowed repositioning for the next day's patrol.

The standard pattern was to surface at or shortly after dark. The captain would order a thorough periscope sweep of the horizon and sky before surfacing — checking for ships, aircraft, and anything suspicious. Radar was used to check for contacts before the bridge watch was sent topside. Only when satisfied that the immediate area was clear would the captain order the ballast tanks blown and the crew to surface stations.

Factors influencing the timing included: remaining battery charge (how urgent was recharging?), the distance to potentially hostile shipping routes (closer = riskier to surface), moon phase (a full moon made the submarine visible; new moon favored surfacing), sea state (rough seas were both uncomfortable and reduced visibility for enemy lookouts and aircraft), and any specific tactical situation. In waters close to Japanese airfields or with known air patrols, surfacing in the early hours before dawn — when Japanese air patrols were least active — was sometimes preferred over surfacing at dusk when evening patrols were possible.

The captain never delegated the surfacing decision itself. However careful the preparation, surfacing in enemy waters was always a calculated risk.

193: What was the escape trunk on a submarine?

The escape trunk (also called the escape hatch) was a small flooding chamber built into the pressure hull of a fleet submarine, designed to allow crew members to escape from a sunken submarine. Fleet submarines had two: one in the forward torpedo room and one in the after torpedo room — the same hatches that also served as access points for loading torpedoes.

The escape trunk worked as an airlock: the inner hatch was sealed, the chamber was flooded with water until the pressure equalized with the outside water pressure, then the outer hatch was opened and the crew member could swim to the surface. Each man wore a rebreather device (the Momsen lung) that recycled exhaled breath through a CO₂ absorbent canister, providing a small air supply for the ascent.

In practice, the escape trunk was of limited usefulness in actual wartime conditions. It required the submarine to be on the bottom in relatively shallow water (not more than 100-150 feet) with at least one compartment intact, pressure below crushing depth, and crew members who had trained on the device and could remain calm enough to use it while the boat was flooding. Most wartime submarine sinkings did not meet these conditions — they occurred at depth, through violent destruction, or in circumstances that left no time or structural integrity for escape. The escape trunk was a last-resort tool, and few men used it successfully during the war.

194: How did sonar operators track enemy ships on a submarine?

The sonar operator — typically a rated sonarman or an experienced petty officer trained in sound listening — worked at the sonar station in the forward torpedo room or control room, wearing headphones and listening to the ocean through the submarine's hydrophone arrays. His job was to detect, identify, classify, and track any acoustic contacts.

Passive listening was the primary technique. The operator would slowly rotate the sonar head through the water, listening for the distinctive sounds of propellers. Different ships made different sounds: a destroyer's high-speed turbines produced a high, whirring scrape; slow merchant diesels produced a lower, rhythmic thudding; propeller blade count and rpm gave a rough indication of ship type and speed. An experienced

sonarman could estimate how many propellers a contact had and whether it was approaching or receding based on pitch change (the Doppler effect).

When tracking an attacking destroyer, the operator became as important as the captain. His reports — "contact bearing 020, getting closer," "contact drawing left, bearing now 010" — gave the captain the information needed to evade. When the sonarman reported "pinging" (active sonar transmissions from an enemy vessel), it meant they had been detected and the counter-attack was imminent. The interval between pings told the captain how far away the attacker was and whether its sonar was in contact with the submarine. Sonarmen who could maintain calm, continuous tracking data during a depth charge attack were invaluable.

195: What was the role of the chief of the boat on a submarine?

The chief of the boat (COB) was the senior enlisted man aboard the submarine — typically a chief petty officer or master chief — and the most important non-commissioned position on the boat. He was the primary link between the enlisted crew and the officers, serving as both the crew's advocate and the enforcer of standards and discipline below the officer level.

The COB's responsibilities included: supervising the enlisted watch bill in coordination with the executive officer, maintaining crew discipline and morale, managing the boat's daily administrative routine for enlisted personnel, monitoring the physical condition of the submarine's systems with the department heads, and advising the captain directly on crew matters. The COB was expected to know every enlisted man — his performance, his personal situation, his strengths and problems — and to bring issues to the officers' attention before they became crises.

During depth charge attacks and battle stations, the COB typically took the diving station in the control room, managing the ballast and trim to execute the captain's maneuvers. He was responsible for the boat's trim — the fore-and-aft balance — more than any other individual on a routine basis. A good COB who ran a tight, professional boat and looked after his crew was one of the defining characteristics of a successful submarine. Captains who had excellent chiefs of the boat were fortunate; those with weak or ineffective COBs often struggled to maintain the standards that a combat-ready submarine required.

196: How did torpedo spreads work on a submarine?

A torpedo spread (also called a fan shot) was the technique of firing multiple torpedoes at a single target, with each torpedo's gyro angle set slightly differently so the weapons fanned out to cover a range of potential target positions. Rather than firing one torpedo at a precise predicted position, a spread sent three or four torpedoes covering slightly different tracks — accounting for uncertainty in the target's exact course and speed.

The standard spread for a large merchant or tanker target was typically three to four torpedoes, with the angles spread several degrees apart. The first torpedo was aimed at the target's expected bow position, the last at its expected stern, and the middle torpedoes covered the area between. If the target had not changed course, at least one or two torpedoes should hit. If the speed estimate was slightly off and the target was farther ahead or behind the predicted position, the spread increased the probability that at least one weapon would still connect.

The decision to fire a spread was the captain's. Firing more torpedoes improved hit probability but consumed the boat's limited supply. With 24 torpedoes total, firing four-torpedo spreads at every target

would leave a submarine with very few shots for subsequent opportunities. Captains had to balance aggressive engagement (increasing damage to the enemy) against conservation (maintaining weapons for the remainder of a 60–75 day patrol). On a successful patrol, a submarine might fire 15–20 torpedoes and return with the remainder.

197: How did submarines work together to avoid colliding with each other?

Deconflicting submarines operating in the same area — avoiding blue-on-blue (friendly submarine vs. friendly submarine) incidents — was a serious, and sometimes fatal, problem during World War II. Several US submarines were fired upon or sunk by other US submarines in cases of mistaken identity.

The primary method of avoiding collision and fratricide was geographic separation. Submarine command at Pearl Harbor assigned each submarine a specific patrol zone, and submarines operating in adjacent zones were on strictly defined boundaries. A submarine was expected to remain in its assigned area; crossing into another submarine's zone without explicit permission and coordination was forbidden.

For submarines operating in coordinated wolf packs, deconfliction was managed by assigning specific attack sectors and timing windows — one submarine attacked from one bearing while another waited, preventing simultaneous convergence. Voice radio on short-range frequencies allowed brief coordination between pack members.

The challenge was that in the dark ocean, a surfaced submarine looking for convoy targets and a friendly submarine transiting the same area could converge rapidly. Emergency identification signals — light blinker codes, specific radar emissions, radio challenges — helped, but only if both parties were alert. Several losses were attributed to friendly fire: USS Dorado (SS-248) was likely sunk by a US aircraft, and USS Grunion's loss has been attributed to a possible own-torpedo circular run. These tragedies drove continuous improvement in identification protocols throughout the war.

198: How did a submarine conduct a periscope attack step by step?

A periscope attack was a carefully choreographed sequence that could unfold over hours or be compressed to minutes, depending on how quickly the target was closing.

Initial approach: The submarine detected the target — by sound, radar, or sighting — and maneuvered to an attack position. The ideal setup was to be ahead of the target's track, roughly 90 degrees off its bow, at a range of 1,500 to 3,500 yards when firing. The captain approached submerged, usually at around 3 knots to minimize noise and periscope wake.

Observation and solution-building: When close enough, the captain began "shooting" periscope bearings — raising the scope for a few seconds, marking the target's bearing, estimating range by stadimeter or by comparing the target's mast height to the scope's scale, then lowering immediately. The XO (or approach officer) fed each observation into the Torpedo Data Computer (TDC), which continuously updated the firing solution: target course, speed, range, angle on the bow.

Firing: When the TDC solution was "generated" — stable and reliable — the captain ordered "stand by." Torpedoes were fired at 6-second intervals to spread the spread across the target. The captain called "Fire!" (or "fire" was triggered automatically from the TDC). Four to six torpedoes per attack was typical.

Evasion: Immediately after firing, the captain went deep and changed course. "Rig for depth charge" — crew braced, secured machinery, prepared for counterattack. The crew listened for hits. A hit sounded like

a metallic "clang" at depth. Multiple hits meant a sinking; silence meant a miss.

199: What was the after engine room on a submarine?

Fleet submarines like the Pampanito had two engine rooms: the forward engine room and the after engine room, separated by the maneuvering room. Together the four Fairbanks-Morse diesel engines (two per room) gave the submarine its surface propulsion and battery-charging power.

The after engine room was located directly aft of the maneuvering room and housed two of the four Fairbanks-Morse Model 38D8-1/8 opposed-piston diesel engines. Each engine drove a generator rather than the propeller shafts directly. The generators fed power either to the main motors (for propulsion) or to the batteries (for recharging), and the maneuvering room controlled which configuration was in use at any given moment.

The after engine room was operated by machinist's mates (MM rating). Conditions were brutally hot — the engines radiated enormous heat, and in tropical patrol areas the temperature in the engine rooms could exceed 120°F. Noise was deafening; communication required shouting or hand signals. Despite this, machinists took great pride in their engines and kept meticulous logs of operating hours, temperatures, and fuel consumption.

In a submerged boat, both engine rooms were shut down. The engines were diesel and required air — running underwater was impossible. The after engine room went quiet and the crew worked in the dark heat, waiting for the next surface period when the engines could be restarted.

200: What were the diving planes on a submarine and how did they work?

Diving planes were the horizontal control surfaces — like horizontal rudders — that controlled a submarine's depth and pitch angle. Fleet submarines used two sets: the bow planes (forward) and the stern planes (aft, near the propellers).

Bow planes: Mounted on the forward part of the hull. When angled down, they pushed the bow downward, helping the boat dive. When angled up, they helped level off or climb. On fleet submarines the bow planes were retractable — they folded flat against the hull when surfaced so waves wouldn't damage them, and were rigged out before diving.

Stern planes: Located at the aft end of the boat near the screws. These were the primary depth-control surface and were used constantly while submerged to maintain ordered depth. The stern planesman sat in the control room next to the helmsman, watching the depth gauge and bubble (inclinometer) and working the planes continuously — it was essentially the same skill as flying an airplane.

The two planesmen worked together: bow planes controlled angle, stern planes controlled depth. On the attack team, the diving officer supervised both and was responsible for keeping the boat at periscope depth — shallow enough to use the scope, deep enough to avoid showing hull.

A full crash dive combined putting the bow planes on full dive, flooding the ballast tanks, and increasing speed. An experienced crew could take a surfaced submarine below periscope depth in under 60 seconds.

201: How did submarines use radar during World War II?

Radar transformed submarine warfare. US fleet submarines were equipped with two types: the SJ surface search radar and the SD air search radar.

SJ surface search radar: The most tactically important. It could detect a large surface ship at 10–12 miles and a small vessel at 4–6 miles, day or night, in rain or fog. The SJ gave precise bearing and range, which could be fed directly into the Torpedo Data Computer for a fire control solution. Night surface attacks using radar alone — without any periscope exposure — became standard practice by 1943–44 and were extremely effective against Japanese convoys. The radar operator became one of the most critical crew members.

SD air search radar: A simpler omnidirectional set that detected aircraft. It gave only range, not bearing, so it couldn't tell you which direction the plane was coming from — only how far away it was. When the SD showed a contact closing, the standing order was to dive immediately. The SD saved many submarines from air attack by giving a warning that visual lookouts might have missed.

Risk of radar: Transmitting radar pulses could be detected by enemy radar warning receivers beginning in 1944. Japanese aircraft were eventually equipped with radar detectors that could pick up SJ transmissions. This forced submariners to use radar intermittently rather than continuously, reducing the advantage. Nonetheless, radar remained one of the greatest tactical advantages US submarines had over Japanese naval and merchant forces.

202: How did submarine crews cope with the heat in the tropics?

The Pacific War was fought in some of the hottest waters on earth, and fleet submarines of the World War II era had no air conditioning — the crews simply endured.

In tropical patrol areas, below-decks temperatures routinely reached 95–110°F, and the engine rooms could exceed 120°F. The air was thick with diesel fumes and humidity. Men working the engine rooms sweated through their clothes within minutes. Condensation ran down the steel hull as warm air met cold seawater temperatures, making surfaces slippery and equipment rust-prone.

Water was rationed strictly. Fresh water was used for drinking, cooking, and battery maintenance. Showers were infrequent — a "Navy shower" meant a 30-second rinse. Men slept and worked in minimal clothing — shorts and T-shirts, or nothing at all in the worst compartments. Significant weight loss over a patrol (five to ten pounds) was common.

The ice cream machine was not a luxury but a morale necessity — cold food was one of the only reliefs from the heat. Submarine crews were known for receiving better rations than other Navy branches partly as compensation for the brutal conditions. The boats also carried more cold storage capacity than other ships of their size, refrigerating food and providing cold drinks that were small but real comforts in 100-degree heat.

203: What qualities made a successful submarine commander?

The submarine commanders who achieved the greatest success — the "aces" — shared a recognizable set of characteristics that went beyond technical skill.

Calculated aggression: The most successful captains were aggressive, pushing into close range and staying on the attack even when the tactical picture was uncertain. But they weren't reckless — they knew when to

break off and when to accept risk. Dudley "Mush" Morton of USS Wahoo and Sam Dealey of USS Harder both exemplified this quality.

Decision speed: An attack unfolded in minutes. A captain who hesitated or second-guessed his TDC solution missed his shot. The ability to commit quickly, with incomplete information, was essential.

Crew confidence: In an enclosed steel tube with no escape, the crew had to trust their captain's judgment completely. A captain who inspired confidence — calm under depth charges, decisive at the scope, honest about dangers — could get more out of his crew than pure technical skill could explain.

Spatial reasoning: The mental geometry of an attack — visualizing a moving target, calculating intercept angles, estimating ranges — came more naturally to some officers than others. The best captains were almost running a parallel solution in their heads alongside the TDC.

Patience: Most of a patrol was boredom, watchful waiting. The captain set the tone. An impatient captain wasted torpedoes on poor setups; a patient one waited for the ideal firing position.

Well-known aces: Richard O'Kane (USS Tang), Dudley Morton (USS Wahoo), Samuel Dealey (USS Harder), Eugene Fluckey (USS Barb), Slade Cutter (USS Seahorse).

204: What was it like inside a submarine during a depth charge attack?

A depth charge attack was one of the most terrifying experiences a submarine crew could endure, and veterans who survived them rarely forgot the sound.

When the first depth charge exploded nearby, the shock came not as a distant rumble but as an instantaneous, skull-rattling CRACK — a physical concussion that shook the entire boat. Cork insulation flew off the interior walls. Light bulbs shattered. Gauges broke their mountings. Men were thrown against equipment. In a close pattern, the boat lurched sideways or pitched bow-down.

The boat went into "silent running": all non-essential equipment shut off, pumps and fans stopped, crew moved on sock feet and whispered. The temperature rose as ventilation stopped. CO₂ accumulated. Men sat motionless at their stations for hours, listening to the sonar operator track the destroyer's propeller sounds — "getting louder" meant another run; "fading" meant a miss.

The persistent dangers during a prolonged depth charge attack were flooding (from cracked seams, burst fittings, or failed gaskets), battery damage (seawater in the batteries generated chlorine gas, potentially fatal), and oxygen depletion from hours submerged without snorkeling. Submarines carried CO₂ absorbent and oxygen canisters for extended submergences.

Some boats absorbed 100 or more depth charges over a single engagement before escaping or being destroyed. The psychological toll on crews who survived repeated attacks was severe, and "depth charge nerves" was a recognized condition by the end of the war.

205: How did submarines evade destroyers and anti-submarine escorts?

Surviving a destroyer counterattack required every tool available — mechanical, tactical, and in some cases luck.

Go deep and go slow: The first response to a depth charge attack was to dive below the estimated depth of the charges and slow to "creep speed" — as quiet as possible. At very slow speeds, the submarine's own

propeller noise dropped dramatically, making it harder for the destroyer's sonar to track. Some captains stopped completely and let the boat drift, relying on neutral buoyancy.

Silent running: All non-essential machinery shut down. Pumps, fans, gyrocompasses — anything that made noise went off. Crew whispered. No one moved unnecessarily. A single dropped wrench could be heard by the enemy sonar.

Use the thermocline: Ocean temperatures drop with depth, and these temperature gradients — thermoclines — create acoustic barriers. A sonar ping from above the layer frequently bent or reflected rather than penetrating to the submarine hiding below. Experienced captains knew the water temperature profile and got below the thermocline when possible.

Maneuver unpredictably: Random course changes disrupted the destroyer's attack pattern, which required tracking the submarine's course ahead to place depth charges in the right spot.

Counter-attack: Some commanders fired a torpedo at an approaching destroyer both to defend themselves and to force the escort to evade, breaking up the attack run. This worked for a few extraordinary captains (Dealey sank five destroyers this way).

Time: Most attacks eventually ran out of depth charges or contact. Staying alive long enough for the enemy to give up was strategy in itself.

206: Why were aircraft such a dangerous threat to submarines?

Aircraft were uniquely dangerous to submarines because they could appear with almost no warning and attack before the submarine could dive to safety.

Speed of approach: A surfaced submarine had lookouts scanning the horizon, but an aircraft at high altitude could close from 10 miles to overhead in less than two minutes. The time from first sighting to crash dive completion was a race — a slow crash dive crew could easily be caught on the surface or at shallow depth where bombs and aircraft-delivered depth charges were most effective.

Depth of attack: Aircraft dropped shallower-set depth charges than surface ships, and they could also drop bombs. Shallow explosions were more effective against submarines caught at periscope depth or during the early phase of a dive than the deep patterns used by destroyers.

Coordination: Allied aircraft in the Atlantic learned to coordinate with surface ships, chasing submarines down while calling in surface escorts for the kill. Japan eventually developed similar coordinated ASW capability.

SD radar: The US developed the SD air search radar specifically to give submarine crews more warning time. When the SD scope showed a closing contact, standing orders were to dive immediately without visual confirmation. The SD saved many boats.

Aircraft losses: Aircraft were responsible for a significant portion of US submarine losses. Many boats vanished without survivors — believed sunk by air attack in shallow water where escape was impossible. The introduction of radar-equipped Japanese patrol aircraft in 1944–45 made the nighttime surface period — previously almost safe — significantly more dangerous.

207: What were wolfpack tactics and how did the US Navy use them?

Wolfpack tactics were coordinated operations by multiple submarines against a single convoy or target area. While Germany's famous Rudeltaktik system used large packs tightly coordinated from shore, the US Navy developed a distinctly American version.

US wolfpacks were typically groups of two to four submarines placed under a task group commander — usually a senior officer embarked on one of the boats. The submarines were assigned overlapping patrol areas designed to catch a convoy no matter which route it took. When one submarine made contact, it tracked and reported while others moved into attack position.

US wolfpacks differed from German ones in key ways: they were coordinated more loosely, with individual captains retaining far more tactical independence; communication was minimal to reduce radio risk; and the primary goal was attacking the same convoy with multiple boats rather than massed night surface attacks.

Notable US wolfpacks included "Ben's Busters" (Cdr. Thomas Oakley, 1944), "Blair's Blasters" (Cdr. Leon Blair), and operations associated with the "Hit Parade" series. The South China Sea and Luzon Strait became primary wolfpack hunting grounds by 1944–45 as ULTRA intelligence identified convoy routes.

Results were mixed: some wolfpack operations sank many ships; others had poor coordination and individual boats got in each other's way. Most US skippers by 1944 were operating in the prime shipping lanes, and whether in a wolfpack or alone, were finding targets.

208: Did US submarines ever sink enemy submarines?

Yes — US submarines did engage and sink enemy submarines, though this was never their primary mission. Fleet boats were tasked with attacking merchant shipping, fleet units, and troop transports; submarine-versus-submarine combat was relatively rare.

The most remarkable example was USS Batfish (SS-310), which sank three Japanese submarines in three consecutive nights in February 1945 in the Luzon Strait — an extraordinary feat that earned her a Presidential Unit Citation. All three kills were made on the surface by gunfire and torpedo.

Most US sub-vs-sub kills occurred when both submarines were surfaced and one sighted the other visually or by radar. Submerged torpedo attacks against submerged submarines were extremely difficult — the attacking boat had to fire on passive sonar contact without active pinging (which would reveal its own position), and a moving submerged target at periscope depth was a very small and elusive target.

Japanese submarines did sink several US submarines in return. I-176 sank USS Grunion, and other sinkings have been attributed to Japanese submarines though some remain uncertain. Neither side made sinking the other's submarines a high strategic priority — Japan because of the IJN's offensive doctrine, the US because of ULTRA intelligence showing that hitting Japanese logistics was far more valuable.

Overall, US submarines sank approximately 20–25 Japanese submarines during the war, a secondary but real contribution to the defeat of Japan's naval forces.

209: What was the difference between a magnetic exploder and a contact exploder on a torpedo?

The exploder was the detonating mechanism in the torpedo's warhead, and its reliability — or lack of it — was one of the greatest scandals of American submarine warfare.

Contact exploder (Mark 5): The older, simpler mechanism. When the torpedo struck the hull of a ship, a firing pin was driven forward by the impact, detonating the explosive charge. It worked exactly as advertised — when the pin functioned correctly. Early in the war, the contact exploder was discovered to have a critical flaw: the firing pin was too fragile to withstand a direct, 90-degree bow-on impact. It bent rather than firing, causing duds even on dead-center hits.

Magnetic exploder (Mark 6): A more sophisticated device designed to detect the change in the earth's magnetic field caused by a ship's steel hull passing overhead. Rather than requiring direct contact, the torpedo was supposed to pass under the keel and detonate from the magnetic influence. The explosion beneath the keel was theoretically more lethal — it could break a ship's back rather than just punch a hole in the side.

In practice, the Mark 6 was a disaster. It detonated prematurely (up to 300 yards short), ran erratically at the wrong depth to trigger, or failed to detonate at all. Commanders in the Pacific were convinced the exploders were defective by early 1942, but the Bureau of Ordnance insisted the weapons were reliable and blamed submarine commanders for firing errors.

It was not until mid-1943 that the Navy officially acknowledged the exploder failures and corrected them. The combination of a faulty magnetic mechanism AND a flawed contact backup meant that many perfectly aimed shots produced duds — a catastrophic loss of opportunity in the critical early years of the submarine campaign.

210: How effective was Japanese anti-submarine warfare against US submarines?

Japanese anti-submarine warfare (ASW) was effective enough to destroy 52 US submarines — about 22% of all US submarines deployed in the war — making it one of the most lethal aspects of the Pacific campaign for submariners.

Japanese capabilities: The IJN equipped its destroyers and convoy escorts with hydrophones, sonar (Type 93 underwater sound equipment), and depth charges comparable in basic capability to Allied equipment. Japanese destroyer crews were skilled and aggressive — many individual attacks were devastatingly well-executed. Some of the most significant US submarine losses happened in the first two years of the war when Japanese ASW patrols were frequently encountered and US crews were still learning.

Japanese weaknesses: Japan was slow to organize a coherent convoy escort system — convoys ran without adequate destroyer protection until relatively late in the war. Japan also never fully coordinated aerial and surface ASW the way the Allies did in the Atlantic. Japanese aircraft rarely carried updated radar, limiting their effectiveness against nighttime surfaced submarines until 1944–45.

Improvement over time: As the war progressed, Japanese ASW improved. More escorts were assigned to precious oil tanker convoys. Aircraft equipped with radar appeared by 1944, making night surface operations more dangerous. Magnetic mines and new depth charge patterns caused increased US losses in 1944.

By war's end, the combination of improved Japanese ASW, minefields, and the sheer operational intensity of US submarine patrols had made the submarine service the highest casualty rate of any branch of the US armed forces in the Pacific.

211: What was a submarine tender and what did it provide to submarines?

A submarine tender was a large support ship that served as a mobile base for a flotilla of submarines. Without tenders, the US submarine campaign in the Pacific would have been impossible — submarines needed constant resupply, repair, and crew rest that no shore base in the combat zone could provide.

What the tender provided: A tender carried everything a submarine needed: torpedoes, fuel, food, spare parts, fresh water, and technical specialists. The machine shops aboard a tender could perform almost any repair short of drydocking — overhauling engines, reconditioning torpedoes, replacing electronics, patching hull damage. Tenders also had medical facilities, including surgeons, since submarines had only pharmacist's mates.

Crew rest: Between patrols, submarine crews lived aboard the tender in relative comfort while their boat was serviced. A proper bunk, hot meals in a real galley, and freedom from the cramped, smelly submarine interior made the tender a significant morale and readiness resource.

Key tenders in the Pacific: USS Fulton (AS-11), USS Sperry (AS-12), USS Euryale (AS-22), USS Holland (AS-3), and USS Proteus (AS-19) supported fleet operations at Pearl Harbor, Midway, Majuro, Guam, and later Subic Bay. The forward staging of tenders — moving them closer to the patrol areas as the war progressed — allowed longer operational time on station and shorter transit time from patrol area to refit.

Organizational hub: The tender was also the administrative home for the squadron. The squadron commander (ComSubRon or ComSubDiv) was usually embarked aboard the tender, where patrol reports were filed, orders were received, and awards were presented.

212: Why did the Mark 14 torpedo have a depth-running failure and run deeper than set?

The Mark 14 torpedo's depth-running failure was one of the most damaging and frustrating equipment failures of the entire war. Submarine commanders saw torpedo after torpedo pass harmlessly under their targets, and for over a year the Bureau of Ordnance refused to believe them.

The problem: The Mark 14 ran approximately 10 to 11 feet deeper than its depth setting. A torpedo set to run at 10 feet — aimed at piercing a destroyer's hull — actually ran at 20 to 21 feet, passing under the keel without making contact. Against shallow-draft vessels, this meant a clean miss every time, regardless of how perfectly the firing solution was calculated.

Why it wasn't caught: Pre-war testing used exercise heads (empty) rather than live warheads. The exercise heads were slightly lighter than war shots, causing the torpedo to run accordingly shallower during testing. The weight difference, and therefore the depth discrepancy, was never measured or reported. The torpedo entered service uncorrected.

Cover-up and resistance: The Bureau of Ordnance insisted the weapons were correct and blamed the submarine captains for poor technique. Reports of duds and missed shots were dismissed as operator error. It took nearly 18 months of combat failures before Admiral Lockwood in 1942 authorized his own tests at Fremantle — using nets strung in a harbor — to prove the depth was wrong.

Fix: Once the depth problem was confirmed in late 1942, the depth mechanisms were corrected. A separate problem — the firing pin in the contact exploder — took until September 1943 to fix. And the magnetic

exploder's failures had already been known since early 1942. The complete set of fixes to the Mark 14 didn't come until mid-to-late 1943 — nearly two years into the war.

213: How did ULTRA codebreaking intelligence help US submarines find targets?

ULTRA was the Allied codebreaking program that broke the Japanese naval cipher system JN-25 and the Maru (merchant shipping) codes. It gave US submarine commanders intelligence of extraordinary value — advance knowledge of convoy routes, departure times, composition, and escort strength.

How it worked operationally: When codebreakers at Pearl Harbor, Brisbane, or FRUPAC (Fleet Radio Unit Pacific Fleet) decrypted a Japanese naval message indicating a convoy departure, that information was relayed to ComSubPac (Commander Submarines Pacific). Patrol orders were issued routing submarines into the convoy's path. To conceal the source, submarines were sometimes told they had received intelligence from "a reliable source" or from a fictional reconnaissance aircraft.

Impact: By 1943–44 ULTRA was enabling an average of 20–25% of all US submarine sinkings. Some analysts credit ULTRA with an even higher fraction of the decisive sinkings in 1944 when the oil convoy routes were targeted. The ability to intercept specific, high-value convoys — including tankers carrying oil from the Dutch East Indies — was directly enabled by ULTRA.

Secrecy: The intelligence was so sensitive that submarine captains were not formally told they were receiving ULTRA-sourced intelligence. Commander John P. Cromwell of USS Sculpin knew the full extent of ULTRA — and when Sculpin was sunk and the crew ordered to abandon ship, Cromwell chose to go down with the boat rather than risk capture and interrogation. He was awarded a posthumous Medal of Honor for this decision.

Long-term effect: Japan never suspected its codes were broken. The JN-25 cipher was periodically updated, but each new version was eventually broken. The most valuable decryption period was 1943–1945, when the submarine campaign was reaching its decisive phase.

214: What was a down-the-throat shot on a submarine?

A down-the-throat shot was a torpedo fired directly at an oncoming enemy warship — a destroyer or escort heading straight toward the submarine at high speed, usually during a depth charge attack.

The tactical situation: After a submarine attacked a convoy, the escort vessels counterattacked. A destroyer tracking the submarine would line up and charge directly at it. The submarine was in a desperate position — it could dive deeper, evade, and absorb depth charges, or in some cases choose to fight back. A down-the-throat shot gave the submarine a chance to stop the attack before it reached firing point.

Why it was difficult and desperate: The geometry was extremely challenging. The target's profile was almost nothing — a ship coming bow-on presents a tiny target cross-section, maybe 30 feet wide. The gyro angle was near zero (torpedoes fired straight ahead), so there was almost no room for error in timing. The range had to be correct. The submarine also had to be at periscope depth to fire — shallow enough to be vulnerable to ramming and gun fire, at exactly the moment the attacker was closest.

Why it worked sometimes: Despite the difficulty, some captains scored down-the-throat kills. Commander Samuel Dealey of USS Harder made a specialty of it — he sank five destroyers, many with close-range bow-

or-stern shots that became legendary. The Japanese destroyer captains who pressed their attacks straight in, line-abreast, gave submarines this opportunity.

Risk and reward: A successful down-the-throat shot eliminated the attacking escort, freeing the submarine to continue with its patrol. Failure meant a close-aboard explosion, possible ramming, or an accurate depth charge pattern at close range. It was a last-resort tactic that required exceptional nerve.

215: How did mines threaten US submarines and how did crews cope?

Mines were a constant, silent threat to US submarines operating in Japanese-controlled waters. Unlike aircraft or destroyers, a mine gave no warning and no chance to evade — the first sign of a mine field was sometimes an explosion.

Japanese minefields: Japan heavily mined the shallow approaches to its home islands, harbor entrances, and key shipping straits. Minefields were placed in the waters around Japan proper, in the Tsushima/Korea Strait, around Okinawa, and in the East and South China Seas. These fields were known to exist but their exact boundaries were uncertain — charts were based on intelligence that was often incomplete.

How submarines avoided them: Submarine headquarters distributed mine-avoidance routing charts, and ULTRA intelligence sometimes revealed newly-laid fields from decrypted Japanese naval messages. Submarines were routed around known fields where possible. In shallow waters, a submarine could try to stay deep (below mine depth) or navigate narrow swept channels. Sound gear could sometimes detect anchored mine cables.

Notable losses: USS Grunion (SS-216) was lost in 1942, probably to a mine or a malfunctioning torpedo. USS Flier (SS-250) was sunk by a mine in the Balabac Strait in August 1944 — eight survivors managed to reach shore and were eventually rescued. USS Tullibee was lost to her own circular-running torpedo, illustrating how fine the line was between enemy weapons and friendly ones going wrong.

The final barrier: Operation Barney (June 1945) sent nine submarines through the heavily mined Sea of Japan as a coordinated penetration, using FM sonar to detect and navigate around mines. One submarine, USS Bonefish, was lost to a surface ASW attack inside the Sea of Japan. The operation proved submarines could operate inside a mined region — at significant risk.

216: Were the sinking claims made by US submarines accurate?

No — US submarine sinking claims during the war were significantly overstated, a finding that emerged after the war when postwar assessment teams compared American claims against captured Japanese records.

The Joint Army-Navy Assessment Committee (JANAC): After the war, JANAC reviewed every US submarine patrol report against Japanese shipping loss records. JANAC's final report (1947) credited US submarines with sinking approximately 1,314 merchant ships totaling 5.3 million tons, plus 201 warships. This was roughly one-third less than what submariners had claimed during wartime.

Why overclaiming happened: In the pressure of an attack, a submarine heard explosions after firing and noted the target had disappeared from the periscope field. These were interpreted as hits and sinkings. In reality, explosions could be depth charges, the target could have turned away or sped up, and a hit was not always a sink. Submarines rarely had the luxury of surfacing to confirm a sinking.

Was it dishonest? No. The claims were made in good faith in highly stressful conditions, with limited information and under immediate threat. Overclaiming was universal in all navies in all theaters — Japanese and German submarine claims were also substantially inflated.

Historical context: Even the JANAC figures represent a devastating strategic achievement. The 1,314 confirmed merchant sinkings destroyed Japan's ability to import the oil, ore, and food required to sustain its war economy and population. By 1945 Japan had essentially no functional merchant marine. US submarines, representing less than 2% of US Navy personnel, accounted for more than 55% of all Japanese shipping losses.

217: What was the overall strategic impact of the US submarine campaign?

The US submarine campaign against Japan was one of the most decisive strategic operations of World War II — arguably the single factor most responsible for Japan's economic and military collapse by 1945.

Shipping losses: US submarines sank approximately 1,314 merchant ships totaling over 5 million tons, plus about 200 warships. Japan's merchant fleet, which it needed to import oil, ore, food, and raw materials from its conquered territories, was reduced from approximately 6 million tons in 1941 to under 2 million tons by mid-1945.

Oil embargo effect: Japan's critical vulnerability was oil. Its entire military machine, from aircraft carriers to fighter planes, ran on oil imported from the Dutch East Indies. By 1944 US submarines were intercepting the tanker convoys on the routes from Southeast Asia to Japan with devastating effect. By late 1944 the Japanese Navy curtailed training flights because aviation fuel was unavailable. Capital ships that sortied to Leyte Gulf in October 1944 barely had enough fuel for a one-way trip.

Warship sinkings: Among the warships sunk by US submarines were the carriers *Shinano* (the world's largest carrier at the time, sunk by USS *Archerfish* four days after commissioning), *Taiho*, *Unyo*, *Chuyo*, *Shokaku*; the battleships *Kongo* and *Musashi* (partially); and scores of cruisers and destroyers.

Human cost: 52 US submarines were lost — about 22% of all submarines that made war patrols. Approximately 3,500 US submariners died, giving the submarine service the highest proportional casualty rate of any branch of the US armed forces in the Pacific. The motto "Run Silent, Run Deep" captures both the method and the sacrifice of these men.

218: How did a torpedo tube work to launch a torpedo?

Firing a torpedo from a submarine was a carefully engineered sequence, not simply pulling a trigger. Each step had to be correct — an error could sink the submarine.

Loading: The torpedo, weighing about 3,000 pounds and 20 feet long, was loaded into the tube breech-first by the torpedo room crew using a hand-cranked loading mechanism that slid it along a track from the stowage rack into the tube. The breech door (rear door) was then sealed and locked.

Pre-firing preparation: The tube was flooded with sea water to equalize pressure with the outside ocean, preventing the outer door from being blown off when it was opened. A hydraulic piston or compressed-air manifold was connected at the rear. The outer door on the hull exterior was then opened. The submarine at periscope depth meant water pressure outside was relatively low.

Firing: When the captain called "Fire!", the fire control panel electrically completed a circuit that opened the high-pressure air impulse valve at the back of the tube. The impulse charge — a pre-set volume of high-pressure air at about 200–400 psi — pushed the torpedo out of the tube in approximately one second. As the torpedo exited, its engine started (steam turbine for Mark 14, battery for Mark 18).

After firing: The impulse air ejected behind the torpedo vented through an inboard vent to prevent an air bubble from rising to the surface and revealing the firing position — called "clearing the air bubble." The outer door was then closed, the tube inboard-vented, drained, and reloaded. Reloading took about 15–20 minutes.

The submarine could fire multiple tubes in rapid succession — the Pampanito and other Balao-class boats had six forward and four aft tubes.

219: What was the forward battery compartment and why did officers live there?

The forward battery compartment served a dual purpose that was unique and somewhat uncomfortable: it was both the home of one of the two main battery banks and the officers' living area.

Beneath the deck: The forward battery contained 126 lead-acid battery cells housed below the wooden deck plating in a watertight compartment that extended the full length of the space. These cells — each weighing several hundred pounds — were the submarine's underwater power source. Battery maintenance required periodic ventilation and produced hydrogen gas, making open flame strictly prohibited throughout the compartment.

Above the deck — officers' country: On the main deck level sat the wardroom (officers' dining room and lounge), the commanding officer's private cabin, the executive officer's stateroom, and four or five additional officers' staterooms. By warship standards these were reasonably private quarters — each officer had a bunk, a small desk, and a curtain. The CO's cabin was the largest, roughly the size of a small closet.

The wardroom: Officers ate meals in the wardroom, which doubled as a meeting room, game room, and planning space. The captain held pre-attack briefings here. Meals were eaten in shifts, and the wardroom steward (often a Filipino mess attendant in pre-war Navy tradition) served the officers' meals.

Battery smell: Despite the wooden deck covering, a faint sulfurous odor was a constant feature of the forward battery compartment. If the battery was damaged — particularly by seawater contamination — it would produce toxic chlorine gas, one of the most feared emergency scenarios on a submarine. "Battery gas" drills were practiced regularly.

220: Were any submarine crew members awarded the Medal of Honor in World War II?

Yes — five US submariners received the Medal of Honor during World War II, and all five stories reflect extraordinary acts of courage, often at the cost of their own lives.

Commander Howard W. Gilmore, USS Growler (posthumous): In February 1943, after ramming a Japanese ship during a night surface engagement, Gilmore was wounded on the bridge and unable to climb down to the hatch as enemy fire continued. Knowing he could not make it inside before the submarine was sunk, he ordered "Take her down" — commanding his crew to dive without him. He went down with the sea. His Medal of Honor was awarded posthumously.

Commander John P. Cromwell, USS Sculpin (posthumous): When USS Sculpin was heavily damaged and sinking in November 1943, Cromwell ordered the crew to abandon ship. He knew the full extent of ULTRA intelligence — convoy schedules, codes, future operations — and refused to be captured for fear of what the Japanese might extract. He went down with Sculpin. Posthumous Medal of Honor.

Commander Samuel D. Dealey, USS Harder (posthumous): Over multiple patrols, Dealey sank five destroyers in close-quarters attacks, including down-the-throat shots that destroyed escort vessels that would have defended valuable convoys. USS Harder was lost in August 1944. Posthumous Medal of Honor.

Commander Richard H. O'Kane, USS Tang: O'Kane pressed USS Tang into extraordinarily aggressive attacks until a circular-running torpedo from his own boat sank Tang in October 1944. O'Kane survived as a Japanese prisoner of war. Medal of Honor awarded on his return.

Commander Eugene B. Fluckey, USS Barb: Fluckey's extraordinary patrol record included operations in the Sea of Japan and a landing party mission to blow up a Japanese train. Medal of Honor awarded in 1945.

221: How did submarine crews send and receive mail during war patrols?

Mail was one of the most powerful morale factors for submarine crews, and the Navy went to significant effort to route it through the forward areas — but the logistics of delivering mail to a submarine on a secret patrol were inherently awkward.

Receiving mail: A submarine's home base fleet post office (FPO) at Pearl Harbor, Brisbane, Fremantle, or later Guam held accumulated mail for the crew while the boat was on patrol. When the submarine returned to base and tied up to the tender, mail was one of the first things delivered. After a 50–60 day patrol, a crewman might receive a stack of dozens of letters in a single delivery. Any mail that had arrived after the submarine departed was held for the next return.

V-mail: Later in the war the Navy used V-mail (Victory mail) — a system in which letters were microfilmed, shipped on microfilm rolls rather than as bulky paper packages, and printed at the receiving end. This allowed vast quantities of correspondence to be moved with minimal cargo weight and volume.

Outgoing mail: Letters written aboard a submarine were held until the boat reached port. Outgoing mail was censored — a censor (usually a junior officer) read all letters and blacked out any information about location, convoy sightings, patrol areas, or recent actions. Sailors quickly learned what could and could not be written. Some developed private codes with their families to convey a general sense of location; if discovered, such codes were forbidden.

Holiday packages: Families mailed cookies, candy, and small gifts. During long patrols approaching the holidays, mail was especially important — and its absence especially hard. A lost boat meant a stack of undeliverable mail accumulating on a tender, which was how families sometimes first suspected something had gone wrong.

222: Why did submarines use their deck guns to attack surface targets instead of torpedoes?

Deck guns — primarily the 4-inch/50 caliber and later the 5-inch/25 caliber guns mounted on the submarines' weather decks — were not just defensive weapons. In the right circumstances, submarines used them aggressively for surface attacks, conserving torpedoes for higher-priority targets.

Against small and shallow-draft targets: Torpedoes were expensive (roughly equivalent to a small house in wartime cost) and carried in limited quantities — 24 aboard a Balao-class submarine. Small Japanese vessels — coastal freighters, sampans, barges, fishing boats, harbor tenders — were simply too small and shallow-draft for torpedoes to work reliably. Gun action was faster, cheaper, and more certain against these targets. Submarines sank hundreds of small vessels by gunfire.

When torpedoes were already expended: Late in a patrol, after firing most torpedoes, a submarine might still have deck gun rounds available and a tempting target. Rather than return with ammunition unused, captains would attack by gun.

Finishing off damaged ships: If a torpedo had damaged but not sunk a ship, a surfaced gun action could finish the job, saving precious torpedoes. This was particularly useful against tankers, which were hard to sink with a single torpedo.

Conditions for deck gun action: A deck gun attack required calm sea conditions (the deck was only feet above the waterline), absence of aircraft, and minimal escort threat. At night, a radar-equipped surfaced submarine had a significant advantage over an unescorted freighter with no fire control equipment. In these conditions, gun attacks were safer than they might appear.

Late-war evolution: By 1944–45 torpedo reliability had improved and targets were more heavily escorted. Pure gun actions became less common, though the guns remained in use.

223: What was the difference between the conning tower and the control room?

The conning tower and the control room were adjacent but distinct spaces with different functions, located one above the other via a hatch and ladder. Visitors often confuse them because both played vital roles in operating and fighting the submarine.

The conning tower: The conning tower was the small cylindrical compartment above the pressure hull, inside the fairwater (the vertical structure also often called the "sail" or "tower"). It contained the primary attack equipment: the two periscopes, the torpedo data computer (TDC), the helm (steering wheel), the sonar operator's position, and the attack periscope's optics. During a submerged torpedo attack, the commanding officer, TDC operator, and helmsman all worked in the conning tower. It was cramped — roughly 8 feet in diameter and maybe 7 feet tall.

The control room: Directly below the conning tower, accessed through a hatch in the conning tower deck, was the larger control room. This was the depth and diving management center. It contained the ballast and trim control manifolds, the diving planes controls (operated by the two planesmen), the depth gauges, and the Christmas tree (the ballast tank indicator panel). The diving officer supervised the control room during dives; the chief of the watch managed the diving panel. The navigation plot table was also in the control room.

Division of command: During an attack, the captain in the conning tower handled the tactical picture — he was looking through the periscope, reading bearings, calling the solution. The diving officer in the control room kept the boat at periscope depth — a few feet too shallow or too deep would take the scope out of the water or submerge the hull too much. The two spaces had to work in constant coordination through the open hatch between them.

224: What was a Presidential Unit Citation and how did the Pampanito earn one?

A Presidential Unit Citation (PUC) was the highest collective military honor the President of the United States could award to a military unit. For a ship or submarine crew, receiving the PUC was the unit equivalent of an individual receiving the Navy Cross — second only to the Medal of Honor in the hierarchy of valor awards.

What it recognized: The PUC was awarded for extraordinary heroism in action against an armed enemy, performed with distinction that set the unit apart from others facing similar challenges. It required combat action, not merely meritorious service.

USS Pampanito's citation: The Pampanito earned her Presidential Unit Citation for her third and fourth war patrols — conducted in 1944 under Commander Paul E. Summers. These patrols included the September 12, 1944 engagement in the South China Sea in which Pampanito, operating with USS Sealion II and USS Growler, attacked Convoy HI-72 and sank multiple vessels. After the attack, Pampanito and Sealion discovered that the convoy's freighters had been carrying British and Australian prisoners of war from Singapore. Pampanito rescued 73 POWs from the water — the most rescued by any US submarine from a single action — and Sealion rescued additional survivors.

The combination of aggressive convoy attack and the extraordinary humanitarian rescue operation, in enemy waters, under threat of counterattack, was the basis for the citation. The crew also received individual combat awards across these patrols.

What it means for visitors: The Pampanito was not just a museum piece — she is a decorated combat vessel with a documented record of valor and one of the most remarkable rescue operations of the submarine war.

225: Why did submarines have two periscopes and what was each one for?

Fleet submarines like the Pampanito carried two periscopes, and each served a different tactical purpose. Having two periscopes allowed the captain to choose the right tool for the situation rather than compromise between opposing requirements.

The attack periscope (No. 1 scope): The attack periscope was designed to be as small and hard to detect as possible. It was very thin — approximately 1.4 inches in diameter at the upper end — with minimal top optics to reduce the visual disturbance ("feather") at the surface. The trade-off was smaller lenses and lower light-gathering capability. The attack scope was used during the actual firing phase, raised only momentarily for last-second observations. Being thinner, it was less likely to be spotted by a sharp-eyed lookout on the target ship.

The search periscope (No. 2 scope): The search periscope was significantly larger in diameter — approximately 5 inches — with superior optics, higher magnification, and better light gathering for use in limited visibility or dusk/dawn conditions. It also carried a radar antenna and could be equipped with bearing-and-elevation scales for firing control. The search scope was used for the approach phase, when the submarine was still distant from the target and needed maximum observation capability. Because it was larger, it produced a more visible feather at the surface and was retired once the attack run began.

Redundancy: Having two periscopes also provided redundancy. If the attack scope was damaged — not uncommon in rough seas or close action — the search scope could take over. During the entire attack

sequence, from initial detection through firing, a submarine captain might alternate between scopes depending on what information he needed at that moment.

226: How many ships did the USS Pampanito sink and what was her war record?

USS Pampanito (SS-383) was commissioned on November 6, 1943, at Portsmouth Naval Shipyard. She conducted six war patrols before the end of World War II, earning a combat record that earned her a Presidential Unit Citation and four battle stars.

Confirmed sinkings: JANAC (Joint Army-Navy Assessment Committee) credited USS Pampanito with sinking five Japanese vessels, including merchant ships and a transport, totaling approximately 27,000 tons of shipping. She also damaged several additional ships.

Key patrols: Pampanito's third and fourth war patrols (1944, under Commander Paul E. Summers) were her most distinguished. She operated as part of a coordinated attack group with USS Sealion II and USS Growler in the South China Sea. On September 12, 1944, the group attacked Convoy HI-72 — a heavily loaded Japanese convoy. Pampanito and her partners sank multiple ships.

The POW rescue: The convoy's cargo ships turned out to be carrying British and Australian prisoners of war taken from Singapore. Many were on deck when the ships were attacked. After the convoy fight, Pampanito surfaced the following morning and found men in the water. She rescued 73 POWs, then sent messages bringing USS Sealion II back to rescue additional survivors. This was one of the most poignant episodes of the entire submarine war — rescuing men whose captors she had just sunk.

Post-war: Pampanito was decommissioned in 1946. After years in the reserve fleet, she was donated to the National Maritime Museum Association and opened as a memorial submarine and museum ship at Fisherman's Wharf, San Francisco, in 1982, where she remains today.